

SOFTWARE REQUIREMENT SPECIFICATIONS (SRS)

FOR

REVAMPING THE BI DASHBOARD WEB APPLICATION DESIGN & DEVELOPMENT

SUBMITTED TO

IVTECH COMPUTER SYSTEMS LLC

Version: 1.0

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Prepared By:

SPEC INDIA

Revision History

Date	Version	Author	Created / Updated / Reviewed / Verified / Approved By
23-05-2025	1.0	Parth Jotangiya	Initial Draft
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Table of Contents

Contents

1.	Introduction.....	5
1.1.	<i>Project Scope and Objectives.....</i>	5
1.2.	<i>Items outside the Project Scope</i>	8
1.3.	<i>Project Deliverables.....</i>	8
1.4.	<i>Definitions, Acronyms, and Abbreviations</i>	9
1.5.	<i>References.....</i>	10
1.6.	<i>Project Approach</i>	10
1.7.	<i>Security, Best Practices & Supported versions</i>	11
2.	User Descriptions.....	11
3.	System Requirements.....	12
3.1.	<i>Settings > Manage Users Page.....</i>	12
3.2.	<i>Settings > Roles & Permissions Page</i>	15
3.3.	<i>Settings > Manage Users > Role Assignments to a Specific User</i>	20
3.4.	<i>Settings > Multisite Setup Page.....</i>	21
3.5.	<i>Settings > General Page</i>	27
3.6.	<i>Settings > License</i>	29
3.7.	<i>Login Page.....</i>	32
3.8.	<i>Menu Items for BI Dashboard Web Application</i>	34
3.9.	<i>Header Section > Profile Page</i>	36
3.10.	<i>Header Section > Change Password Page</i>	37
3.11.	<i>Header Section > Preferences Page</i>	38
3.12.	<i>Header Section > Notifications</i>	39
3.13.	<i>Configurations > Manage Devices Page</i>	41
3.14.	<i>Configurations > Floor Plans & Zones Page.....</i>	47
3.15.	<i>Events > Event Logs Page.....</i>	54
3.16.	<i>Reports > My Reports Page</i>	57
3.17.	<i>Monitoring > Monitoring Page.....</i>	63
3.18.	<i>Dashboards > Dashboard Creation</i>	66
3.18.1.	<i>Dashboards > Dashboard Page > Add Widget Screen</i>	67
3.18.2.	<i>Dashboards > Dashboard Page > Filters & Export Option.....</i>	70

3.18.3.	<i>Dashboards > Dashboard Page > Widgets</i>	73
3.18.4.	<i>Dashboards > Dashboard Page > Widget > Setup Page</i>	143
3.18.5.	<i>Dashboards > Dashboard Page > Widget > Export as CSV Option</i>	145
3.18.6.	<i>Dashboards > Dashboard Page > Widget > Duration Selection Filter Option</i>	146
3.18.7.	<i>Dashboards > Dashboard Page > Widget > Expand Option</i>	147
4.	System Architecture	149
5.	Assumptions, Dependencies and Constraints	151

1. Introduction

1.1. Project Scope and Objectives

- The purpose of this project is to **revamp an existing BI Dashboard Web Application** by enhancing the user interface & experience which will require minimal user's inputs to perform the required steps.
- This requirement includes enhancing the current application with existing features and functionalities as specified by the client.
- These enhancements will help their customers to view widgets & manage their site wise configurations effectively.
- This application is tailored for their customers. It will be web-based and accessible exclusively through web browsers. Additionally, it will provide the capability to be installed on a customer's local machine via a single executable file (e.g.: **.exe / .msi**).
- The **executable file** will **host the application on IIS** of the **local machine** for the specific client environment. The application will then be **accessible within the client's environment via the LAN network**.
- The **.exe** or **.msi** file will handle the automatic installation of all necessary files, configurations, and dependencies on the customer's local machine. Once installed, the BI Dashboard Web Application will be executed through the customer's web browser, operating within their local environment only.
- This document outlines the requirements for **BI Dashboard Web Applications** to be developed for **IVTECH COMPUTER SYSTEMS LLC**.
- The Requirement is present in the "**Technical Requirements Doc R2**" document shared by **IVTECH COMPUTER SYSTEMS LLC**, which contains information for **BI Dashboard Web Application** modules related to the following:
 - User Login:
 - Users can log in using their credentials.
 - An admin user can reset any user's password (Forgot password).
 - Admin users can change passwords for any specific user.
 - User Management:
 - Add/Update/Delete users.
 - Assign Role - Permissions to Users to access camera/floor/zone/reports.
 - Camera Management:
 - Add/Update/Delete camera.
 - Listing of Cameras with search/filter.
 - Settings:
 - Upload Customer Logo.
 - License Management.
 - Site and Zone Management:
 - Add/Update/Delete Site.
 - Add/Update/Delete Floor Plan.
 - Add/Update/Delete Zone on Floor Plan.
 - Add cameras to zones.
 - Camera rotation.

- Map integration with static map images.
 - **Imp Note:**
 - For the Map Integration, the user needs to be online once, maybe while creating the Site to download the map images. Otherwise, we cannot identify which country/region/area images need to be downloaded to make the map work with static images.
 - For now, we will go for Open source to achieve the functionality of Offline maps. This is the subject of exclusive R&D; hence the final selection of the open-source solution will be done post R&D and discussion with the client team.
- Dashboard:
 - Report Filtering
 - Export to PDF/excel/csv.
 - Camera – No. of Camera Installed
 - Camera – No. of Camera Online/Offline
 - Camera - Online vs Offline Ratio
 - Camera – No. of Camera by Model Type
 - Camera – No. of Camera by Feature type
 - Site - Total Capacity
 - Site - Capacity Utilization (%)
 - Site - Most vs. Least Day for Capacity Utilization
 - Site - Zone-wise details for Max Capacity, Utilization, % (Table)
 - People Counting - People In
 - People Counting - People Out
 - People Counting - Avg. People In/Out in Day with Min and Max Count
 - People Counting - Cumulative People Count for Time
 - People Counting - Zone wise.
 - People Counting - Gender-wise.
 - People Counting - People with and without Mask/Helmet
 - People Counting - Most and Least Count Day for Specific Male and Female
 - People Counting - Slip and Fall Count for People with Gender
 - People Counting - New vs Total Visitors
 - Retail & Business AI Pack - Shopping Cart Counting
 - Retail & Business AI Pack - Heatmap for shopping cart
 - Retail & Business AI Pack - Queue events for shopping cart
 - Retail & Business AI Pack - Queue events for people
 - Retail & Business AI Pack - Blocked exit detection
 - Vehicle (Traffic (ITS) AI pack) - Vehicle Count (Min, Max, Avg. per Day)
 - Vehicle (Traffic (ITS) AI pack) - Vehicle by Type
 - Vehicle (Traffic (ITS) AI pack) - Vehicle In/Out
 - Vehicle (Traffic (ITS) AI pack) - Vehicle in Wrong Direction
 - Vehicle (Traffic (ITS) AI pack) - Stopped Vehicle Count Time

- Vehicle (Traffic (ITS) AI pack) - Speed Detection by Vehicle
- Vehicle (Traffic (ITS) AI pack) - Avg. Speed for Vehicle
- Vehicle (Traffic (ITS) AI pack) - Traffic Jam by Day
- Vehicle (Traffic (ITS) AI pack) - Day-wise Traffic (Most and Least)
- Vehicle (Traffic (ITS) AI pack) - Speed Violation by Vehicle
- Vehicle - Vehicle U Turn detection
- Vehicle (Traffic (ITS) AI pack) - Pedestrian detection
- Vehicle (Traffic (ITS) AI pack) - Vehicle Turning Movement counts
- Vehicle (Traffic (ITS) AI pack) - Vehicle Queue Analysis
- Vehicle (Traffic (ITS) AI pack) - Vehicle Detection Heatmap
- Factory & Safety AI pack - Detect Forklifts
- Factory & Safety AI pack - Proximity detection (between people, forklifts, between a person and a forklift)
- Factory & Safety AI pack - Forklift speed detection
- Factory & Safety AI pack - Counting & Heatmap for forklift and people
- Factory & Safety AI pack - Queue events for forklift and people
- Factory & Safety AI pack - Blocked exit detection
- The list of dashboard widgets specified above includes existing APIs related to Crowd Management, Queue Management, Traffic (Vehicle), Retail, and Factory AI Pack. Which will be provided by the client only.
- Package creation:
 - Create a package to install the web application on Windows PC only.
- Data Ingestion:
 - Scheduled calling of Camera APIs to ingest data in the database.
- Camera API Understanding and Integration:
 - Details of the Solar Power and Battery Utilization will be shown on the dashboard through SNMP/API integration with the BMS and MPTT Controller (***calling of camera API and just displaying the data available from API***).
- Configuration:
 - Standard Backup and Restore.
- AI Box Integration:
 - AI Box integration, considering the addition of multiple cameras in a single AI Box and considering the cameras added in the AI Box to display Dashboard widgets data.
- Alexa Integration:
 - Integration with Alexa, so that when the Alexa device is connected, it will open the specific widget on voice command.
- 7 other key features:
 - Provide data for multiple lanes (8 lanes).
 - Show heat maps on the floor plans and particular zones.
 - Multisite-wise dashboard widget reports.
 - For Multisite should be filtered to select the widgets and show the difference between Full site, single-zone, and Multizone.

- SSL / HTTPS set up, (***The SSL certificate will be purchased by client only***).
- Need to have Event reporting details with filtration with separate widgets or separate pages for this, and when the event is reported, need to have pre- and post-video of 5 seconds each.
- SMTP server setup for email reporting with scheduled job (***Required credentials will be provided by client only***).

➤ **The technology used to develop web applications will be:**

Front-end	React.js
Back-end	.NET Core + Web Api (using C#)
Database	MongoDB

1.2. Items outside the Project Scope

- **Creation of APIs which will be directly interacted with AI camera managed by Hanwha.**
- **SPEC INDIA will only utilize camera APIs managed by Hanwha** and will replicate/ingest the necessary information in the database (if required) & display that required information on the web application.
- **Multilingual support** for web application. Application will be only in English language.
- **Responsiveness to Tablet & Mobile.** Application will be supported to desktop computer screen only.
- **The development of Mobile Application** for Android and iOS native integration will be out of scope.
- **QA Testing is out of scope** because client team will do it from their end.
- **Failover plan** for the database in case of any future failures.

1.3. Project Deliverables

- SPEC INDIA will be delivering below items as part of project deliverables:
 1. System Requirements Specification (SRS) Document.
 2. UI/UX (Figma) Design.
 3. Source Code files.

1.4. Definitions, Acronyms, and Abbreviations

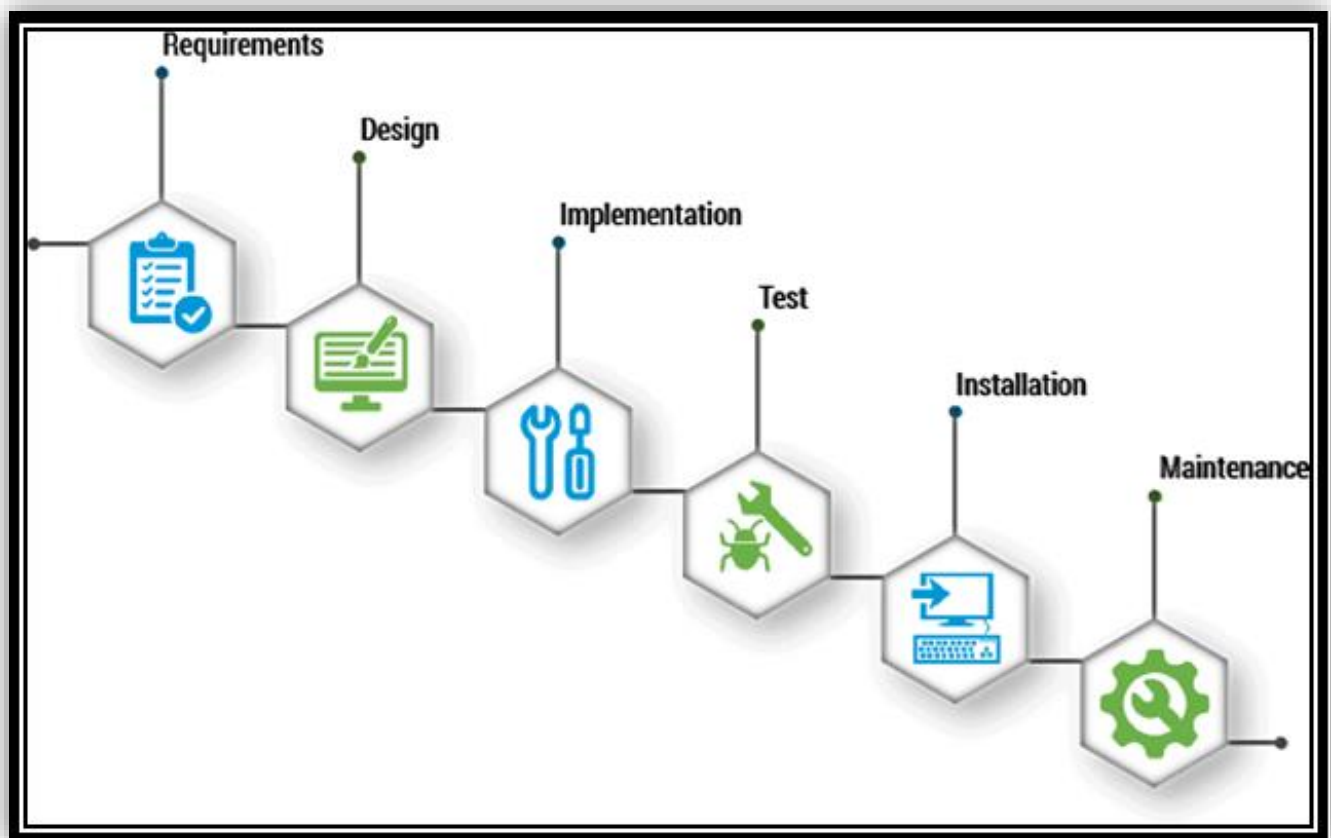
Definitions, Acronyms and Abbreviations	
React.js	A JavaScript library for building user interfaces, particularly single-page applications (SPAs) with component-based architecture.
JSX	JavaScript XML, a syntax extension for JavaScript used in React to describe UI elements in a syntax like HTML.
Component	A reusable UI element in React that can be a class-based or functional component.
State	An object that stores dynamic data in a React component, influencing its rendering.
Props	Short for "Properties," used to pass data between React components.
MongoDB	A NoSQL database that stores data in JSON-like documents, known for its flexibility and scalability.
Collection	A group of MongoDB documents, like a table in relational databases.
Document	A single record in a MongoDB collection, stored in BSON format.
Schema	A structure defining the shape of documents in MongoDB, typically enforced using Mongoose in Node.js applications.
.NET Core Web API	A cross-platform, open-source framework for building RESTful web services using C# and .NET Core.
Controller	A class in .NET Core Web API that handles HTTP requests and responses.
Middleware	Software components in .NET Core that process HTTP requests and responses in a pipeline.
Entity Framework (EF) Core	An Object-Relational Mapper (ORM) for .NET Core, enabling database operations using C#.
DTO (Data Transfer Object)	A design pattern in .NET used to transfer data between layers of an application.
REST API	Representational State Transfer API, a web service architecture used for communication between clients and servers.
JWT (JSON Web Token)	A compact, URL-safe token used for authentication and authorization in web applications.

1.5. References

- Based on discussions and conclusions regarding the **BI Dashboard Web Application** with **Bala & Team**, the SPEC Team has applied their expertise to create a fresh and modern **Figma design** which is also approved by Bala & Team.
- Please find the attached link below for the **Light Theme**:
 - [Click here to Open newly created Figma for Light Theme](#)
- Please find the attached link below for the **Dark Theme**:
 - [Click here to Open newly created Figma for Dark Theme](#)

1.6. Project Approach

- The proposed approach for executing this project is the Waterfall approach, in which requirements will be gathered, analyzed, and baselined to form the system's scope:



1.7. Security, Best Practices & Supported versions

- The BI Dashboard Web Application will be protected by **cyber security and vulnerability**.
- The BI Dashboard Web Application will work on **desktop and laptop screens**; it is not mobile & tablet responsive. It will work on the latest versions of Google Chrome, Mozilla Firefox & Safari.
- The browser version at the time of UAT release will be considered as the latest support version.

2. User Descriptions

Sr. No	User Type	Activities in the system
1	System User	- This user will exist only in the database, and the system will utilize this user's ID for data insertion or modification from the backend only. - This user will not display in the BI Dashboard Web Application. Example: Data Ingestion
2	Super Admin	- This user will be the main user for the BI Dashboard Web Application. The source code will automatically configure the required information for this user, including roles and permissions, when deploying the application in a specific environment. - This user will have full access.
3	Sub-user	- A sub-user is a user created by an authorized user within the system. - This user can be assigned any existing or newly created role by an authorized user. - This user will have access to the application and its features based on the assigned permissions.

3. System Requirements

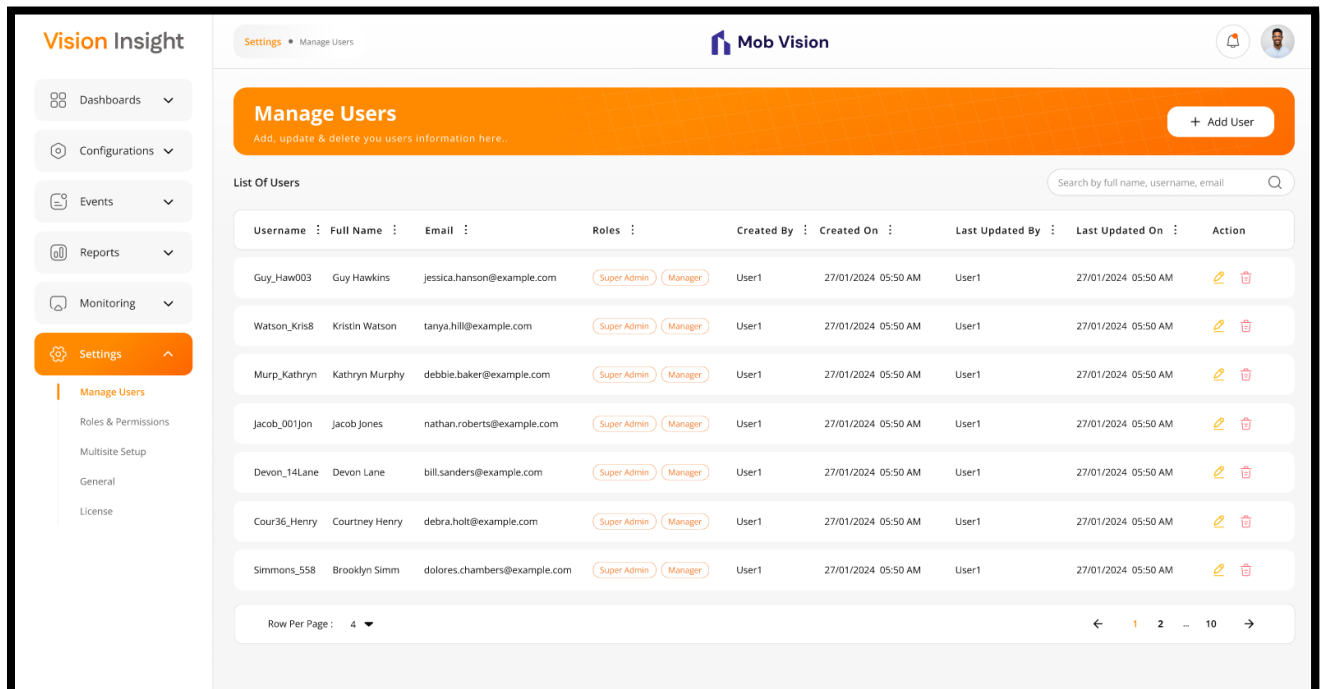
3.1. Settings > Manage Users Page

There is a provision in the system for the **Super Admin or authorized users** to **create and manage users**. The **Super Admin** has **system-level access** and can manage all **sub-users**, their **roles**, and **assigned permissions**.

The system provides functionality for **Super Admin or authorized users** to **create and update sub-user profiles and their associated information**. This includes all **basic and essential details** required for managing sub-user access within the system.

Prototype:

➔ The screen below illustrates the list of all created users, including options for Search, Add, Edit, and Delete:



Manage Users
Add, update & delete you users information here..

+ Add User

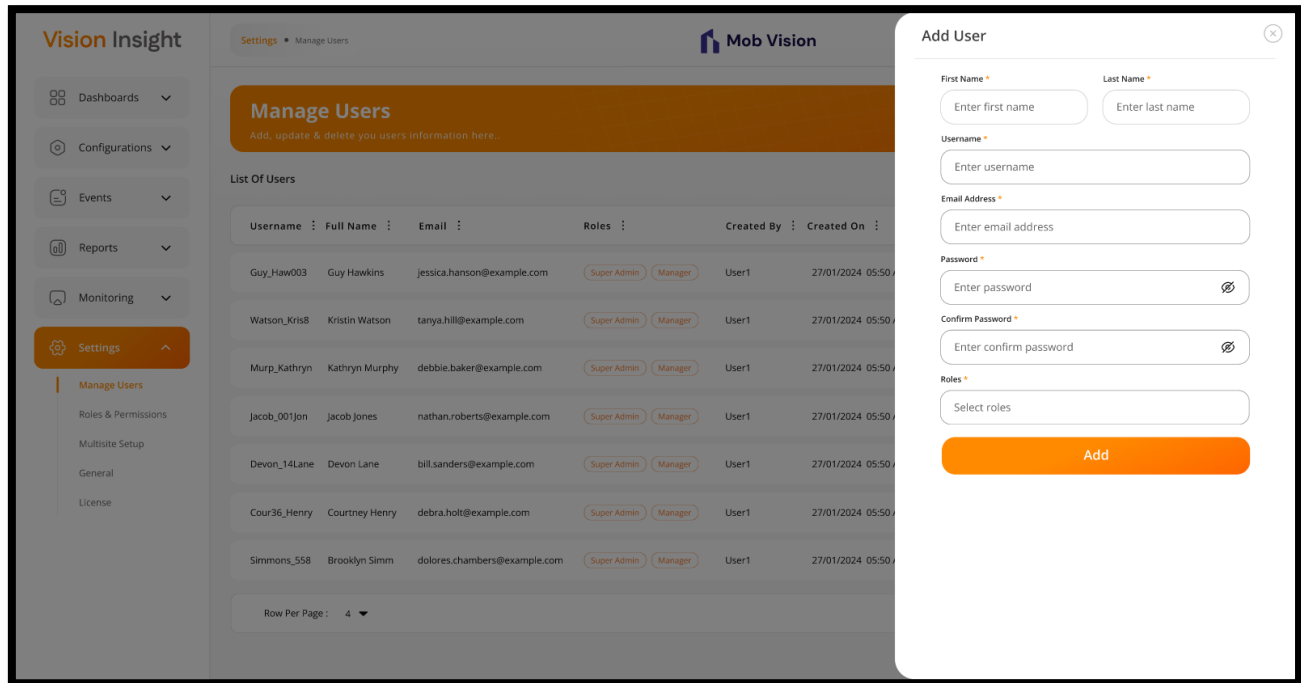
Search by full name, username, email

Username	Full Name	Email	Roles	Created By	Created On	Last Updated By	Last Updated On	Action
Guy_Haw003	Guy Hawkins	jessica.hanson@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Watson_Kris8	Kristin Watson	tanya.hill@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Murp_Kathryn	Kathryn Murphy	debbie.baker@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Jacob_001Jon	Jacob Jones	nathan.roberts@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Devon_14Lane	Devon Lane	bill.sanders@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Cour36_Henry	Courtney Henry	debra.holt@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	
Simmons_558	Brooklyn Simm	dolores.chambers@example.com	Super Admin Manager	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	

Row Per Page: 4

1 2 10

- The screen below represents the form for creating a new user in the BI Dashboard Web Application, allowing users to enter relevant details and submit the information:



Manage Users
Add, update & delete you users information here.

List Of Users

Username	Full Name	Email	Roles	Created By	Created On
Guy_Haw003	Guy Hawkins	jessica.hanson@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Watson_Kris8	Kristin Watson	tanya.hill@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Murp_Kathryn	Kathryn Murphy	debbie.baker@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Jacob_001Jon	Jacob Jones	nathan.roberts@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Devon_14Lane	Devon Lane	bill.sanders@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Cour36_Henry	Courtney Henry	debra.holt@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Simmons_558	Brooklyn Simm	dolores.chambers@example.com	Super Admin, Manager	User1	27/01/2024 05:50

Row Per Page: 4

Add User

First Name *
Enter first name

Last Name *
Enter last name

Username *
Enter username

Email Address *
Enter email address

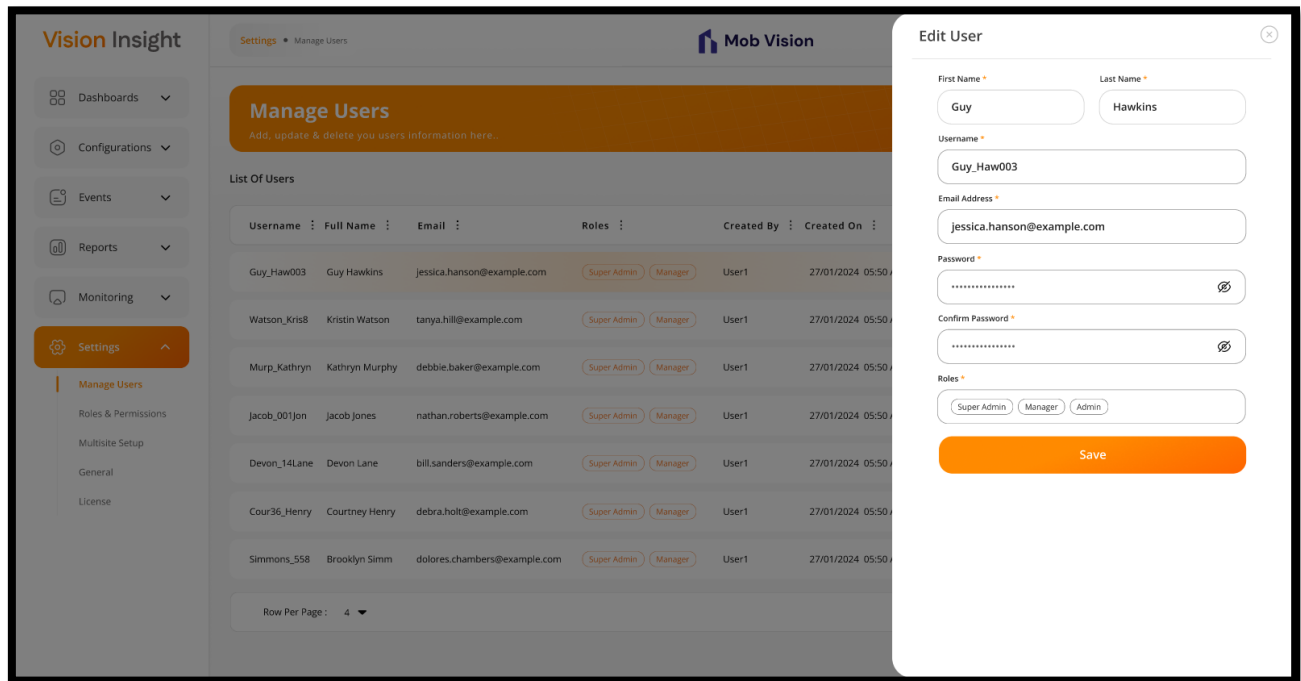
Password *
Enter password

Confirm Password *
Enter confirm password

Roles *
Select roles

Add

- The screen below displays the form for editing an existing user in the BI Dashboard Web Application, allowing users to modify relevant details and update the information. **Since** all users cannot change/reset their passwords independently, the "Password" and "Confirm Password" fields will be visible while editing user details & the authorized user can edit the information of specific user:



Manage Users
Add, update & delete you users information here.

List Of Users

Username	Full Name	Email	Roles	Created By	Created On
Guy_Haw003	Guy Hawkins	jessica.hanson@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Watson_Kris8	Kristin Watson	tanya.hill@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Murp_Kathryn	Kathryn Murphy	debbie.baker@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Jacob_001Jon	Jacob Jones	nathan.roberts@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Devon_14Lane	Devon Lane	bill.sanders@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Cour36_Henry	Courtney Henry	debra.holt@example.com	Super Admin, Manager	User1	27/01/2024 05:50
Simmons_558	Brooklyn Simm	dolores.chambers@example.com	Super Admin, Manager	User1	27/01/2024 05:50

Row Per Page: 4

Edit User

First Name *
Guy

Last Name *
Hawkins

Username *
Guy_Haw003

Email Address *
jessica.hanson@example.com

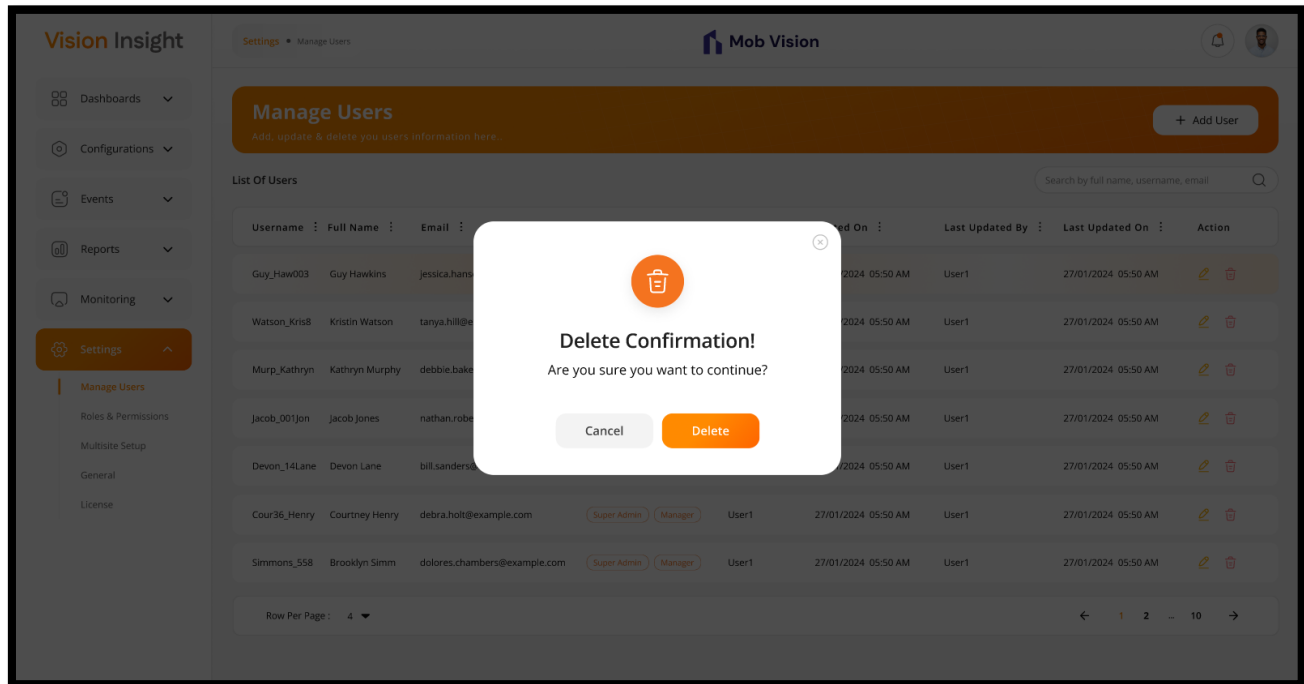
Password *

Confirm Password *

Roles *
Super Admin, Manager, Admin

Save

→ The system will request confirmation when a logged-in user attempts to delete any authorized user:



Business Rules:

- The **authorized user** will have the ability to **create** and **update** sub-users.
- Password field will include following criteria:
 - Minimum password length: 8 characters
 - At least 1 uppercase letter
 - At least 1 lowercase letter
 - At least 1 numeric digit
 - At least 1 special character (excluding *)
- Authorized users will be able to **assign multiple roles** to the specific user.
- The "Add User" & "Edit User" screen will include following fields:
 - First Name*
 - Last Name*
 - Username*
 - Email Address*
 - Password*
 - Confirm Password*
 - Roles*
- Since the super user or authorized user can change the password of any specific sub-user, the **"Password"** and **"Confirm Password"** fields should be displayed when editing user details for them only.
- The asterisk (*) indicates a mandatory field only.
- The system will **not allow duplicate email addresses or usernames**, as they must always be unique. These identifiers are essential for user authentication within the system.
- When a logged-in user (super user or authorized) attempts to delete another user, the system will

prompt for confirmation. Once confirmed, the system will **soft delete** the user, meaning the record will remain in the database but will no longer be visible in the web application.

- The system will offer filter options (**like Sort By ASC, Sort by DESC, Filter, Hide Column, Manage Columns**) on specific columns using the MUI library, which is integrated into the entire web application. These filters will reset when the user reloads the page or navigates to another page.

Data Dictionary:

userMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	username	String	No	Admin	Username is use for login
3	firstname	String	Yes	Parth	First name of the user
4	lastname	String	Yes	Jotangiya	Last name of the user
5	email	String	No	xxxxx@xxxx.xxx	Email address of the user
6	password	String	No	aasdshfbshdf	It will be stored in encrypted format only
7	roleIds	ObjectId[]	Yes	[674daaf7d34a2b901c847454]	Unique Id(s) column of the "roleMaster" table, allow multiples.
8	ProfileImage	String	Yes	base64://	User profile picture in base64 format
9	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
10	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
11	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
12	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
13	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.2. Settings > Roles & Permissions Page

The system allows **super admins** or **authorized users** to **create and manage** sub-user roles and permissions via the **BI Dashboard Web Application**.

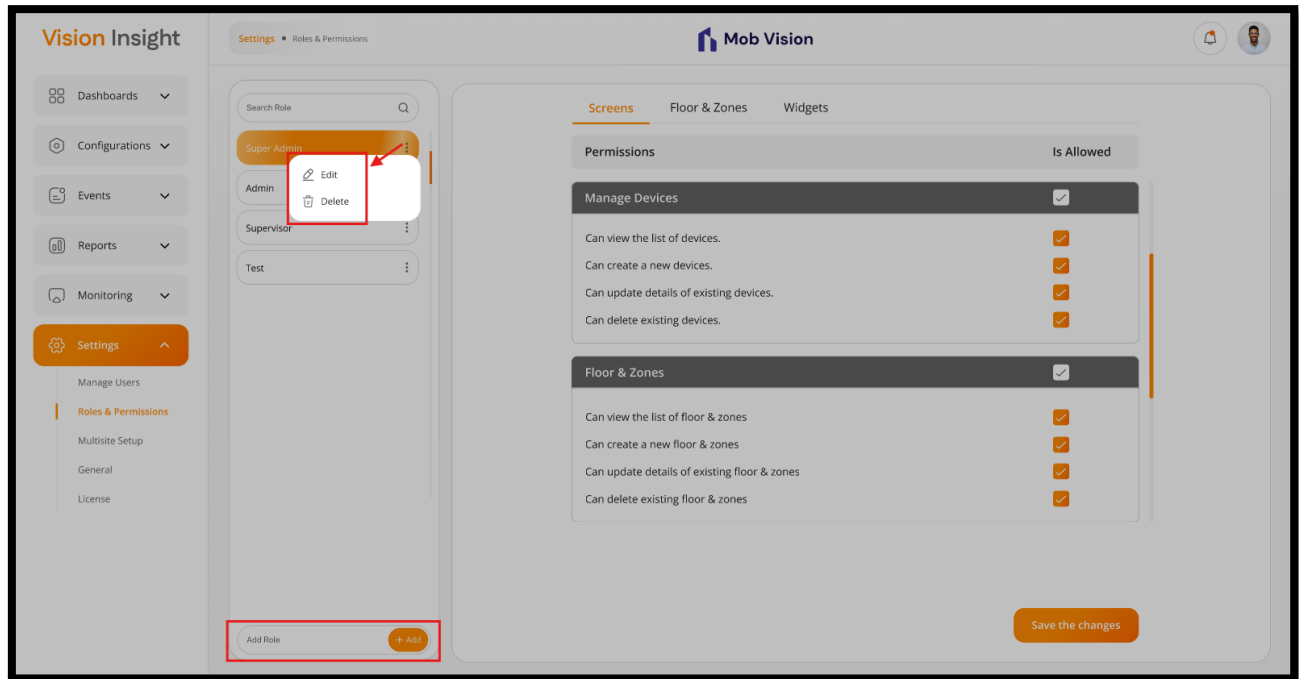
Permissions can be assigned at the **screen level** with access types such as **add, update, delete, and view**, as well as for **zones, floors, and widgets**. **Authorized users** can define **roles with specific permissions** and assign them to **sub-users**. Upon login, users will be able to access **only those functionalities permitted** by their assigned role. This screen enables authorized users to **update role-based permissions** for **screens, actions, zones, floors, and widgets**.

Regarding Floor Plans & Zones Permissions:

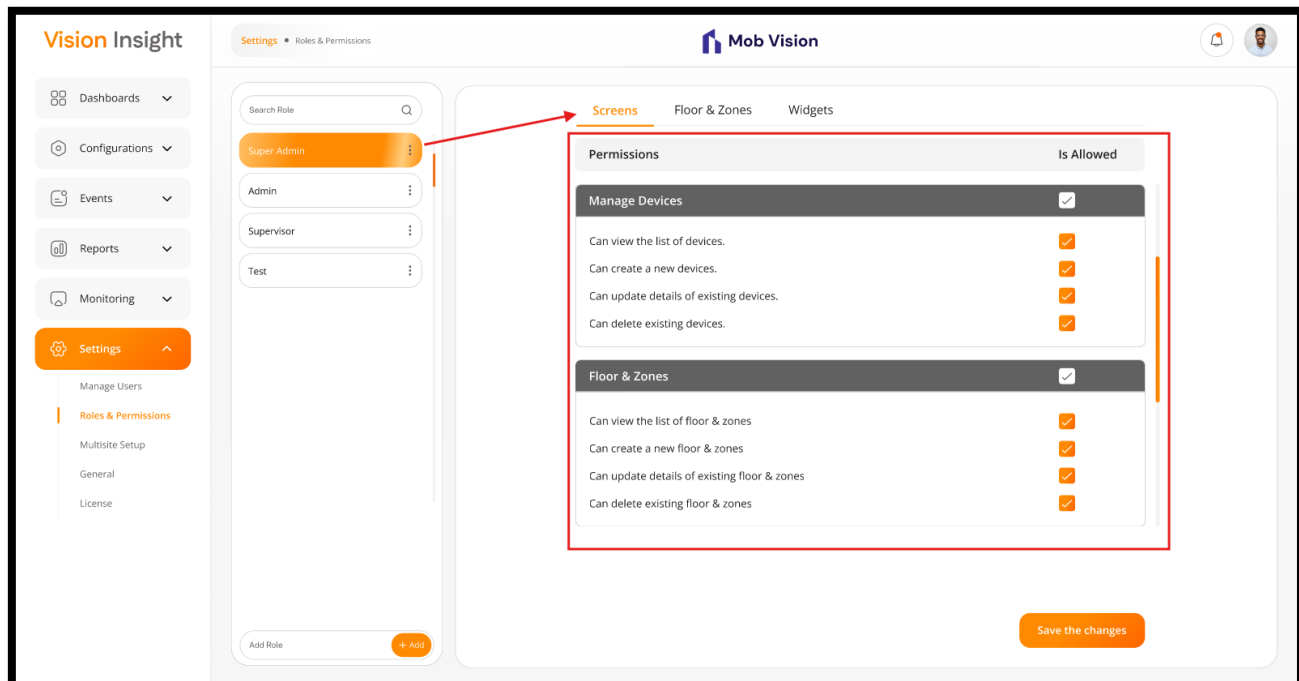
- The system will maintain **permissions at the Floor Plan and Zone levels only**.
- It will be the responsibility of the **admin user** (who configures the application, users, and user permissions) to assign **appropriate permissions** for specific **floor plans and zones** to each user - especially in cases where a **single camera is configured in multiple zones**.
- As per **Bala's request**, the system will **not handle this scenario automatically**.
- The system will display information **strictly based on the assigned permissions** at the **Zone and Floor Plan** levels.

Prototype:

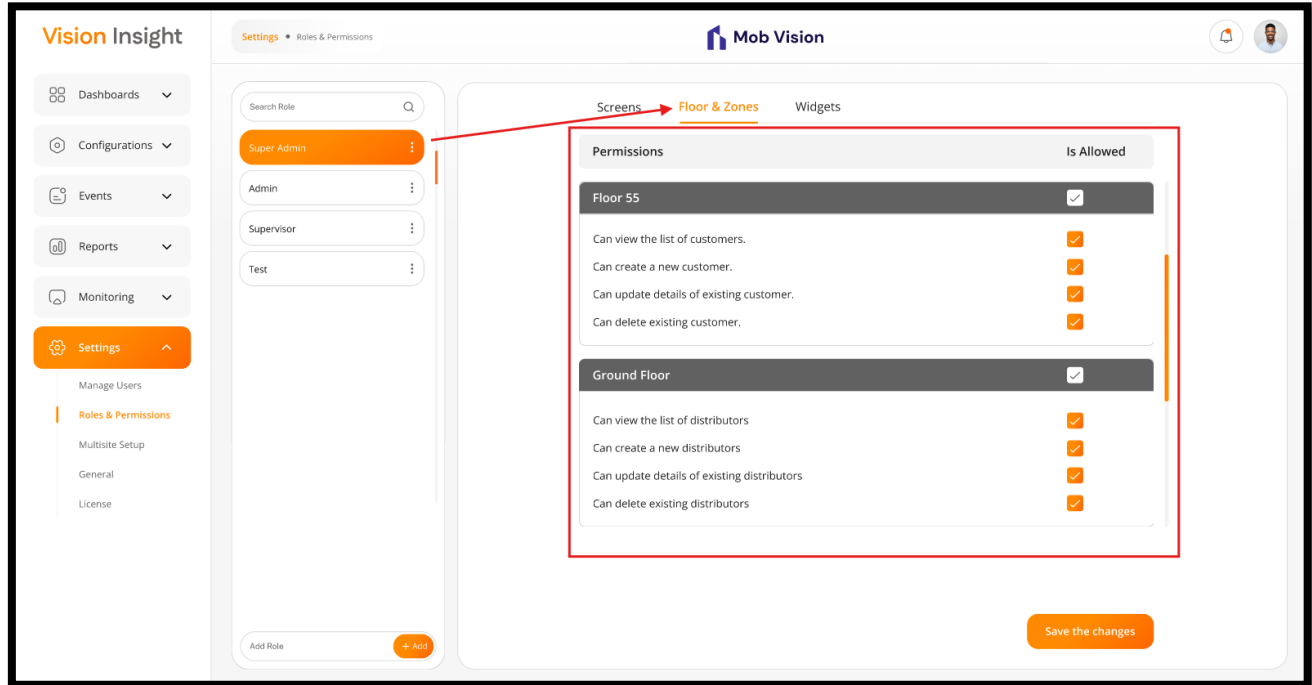
→ The screen below displays the list of all created roles, with options for Search, Add, Edit, and Delete:



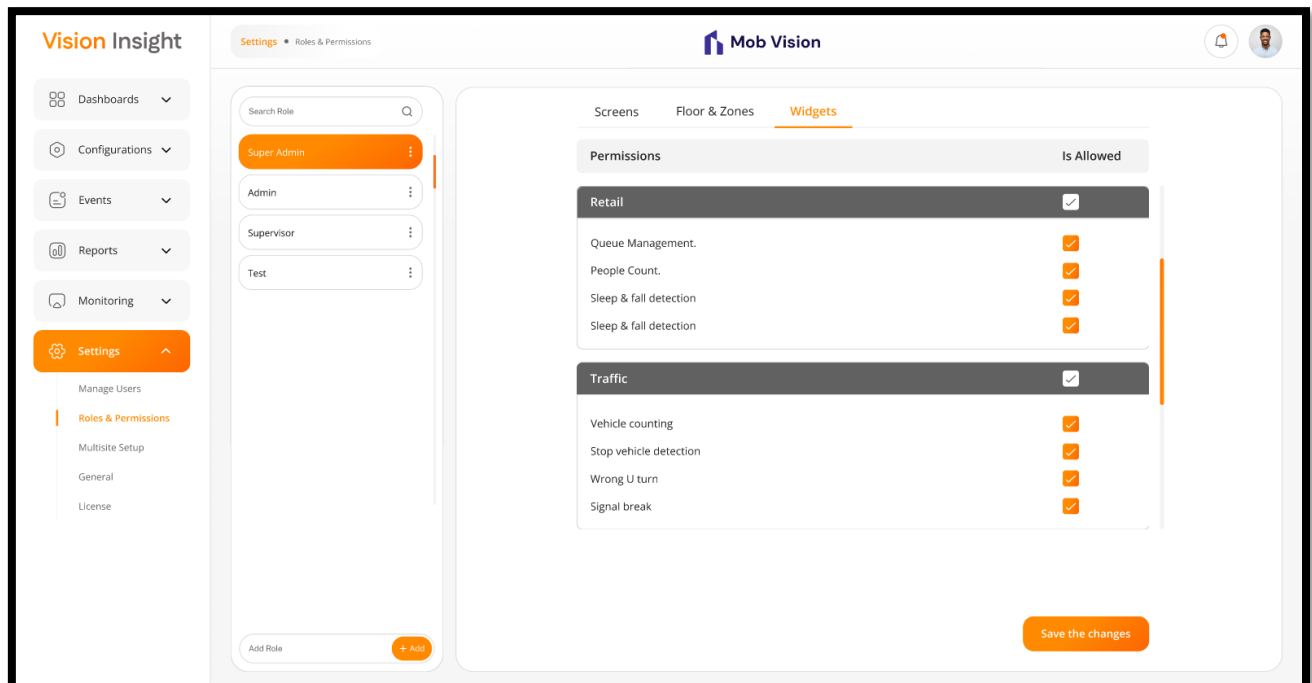
→ The screen below represents the **screen-level** permissions for the selected role:



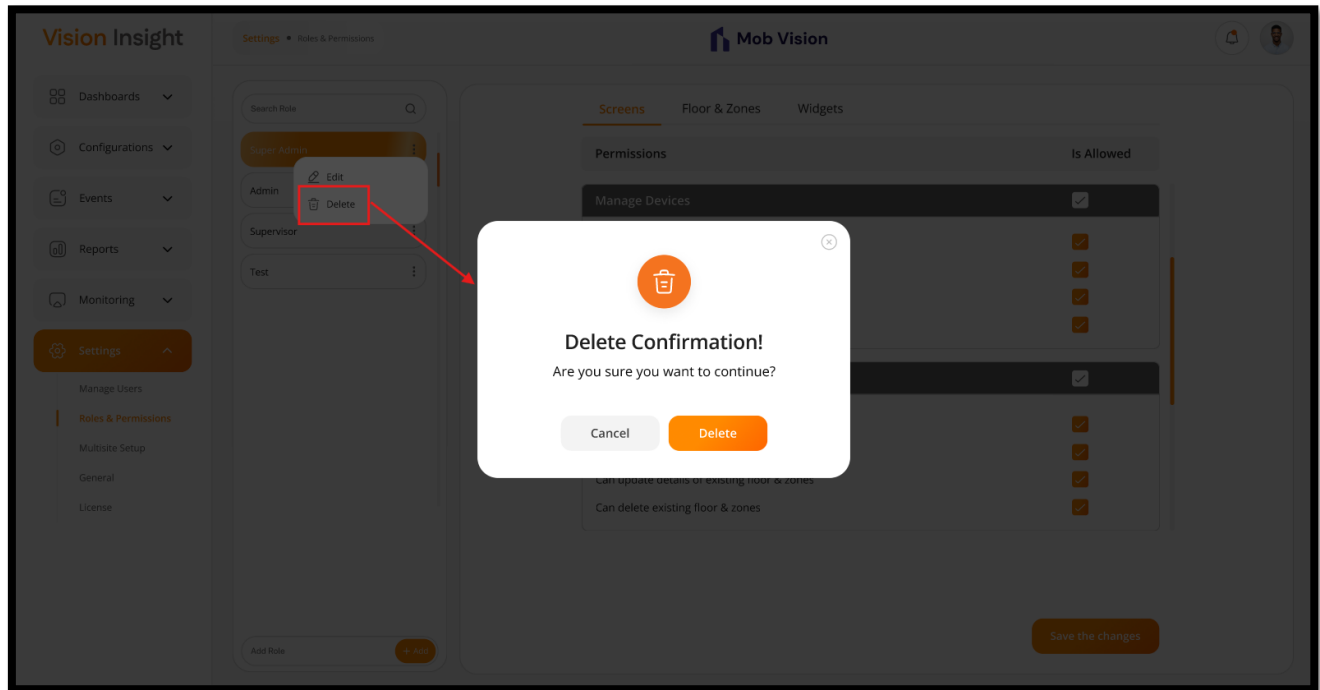
→ The screen below represents the **Floor & Zone-level** permissions for the selected role:



→ The screen below represents the **widgets-level** permissions for the selected role:



→ The system will request confirmation when a logged-in user attempts to delete any role:



Business Rules:

- Authorized users will have the ability to **create, update & delete the roles** within the system.
- The system will **not allow the deletion of a role** if it is already associated with any user.
- The system will **prevent duplicate entries** for roles or user-role assignments.
- When a logged-in user attempts to delete role, the system will prompt for confirmation. Once confirmed, the system will **soft delete** the user, meaning the record will remain in the database but will no longer be visible in the web application.
- The “**Add Role**” & “**Edit Role**” will include below field:
 - Role Name*
- The asterisk (*) indicates a mandatory field only.
- Authorized users can assign or update permissions for specific roles, defining access to various system functionalities.
- The system will ensure that permissions are role-based and cannot be assigned to individual users directly.
- The system should allow access only on those screens / actions / floor plans / zones / widgets which are allowed against the assigned role for logged in sub user.
- By Default, the super admin user will have full access to the admin application.
- The super admin / authorized user can assign screen-level permissions and corresponding actions (e.g., add, update, delete, view) to roles. However, the super admin / authorized user cannot assign permissions to actions that are not applicable to specific screens (**For example, Insert/Update/Delete rights are not applicable to screens like Profile, Change Password and Preferences. Therefore, the system should restrict the super admin / authorized user from assigning these permissions to roles for such screens**).
- The system will maintain the history of each change in the permissions.
- Any updates to role permissions will take effect only after users associated with that role login again.

Data Dictionary:

roleMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	roleName	String	No	Admin	Name of the role
3	description	String	Yes	Admin	description of the role
4	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
7	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

screenMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	screen_name	String	No	Customer Creation	Name of the screen
3	is_active	Boolean	No	TRUE	Active or Inactive
4	parent_screen_id	ObjectId	Yes	674daaf7d34a2b901c847454 / null	Unique Id of this table. To identify reference of parent row.
5	sequenceNo	Int32		1	To identify the sequence of child permissions.
6	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
7	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
9	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

➔ An example of the **screenMaster** table is attached below to provide better clarity on the backend implementation:

	Screen_Name String	Parents_Screen_Id Mixed	Is_Active Boolean
1	"User Master"	null	true
2	"Can View User"	"60c72b2f9b1d8e001f8e4b1..."	true
3	"Can Add Or Update User"	"60c72b2f9b1d8e001f8e4b1..."	true
4	"Can Delete the User"	"60c72b2f9b1d8e001f8e4b1..."	true
5	"Client Master"	null	true
6	"Can Add Or Update Clie..."	"60c72b2f9b1d8e001f8e4b2..."	true
7	"Can View Client"	"60c72b2f9b1d8e001f8e4b2..."	true
8	"Can Delete the Client"	"60c72b2f9b1d8e001f8e4b2..."	true
9	"Role Master"	null	true
10	"Can View Role"	"60c72b2f9b1d8e001f8e4b3..."	true
11	"Can Edit Role"	"60c72b2f9b1d8e001f8e4b3..."	true
12	"Can Add Or Update Role"	"60c72b2f9b1d8e001f8e4b3..."	true
13	"Can Add Or Update Permi..."	"60c72b2f9b1d8e001f8e4b3..."	true
14	"Can View Permission"	"60c72b2f9b1d8e001f8e4b3..."	true

The above table lists all permissions available in the system, categorized by pages and their respective actions, floors and zones, and associated widgets.

For example, the **User Creation** screen will have **View, Add, Edit, and Update** actions. Therefore, the following entries will exist in the table for the **Users** screen:

User Master
-> Can view user list
-> Can edit user details
-> Can add or update user details
-> Can delete user

Role & Screen Mappings: The table below provides details on screen-wise, floor & zone-level, and widget-wise permissions assigned to specific roles:

roleScreenMapping					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	Objectid	No	674daaf7d34a2b901c847454	Autogenerated id
2	roleId	Objectid	No	674daaf7d34a2b901c847454	Role id for screen mapping
3	screenMappings	Objectid[]	No	{{"screenId": "674daaf7d34a2b901c847454", "access_allowed": true}}	Contains the array of screenIds & allowed permissions.
3.1	> screenId	Objectid	No	674daaf7d34a2b901c847454	ScreenId is mapped with 'screenMaster' collection
3.2	> access_allowed	Boolean	No	TRUE	Boolean value for allowing or restricting screen feature
4	dataAccessPermissions	Object[]	No	{{"floorId": "67bc339394b4aeb0ddf5d52a", "zoneIds": ["67bc339394b4aeb0ddf5d526"]}}	List of floor and zones as for data access permission
4.1	> floorId	String	No	674daaf7d34a2b901c847454	Floor id for the data access permission
4.2	> zoneIds	String[]	No	[674daaf7d34a2b901c847454]	Zones for data access permission
5	widgetAccessPermissions	Object[]	No	{{"widgetCategoryId": "67bc339394b4aeb0ddf5d52a", "widgetIds": ["67bc339394b4aeb0ddf5d526"]}}	List of widget category and their widgets for widget access permission
5.1	> widgetCategoryId	String	No	674daaf7d34a2b901c847454	Widget category id for widget access permission
5.2	> widgetIds	String[]	No	[674daaf7d34a2b901c847454]	Zones for data access permission
6	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
7	createdBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
9	updatedBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.3. Settings > Manage Users > Role Assignments to a Specific User

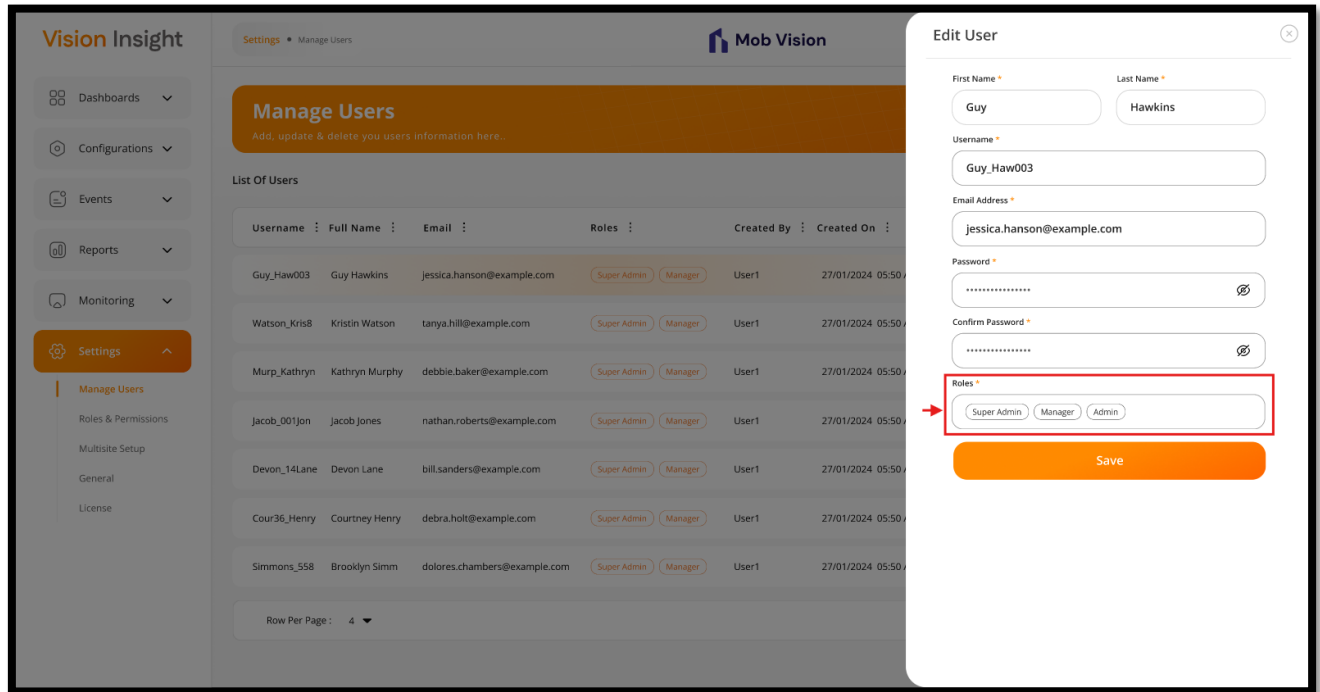
There is a provision in the system for **Super Admin** or **authorized users** to **manage role assignments** for specific users through the **BI Dashboard Web Application**.

Once a **role is defined**, it can be **assigned to users** based on their **designation and responsibilities**. This **role assignment** process will be handled by **Super Admin** or **authorized users**.

The system will allow access **strictly based on the role assigned** to the **logged-in user**, ensuring that users can access only the **functionalities permitted** by their assigned role.

Prototype:

- ➔ The screen below illustrates how an authorized user can assign one or multiple roles to a specific user while creating or editing a user from the Users page:



Business Rules:

- Authorized users can select and assign specific roles from a dropdown list to any sub user.
- Authorized users can **assign one or multiple roles** to a specific user from the **Users Page**.
- Authorized users can **update or modify user roles assignment** from the **Users Page only**.
- Any changes to the role assignment will immediately be reflected after relogging to the user after they log out and log back into the system.

Data Dictionary:

- ➔ The system will store the selected unique **roleId(s)** from the **roleMaster** table in the **userMaster** table.

3.4. Settings > Multisite Setup Page

The system allows the **parent site's authorized / super admin users** to manage **single or multiple child and sub-child site configurations** through the **BI Dashboard Web Application**.

The system will support multisite configuration up to three levels (**Parent, Child & Sub-Child only**).

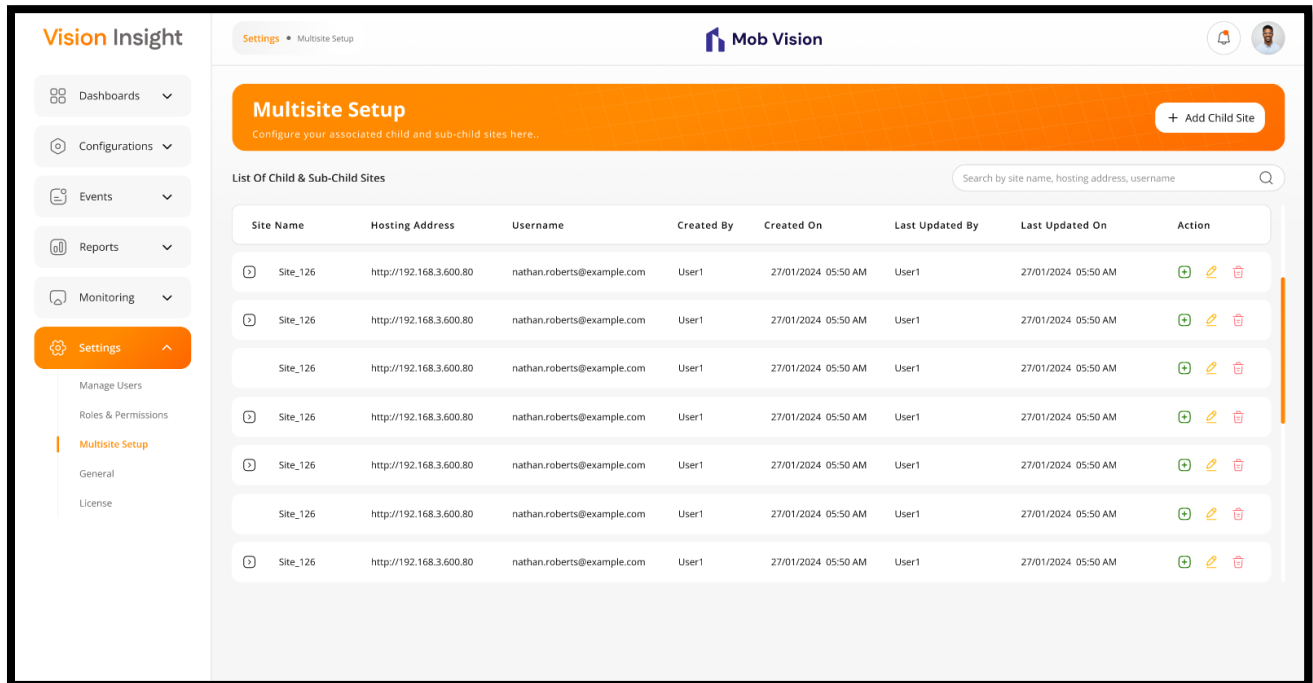
This screen will be accessible by the parent site's authorized user only. This means that a Child or sub-child site users **cannot** configure their own child sites.

Regarding Permissions for this Page:

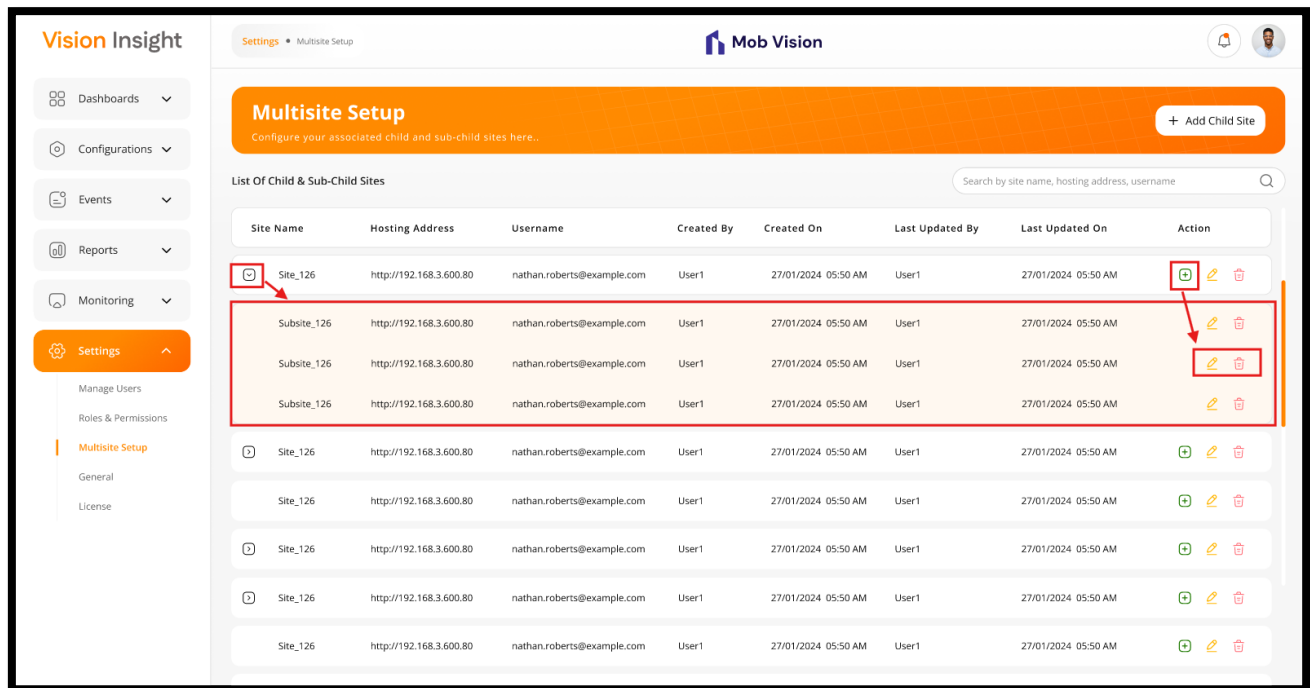
- It is entirely the **responsibility of the Super Admin user to manually grant screen-level permissions** to the **parent site's user** during the **initial configuration of the application** on the customer site. The system will **not handle this scenario automatically**.

Prototype:

→ The screen below illustrates the list of all created **child sites**, along with options to search, add, edit, and delete them:



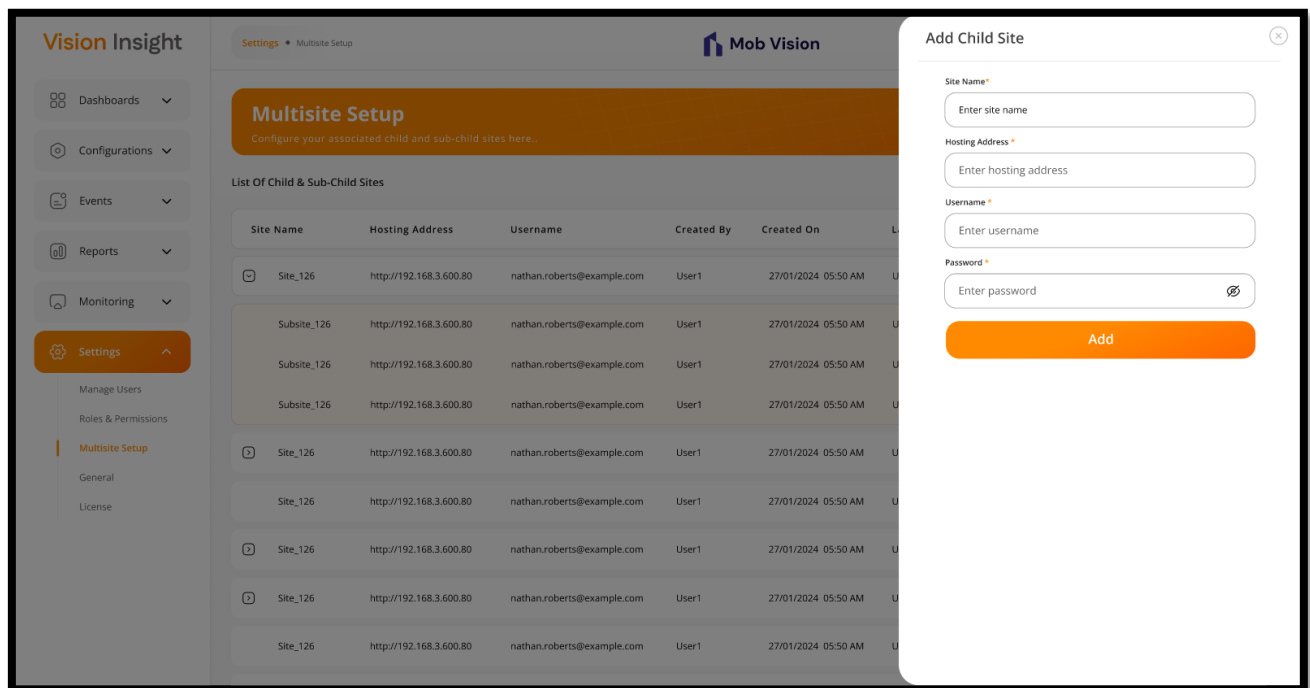
→ The screen below illustrates the list of all created **sub-child sites**, along with options to add, edit, and delete them:



The screenshot shows the 'Multisite Setup' page in the Vision Insight application. The page has a sidebar with navigation options: Dashboards, Configurations, Events, Reports, Monitoring, and Settings. The 'Settings' section is expanded, showing 'Manage Users', 'Roles & Permissions', 'Multisite Setup', 'General', and 'License'. The 'Multisite Setup' section is active, displaying a 'List Of Child & Sub-Child Sites' table. The table has columns: Site Name, Hosting Address, Username, Created By, Created On, Last Updated By, Last Updated On, and Action. The table lists several sites, including 'Site_126' and 'Subsite_126'. Red boxes highlight the 'Add', 'Edit', and 'Delete' icons in the 'Action' column for the first row.

Site Name	Hosting Address	Username	Created By	Created On	Last Updated By	Last Updated On	Action
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]

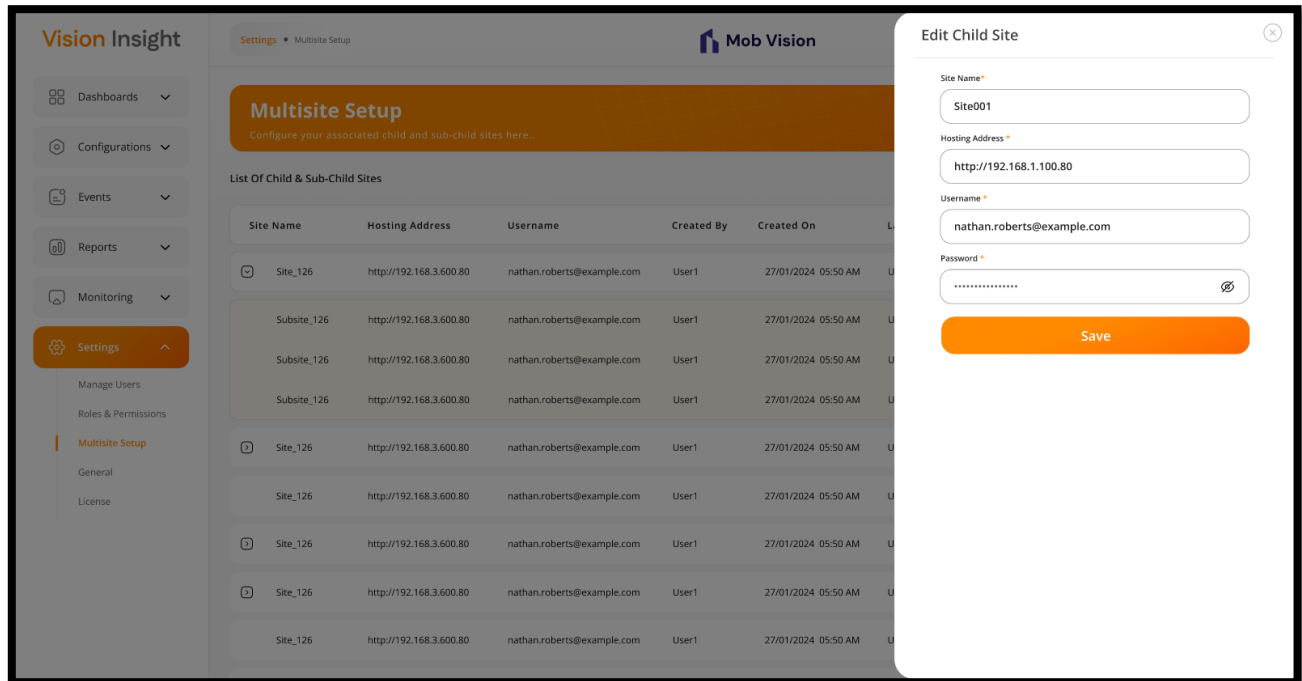
→ The screen below represents the form for **creating a new child site**, allowing users to enter relevant details and submit the information:



The screenshot shows the 'Multisite Setup' page with the 'Add Child Site' form open. The form has fields for Site Name, Hosting Address, Username, and Password. The 'Add' button is at the bottom of the form. The background shows the same table as the previous screenshot.

Site Name	Hosting Address	Username	Created By	Created On	Last Updated By	Last Updated On	Action
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	User1	27/01/2024 05:50 AM	[Add] [Edit] [Delete]

- The screen below displays the form for **editing an existing child site**, allowing users to modify relevant details and update the information:

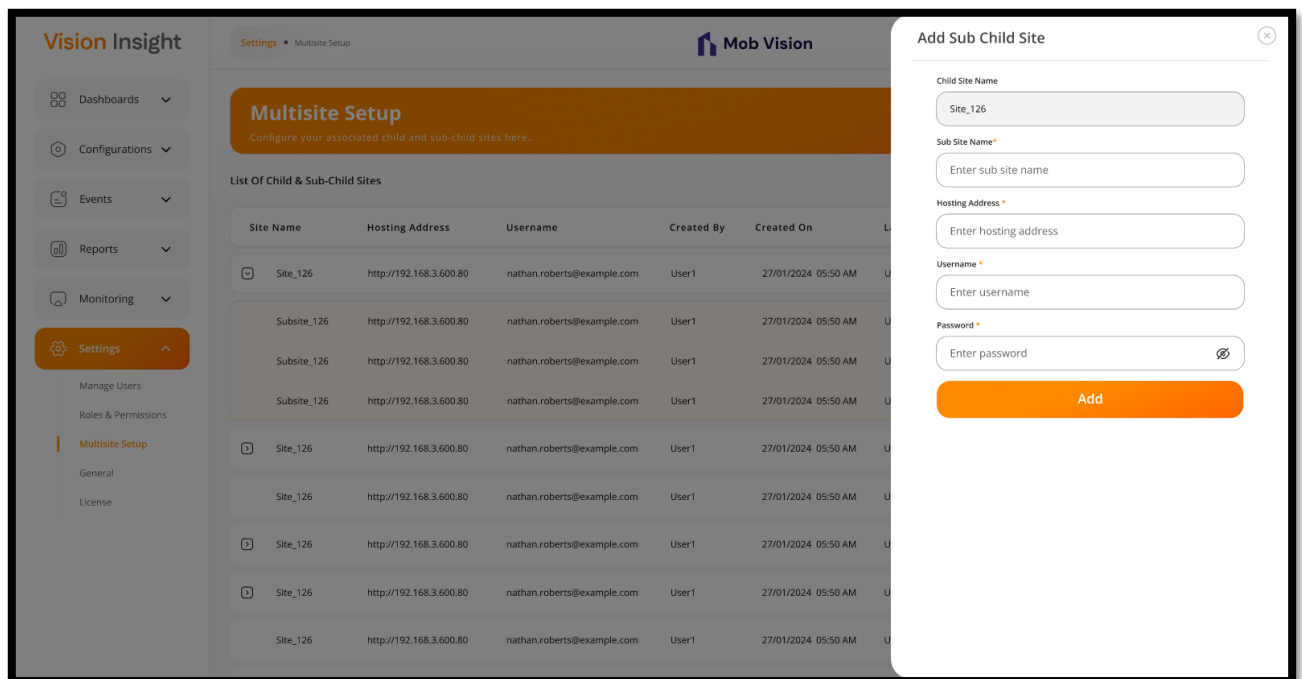


The screenshot shows the 'Vision Insight' dashboard with the 'Multisite Setup' section active. A modal form titled 'Edit Child Site' is open, allowing users to edit the details of an existing child site. The form includes the following fields:

- Site Name***: Text input field with the value 'Site001'.
- Hosting Address ***: Text input field with the value 'http://192.168.1.100.80'.
- Username ***: Text input field with the value 'nathan.roberts@example.com'.
- Password ***: Password input field with a masked value '*****' and a toggle icon.

A blue 'Save' button is located at the bottom of the modal. The background shows a table titled 'List Of Child & Sub-Child Sites' with columns: Site Name, Hosting Address, Username, Created By, and Created On. The table contains several rows of data, including 'Site_126' and 'Subsite_126'.

- The screen below represents the form for **creating a new sub-child site**, allowing users to enter relevant details and submit the information:

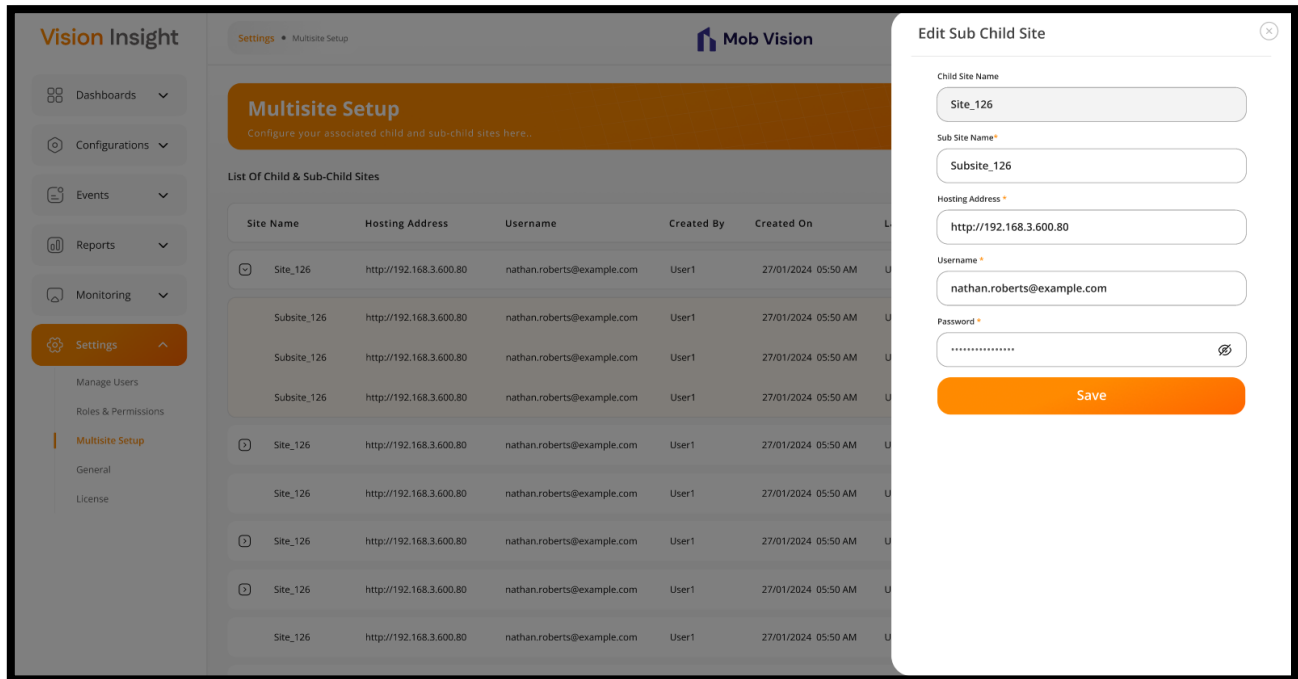


The screenshot shows the 'Vision Insight' dashboard with the 'Multisite Setup' section active. A modal form titled 'Add Sub Child Site' is open, allowing users to create a new sub-child site. The form includes the following fields:

- Child Site Name**: Text input field with the value 'Site_126'.
- Sub Site Name***: Text input field with the placeholder 'Enter sub site name'.
- Hosting Address ***: Text input field with the placeholder 'Enter hosting address'.
- Username ***: Text input field with the placeholder 'Enter username'.
- Password ***: Password input field with a masked value 'Enter password' and a toggle icon.

A blue 'Add' button is located at the bottom of the modal. The background shows the same 'List Of Child & Sub-Child Sites' table as the previous screenshot.

- The screen below displays the form for **editing an existing sub-child site**, allowing users to modify relevant details and update the information:



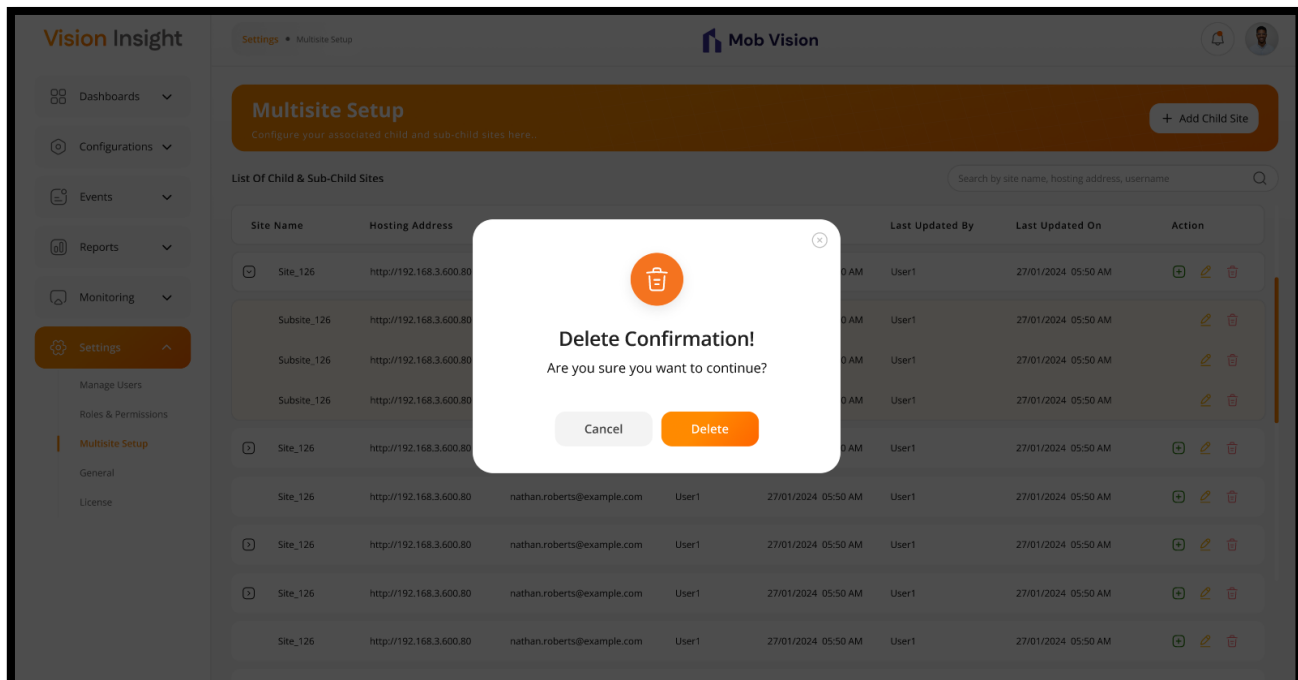
The screenshot shows the 'Vision Insight' dashboard with the 'Multisite Setup' section active. A modal titled 'Edit Sub Child Site' is open, allowing users to edit details for a selected site. The modal contains the following fields:

- Child Site Name:** Site_126
- Sub Site Name:** Subsite_126
- Hosting Address:** http://192.168.3.600.80
- Username:** nathan.roberts@example.com
- Password:** (masked with dots)

A 'Save' button is located at the bottom of the modal. In the background, a table lists existing child and sub-child sites.

Site Name	Hosting Address	Username	Created By	Created On
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM

- The system will request confirmation when a logged-in user attempts to delete any child or sub-child site:



The screenshot shows the 'Vision Insight' dashboard with the 'Multisite Setup' section active. A modal titled 'Delete Confirmation!' is open, asking the user if they are sure they want to continue. The modal contains the following text:

Delete Confirmation!
Are you sure you want to continue?

Buttons: Cancel, Delete

In the background, a table lists existing child and sub-child sites, and a search bar is visible at the top right of the table area.

Site Name	Hosting Address	Username	Last Updated By	Last Updated On	Action
Site_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	[Edit] [Delete]
Subsite_126	http://192.168.3.600.80	nathan.roberts@example.com	User1	27/01/2024 05:50 AM	[Edit] [Delete]

Business Rules:

- Only the parent site's authorized / super admin users will have the ability to configure this information. This means that a child site's user cannot configure their sub-child; only the parent site's authorized user will have that ability.
- Once the above configurations are completed, the system will allow the parent site's authorized / super admin users to view data from all the configured child & sub-child sites only.
- The parent site's authorized / super admin users will be able to add, edit, and delete the following details for each child or sub-child sites:
 - ➔ Site Name
 - ➔ Hosting Address / URL
 - ➔ Username
 - ➔ Password
- When the user enters the required details while adding or updating a child or sub-child site, the system will **validate the connection from the backend** and **display an appropriate message** on the screen, allowing the user to determine whether the entered information is valid or invalid.
- The system will provide the following options for managing child and sub-child sites:
 - ➔ Add Child Site
 - ➔ Edit Child Site
 - ➔ Delete Child Site
 - ➔ Add Sub-Child Site
 - ➔ Edit Sub-Child Site
 - ➔ Delete Sub-Child Site
- This screen will display a list of all configured child and sub-child sites, with the following options for each:
 - ➔ Edit
 - ➔ Delete
- The system will group child and sub-child sites and display them accordingly in a structured list.
- Each child site will have an option to add a sub-child site by clicking the "+" icon.
- When adding or updating a sub-child site, the parent **child site's name** will be displayed on the "Add Sub-Child Site" & "Edit Sub-Child Site" screens as **non-editable**.
- The authorized user can search for a specific record using **Site Name** and **Hosting Address**.
- This screen will **not support pagination** but will allow **vertical scrolling**, enabling users to view all records smoothly.
- If at least one sub-child site exists for a child site, the system will allow users to **expand and view sub-child sites** within the list.

Data Dictionary:

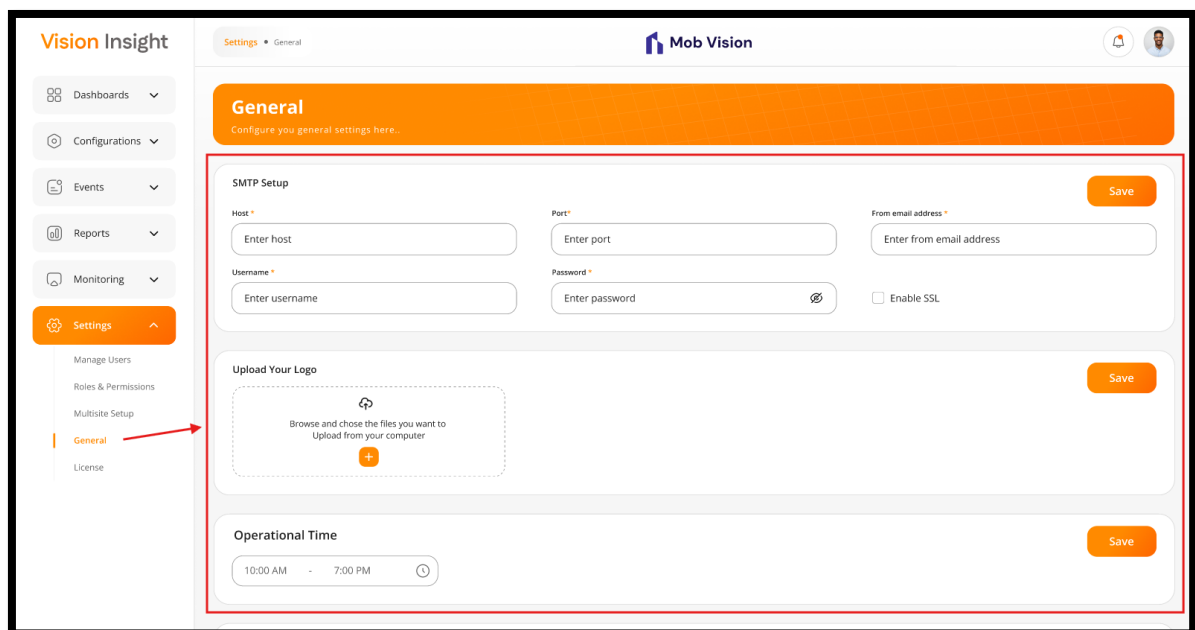
siteMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	siteName	String	No	Headquter2 Ltd	Name of the site
3	hostingAddress	String	No	192.168.1.1:8081	Hosting address where application is running
4	username	String	No	shareduser	User name
5	password	String	No	Admin@123	Password
6	childSites	Object[]	Yes	[{}, {}]	List of child sites available in parent compant
6.1	> siteName	String	No	Sub Company Ltd	Name of the site
6.2	> hostingAddress	String	No	192.168.2.2:8081	Hosting address where application is running
6.3	> username	String	No	shareduser	User name
6.4	> password	String	No	Admin@123	Password
7	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
8	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
10	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
11	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.5. Settings > General Page

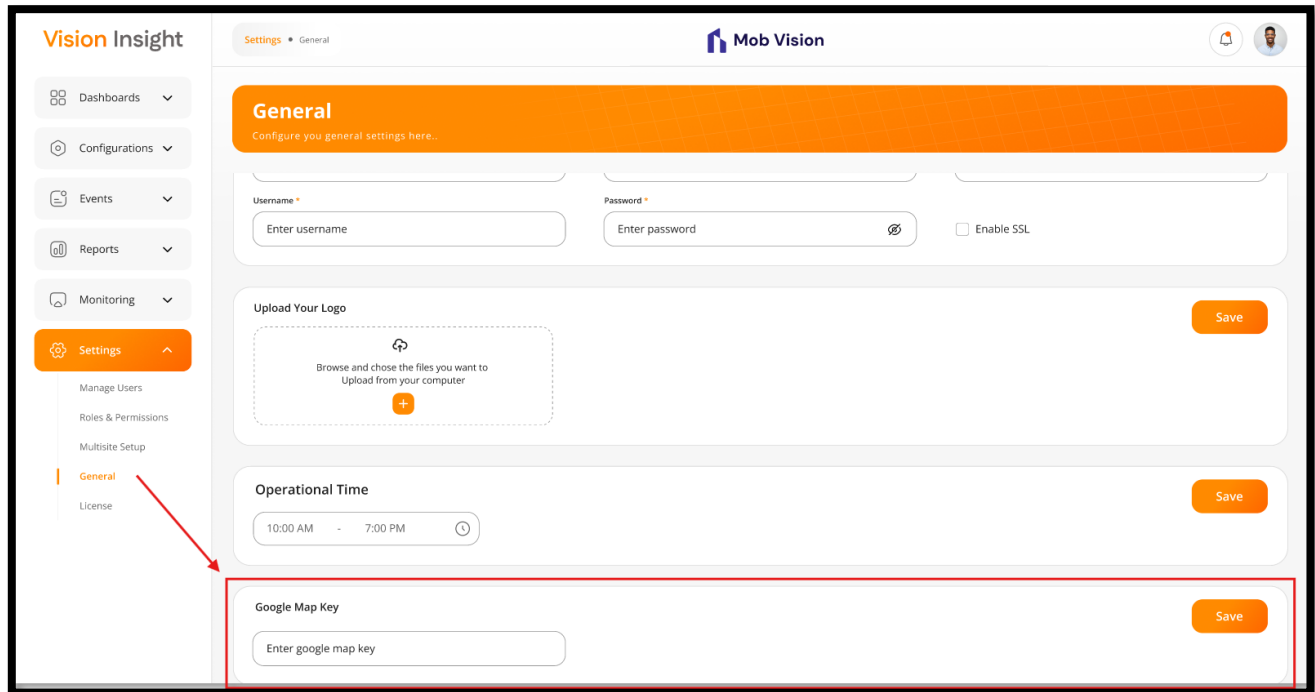
The system allows the **Super Admin** or **authorized user** to manage **general settings** such as **SMTP setup**, **uploading the company logo**, **configuring operational time**, and **setting the Google Map API key** from this page in the **BI Dashboard Web Application**. This screen will be **accessible only to authorized users**.

Prototype:

➔ The screen below represents the **form** for configuring **SMTP details**, **company logo**, and **operational time**, allowing users to **enter** and **submit** the relevant information:



→ The screen below represents the **form** for configuring **Google Map Key**, allowing users to **enter** and **submit** the relevant information:



Business Rules:

- **SMTP Setup:**
 - **Authorized users** will be able to configure their **SMTP credentials** by entering the following fields from this section:
 - Host*
 - Port*
 - From email address*
 - Username*
 - Password*
 - Enable SSL (checkbox)
 - The system will send email notifications to users only when SMTP is configured only.
- **Upload Your Logo:**
 - Authorized users will be able to configure their company logo from this section.
 - Users will be able to select and upload their company logo.
 - The logo size will be fixed to maintain design consistency.
 - The uploaded logo will be displayed at the top center of the header section & it will be displayed when the logo is configured only.
- **Operational Time:**
 - Authorized users will be able to configure operational time by entering the following values:
 - Start Time*
 - End Time*

- ➔ The system will calculate the **day start and end** based on the **configured operational time** for the specific site.
 - If no operational time is configured, the system will **default to a 24-hour operational time**.
- **Google Map Key:**
 - ➔ **Authorized users** will be able to **configure the Google Map Key** by entering it into this section.
 - ➔ The **map feature will be enabled** for the specific site **only if the key is configured**. Otherwise, the **map plan will not be accessible** for those users or sites.
 - ➔ The **Google Map Key** will be **purchased & managed** by the **specific client** only.
- Each of the **above-mentioned sections** will be managed **independently**. The user must click the **“Save”** button to **update the values** of each section **separately**.
- The system will display an appropriate **“Success”** or **“Error”** message when the user clicks the **“Save”** button.

Data Dictionary:

clientSettings					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	Logo	String	Yes	base64://	Client company logo
3	smtpSettings	Object	Yes	{}	for send email notifications
3.1	> host	String	No	smtp.demo.com	host name of the smtp server
3.2	> port	Int32	No	8000	port name of the smtp server
3.3	> enableSsl	Bool	No	TRUE	SSL enabled or not
3.4	> username	String	No	xyz	username of the smtp server
3.5	> password	String	No	xyz@123!	password of the smtp server
3.6	> fromEmail	String	No	mysmtp@gmail.com	from email address to send an email notification
4	operationalTiming	ObjectId	Yes	{}	object
4.1	startTime	Int32	No	10.00 AM	Start time of the site
4.2	endTime	Int32	No	7.00 PM	end time of the site
5	googleApiKey	DateTime	Yes	674daaf7d34a2b901c847454674daaf7d34a2b901c847454	Google Map Key, maintained by client only
6	isDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
7	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
9	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

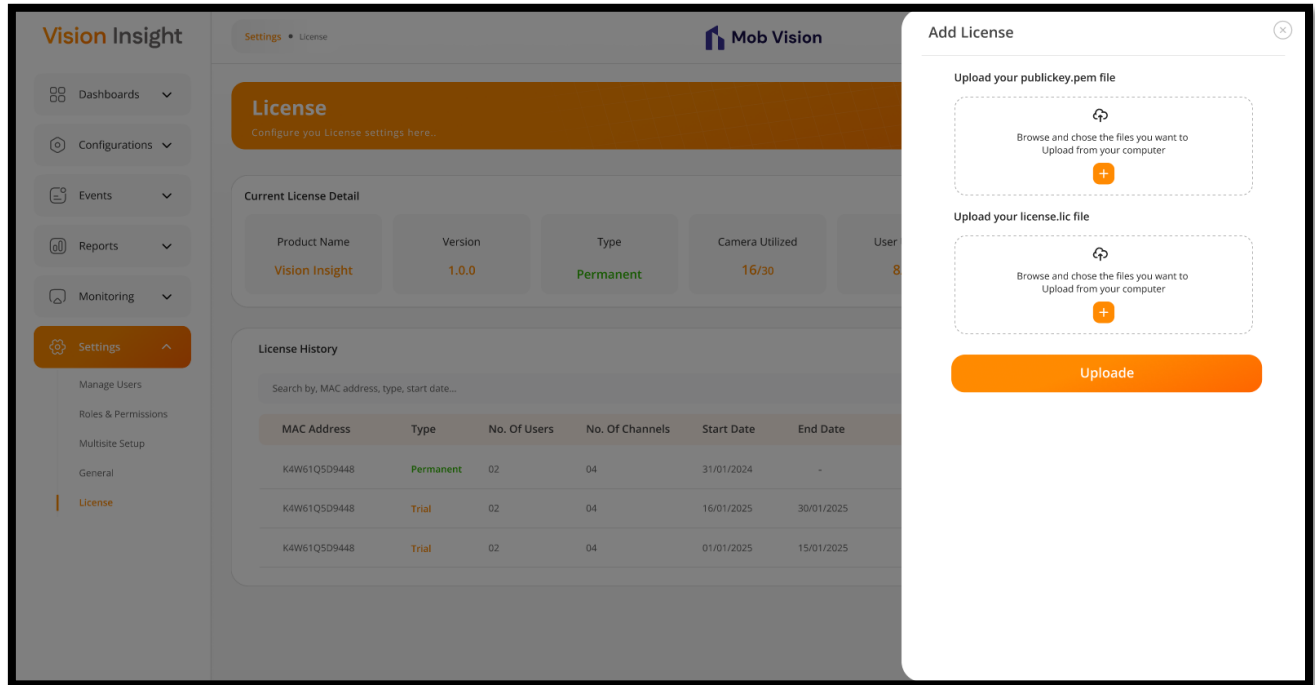
3.6. Settings > License

The system allows the **Super Admin** or **authorized users** to **upload and manage license details** from this page in the **BI Dashboard Web Application**.

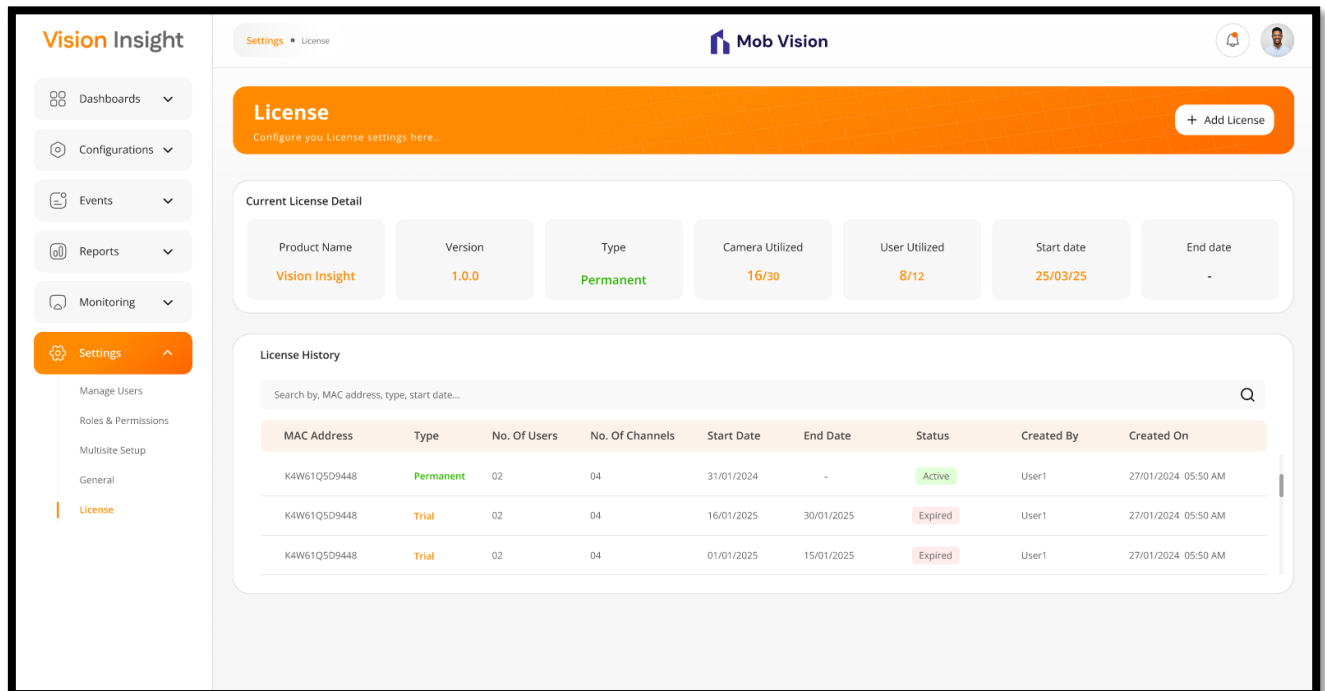
This screen will be **accessible only to authorized users**.

Prototype:

- The screen below represents the form to **upload a license file** for the **specific site only**, allowing users to **upload a publickey.pem & license.lic file** and **submit it**:



- The screen below represents the **details of currently active license** and their **history**, including an option to **search the specific record** from the **history table**:



MAC Address	Type	No. Of Users	No. Of Channels	Start Date	End Date	Status	Created By	Created On
K4W61Q5D9448	Permanent	02	04	31/01/2024	-	Active	User1	27/01/2024 05:50 AM
K4W61Q5D9448	Trial	02	04	16/01/2025	30/01/2025	Expired	User1	27/01/2024 05:50 AM
K4W61Q5D9448	Trial	02	04	01/01/2025	15/01/2025	Expired	User1	27/01/2024 05:50 AM

Business Rules:

- **Authorized users** will be able to upload the following files to authenticate the license:
 - ➔ **publickey.pem** – Contains the customer's **public key**, required for **authentication** during license configuration at the customer's site.
 - ➔ **license.lic** – Contains the **generated license details** for the customer, used to **validate and authenticate** the license during configuration.
- The system will **validate the associated MAC Address** against the **PC's MAC Address** when the authorized user attempts to upload the license from the **"Add License"** popup.
- If the **MAC Address does not match**, the system will display an **"Unauthorized" error message**, and the user will be instructed to **contact their administrator** for further assistance.
- If the **MAC Address matches**, the system will:
 - ➔ **Successfully upload** the license.
 - ➔ Display the following **license details** on the screen:
 - **Product Name** (e.g., *Vision Insight*)
 - **Version** (e.g., *1.0.0*)
 - **Type** (e.g., *Permanent / Trial*)
 - **Camera Utilized** (e.g., *16/30*)
 - **User Utilized** (e.g., *8/12*)
 - **Start Date** (e.g., *25/03/2025*)
 - **End Date** (e.g., *10/04/2025* or *"–"*)
 - ➔ Show an appropriate **success message** to the user.
- At any given time, **only one license** will be **active** in the system.
- The system will **maintain a history** of all uploaded licenses, which will be displayed under the **"License History"** section with the following columns:
 - ➔ MAC Address
 - ➔ Type
 - ➔ No. Of Users
 - ➔ No. Of Channels
 - ➔ Start Date
 - ➔ End Date
 - ➔ Status
 - ➔ Created By
 - ➔ Created On
- Users will be able to **search licenses** by:
 - ➔ MAC Address
 - ➔ Type
 - ➔ Start Date

Data Dictionary:

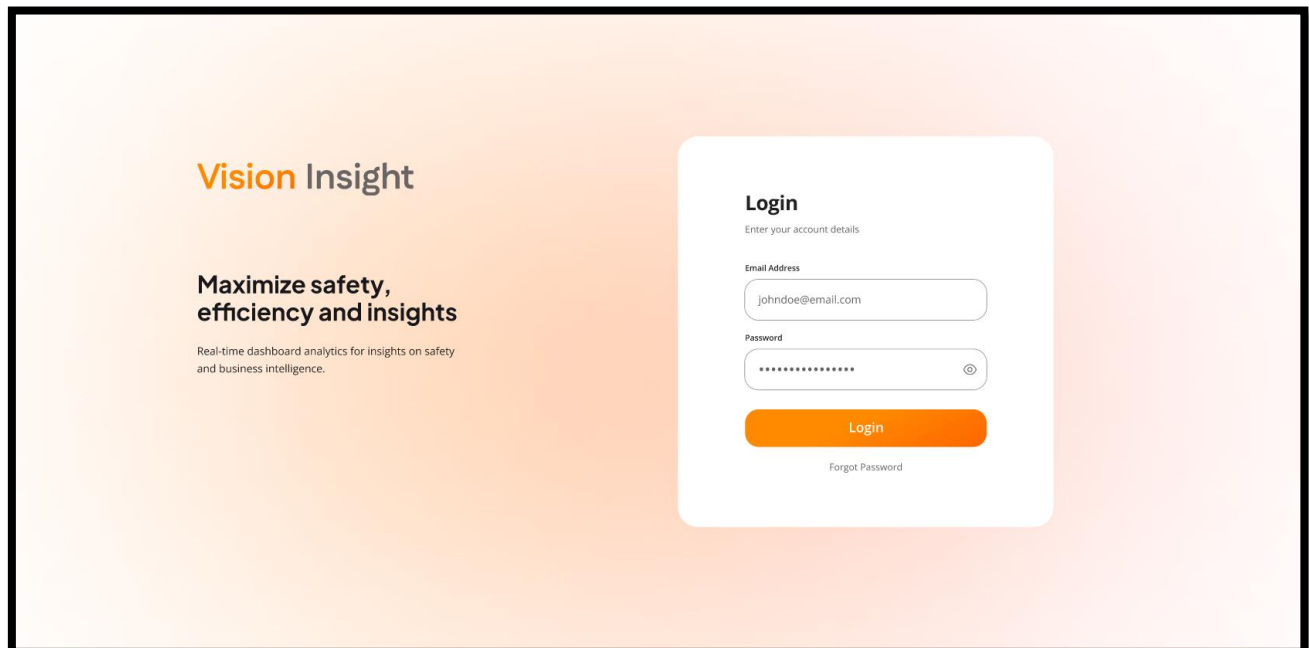
licenseHistory					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	licenseType	String	No	Permanent	Types of license
3	startDate	DateTime	No	2025-04-24T18:30:00.000Z	Start date of the license
4	expiryDate	DateTime	No	2025-04-24T18:30:00.000Z	end date of the license
5	macAddress	String	No	6C-4B-90-AA-AA-AA	MAC address of the client machine
6	numberOfUsers	Int32	No	50	Number of users assigned to the license
7	noOfChannel	Int32	No	200	Number of channels assigned to the license
8	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
9	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
11	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
12	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.7. Login Page

There is a provision in the system for **all users** to **login to the BI Dashboard Web Application**.

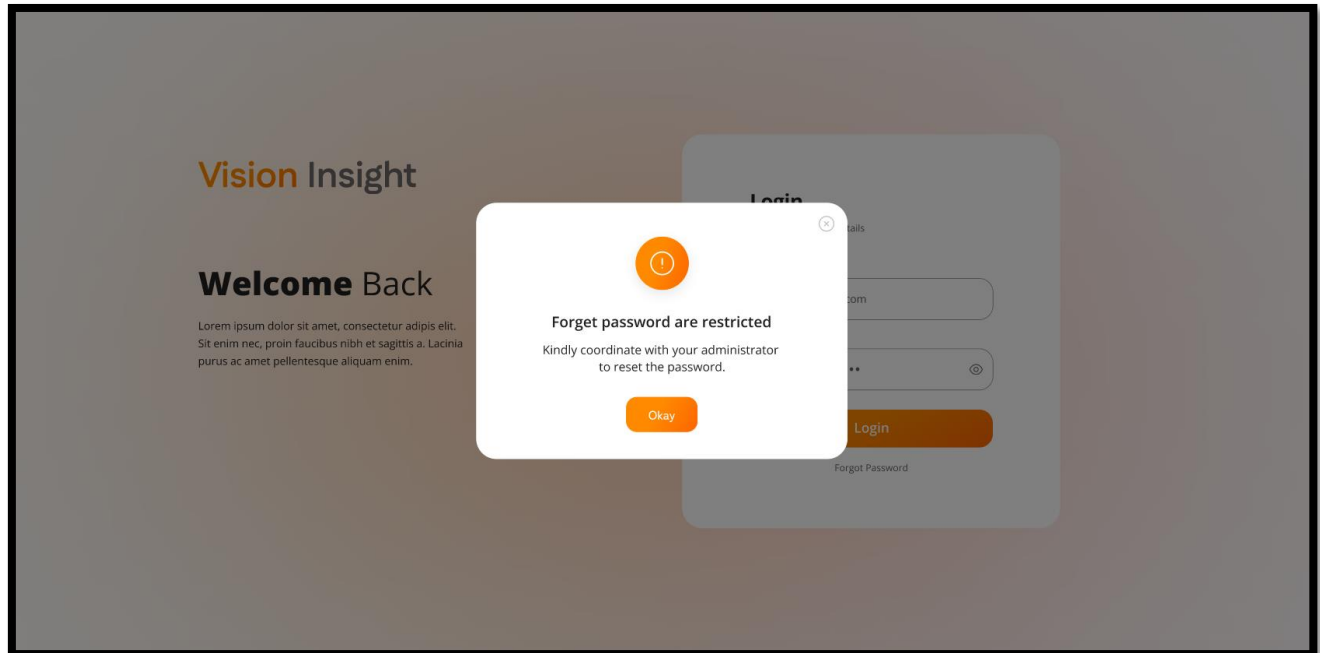
This page will **validate the user's credentials** whenever a user attempts to login by entering their **own login details**.

Prototype:



The prototype shows a login page for 'Vision Insight'. On the left, the logo 'Vision Insight' is displayed above the tagline 'Maximize safety, efficiency and insights' and a sub-tagline 'Real-time dashboard analytics for insights on safety and business intelligence.' On the right, a white login card is centered. It has a title 'Login' and a subtitle 'Enter your account details'. Below this are two input fields: 'Email Address' with the value 'johndoe@email.com' and 'Password' with masked characters '*****'. A blue 'Login' button is positioned below the password field, and a 'Forgot Password' link is at the bottom of the card.

- The screen below represents the **message** which will be displayed when the user **clicks on the “Forgot Password” link** located just below the **Login button**, since the system **does not allow users to reset their own password**:



Business Rules:

- The **login screen** will include a logo and text fields for entering the **username or email address** and **password**, along with a **Login (Submit)** button as well as “Forgot Password” link.
- Upon clicking the "Forgot Password" link, the user will see the following message, as sub-users cannot reset their own passwords. Only the super admin or an authorized user can change the sub-user's password:

“Kindly coordinate with your administrator to reset the password”
- Upon clicking the “Login” button, the application will validate the **entered username or email address** and **password**.
- If the credentials are successfully validated:
 - The application will check the user's authorizations and redirect them to the **Dashboard screen**.
- If the entered credentials are invalid:
 - The system will display an appropriate error message on the login screen.
- If the user is marked as **Inactive/Deleted** in the system:
 - They will not be able to login using their existing credentials.
 - In such cases, the user must contact the **super user & authorized users** for further assistance.
- The system will validate below highlighted fields while logging into the system.

Data Dictionary:

➔ The system will utilize **below highlighted fields** from the **userMaster** table:

userMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	Objectid	No	674daaf7d34a2b901c847454	Autogenerated id
2	username	String	No	Admin	Username is use for login
3	firstname	String	Yes	Parth	First name of the user
4	lastname	String	Yes	Jotangiya	Last name of the user
5	email	String	No	xxxxx@xxxx.xxx	Email address of the user
6	password	String	No	aasdshfbshdf	It will be stored in encrypted format only
7	roleids	Objectid[]	Yes	[674daaf7d34a2b901c847454]	Unique Id(s) column of the "roleMaster" table, allow multiples.
8	ProfileImage	String	Yes	base64://	User profile picture in base64 format
9	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
10	createdBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
11	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
12	updatedBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
13	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.8. Menu Items for BI Dashboard Web Application

The system allows all users to log in to the BI Dashboard Web Application and access the following menus and options based on the specific assigned permissions:

The following options/menus will be displayed on the sidebar:

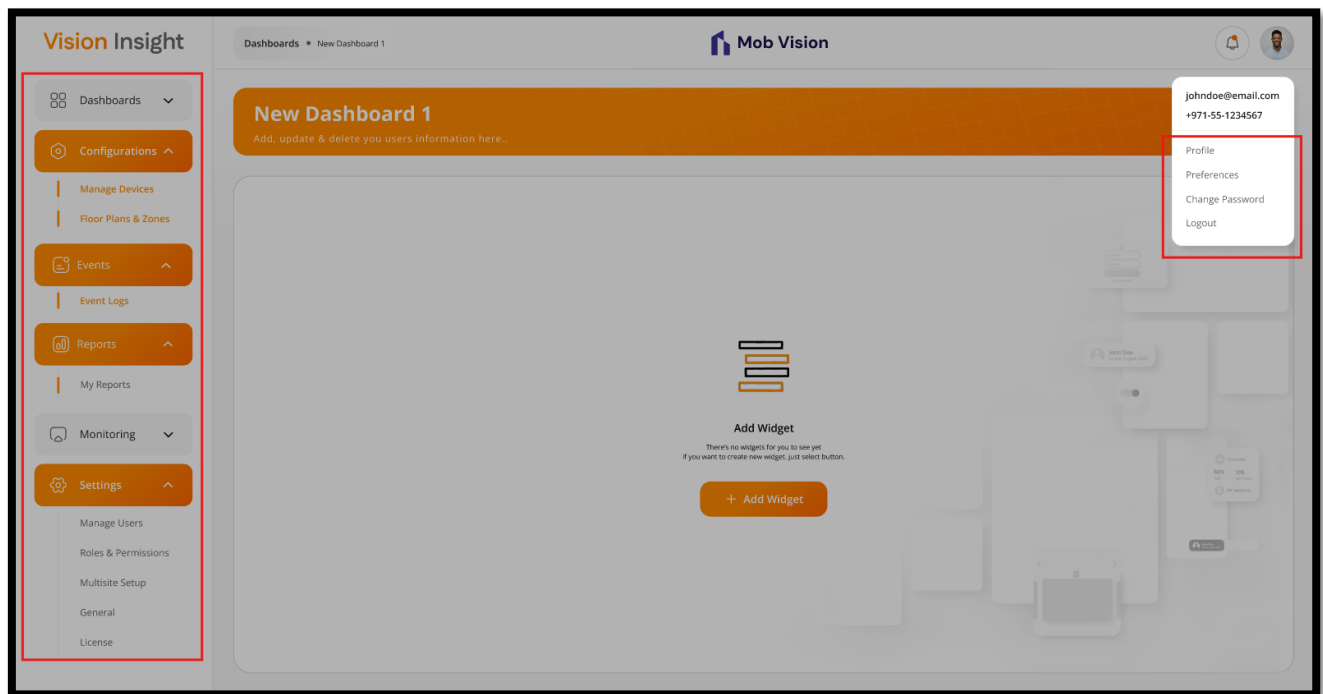
- Dashboards
 - ➔ The **user** will be able to configure **any number of dashboards**, each with a **custom name**.
- Configurations:
 - ➔ Manage Devices
 - ➔ Floor Plans & Zones
- Events
 - ➔ Event Logs
- Reports
 - ➔ My Reports
 - Site Performance Comparison Report
 - Zone Performance Comparison Report
- Monitoring
 - ➔ The **user will be able to configure any number of monitoring**, each with a **custom name**.
- Settings:
 - ➔ Manage Users
 - ➔ Roles & Permissions
 - ➔ Multisite Setup
 - ➔ General
 - ➔ License

The following options/menus will be displayed when the user clicks on their profile image in the top right-hand corner:

- Profile
- Preferences
- Change Password
- Logout

Prototype:

- ➔ The screen below illustrates above mentioned menu items & options for Bi Dashboard Web Application.



Business Rules:

- The **logged-in** user will be able to access the above highlighted menus or options based on the specific assigned permissions only.
- All users will be able to access the following options:
 - ➔ Profile
 - ➔ Preferences:
 - ➔ Change Password
 - ➔ Logout
- The system will automatically hide any menus or specific actions on screens that the logged-in user is not authorized to access based on their assigned permissions.

Data Dictionary:

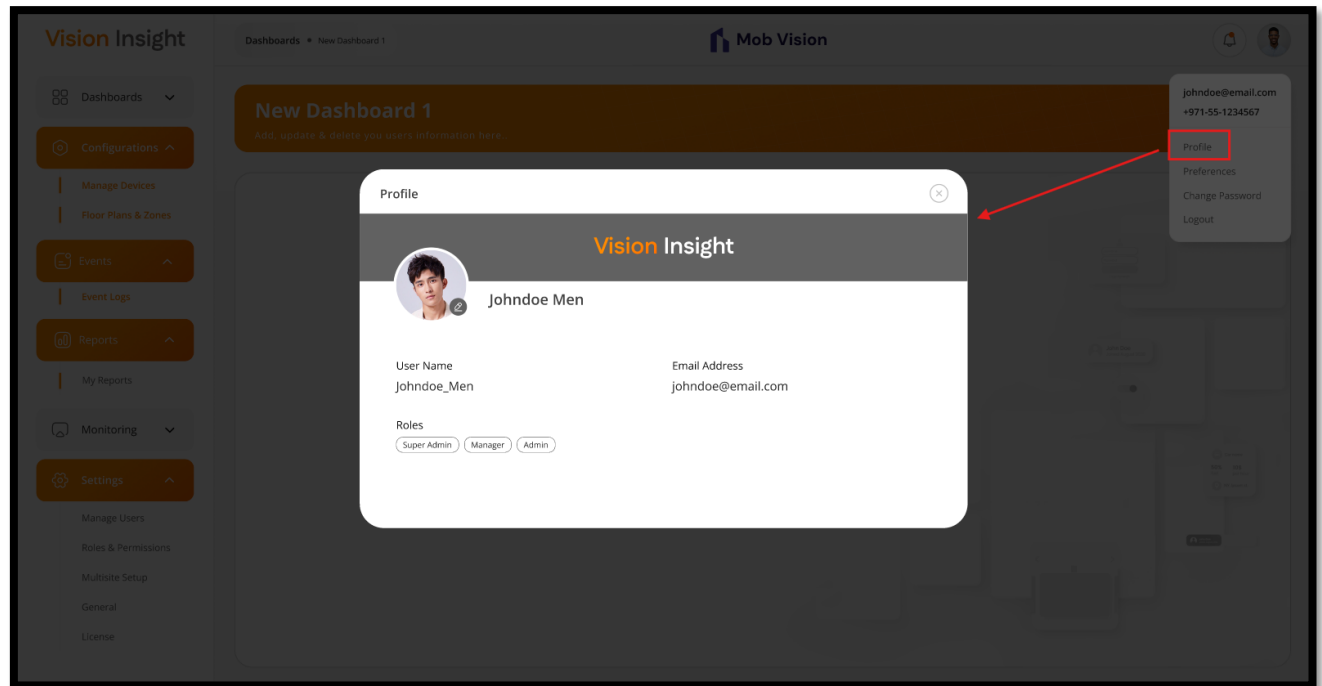
- *It is not required to manage any data dictionary for menu items, as it will be handled directly in the frontend code only.*

3.9. Header Section > Profile Page

The system provides a feature for **all users to log in** to the **BI Dashboard Web Application** and **access their Profile page**.

This page displays **details specific to the logged-in user only**. The screen can be **accessed by clicking on the Profile link** located at the **top-right corner of the top bar**.

Prototype:



Business Rules:

- The system will display information related to the logged-in user on this page. Which are enlisted below:
 - Profile Image
 - Full Name (*First & Last Name*)
 - Username
 - Email Address
 - Roles (*which is assigned to logged-in user from the **Manage Users** page*)
- This screen will allow logged-in users to only view their details except updating their profile image only.
- The user can update their **profile picture** by clicking on the current profile image and selecting a new one & the updated profile picture will be reflected across the entire system **after re-logging in**.

Data Dictionary:

- This page will display registered information for the specific logged-in user from the **userMaster** table only.*

3.10.Header Section > Change Password Page

The system will **not allow any users to change their own password** from the **BI Dashboard Web Application**.

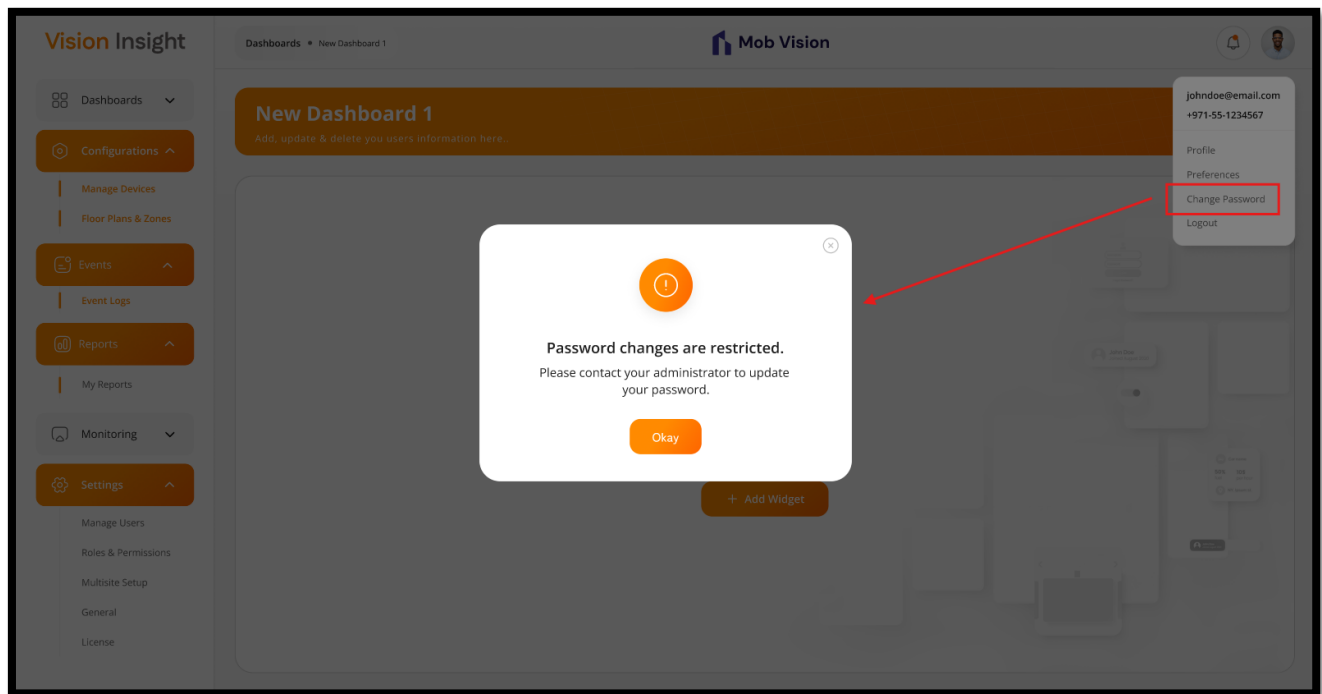
To **reset or change their password**, users must **contact a Super User or authorized user**.

Only a **Super User or authorized user** can **update or reset passwords** from the **Manage Users page** for any user.

This page will be **accessible when the user clicks on their profile image** located at the **top right-hand corner** of the screen.

Prototype:

→ The screen below illustrates the **message** that will appear once users **click on the “Change Password”** option:



Business Rules:

- All users will have access to this option.
- Upon clicking this option, the system will display the following message:
“*Kindly coordinate with your administrator to change your password*”
- If the **Super Admin user forgets their password**, they will need to **contact the technical development team to reset or change their password**.

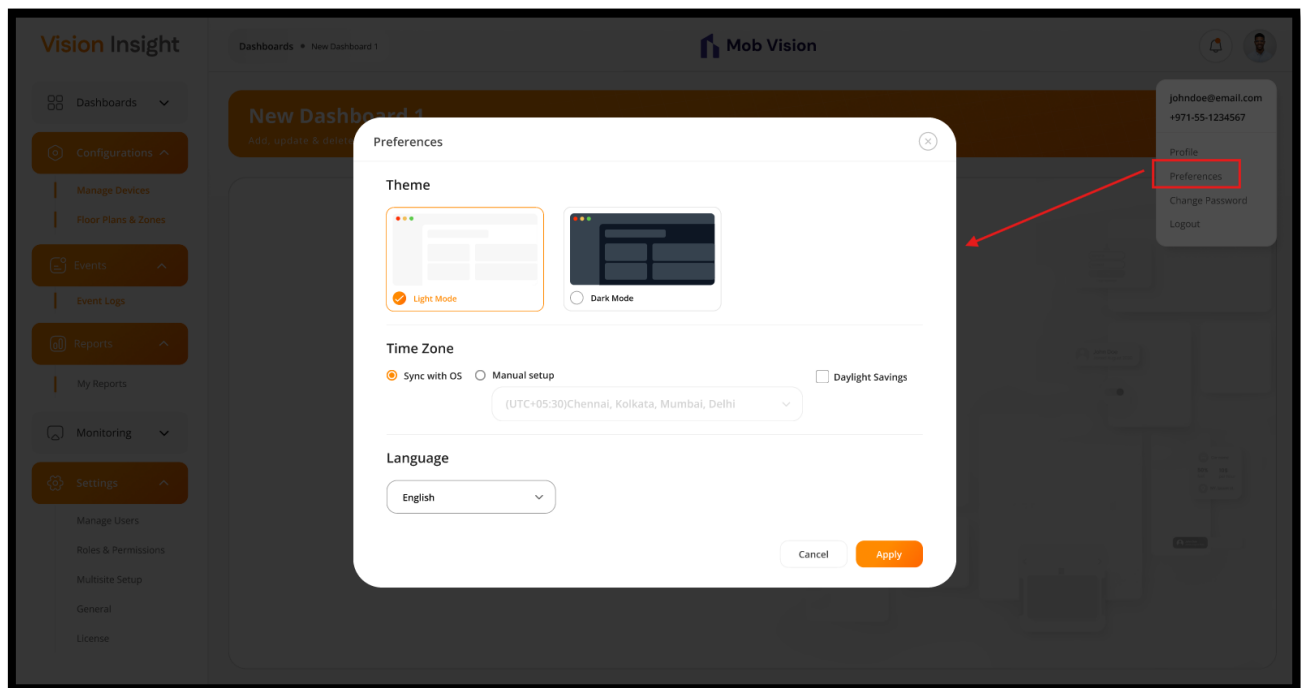
3.11. Header Section > Preferences Page

The system allows the **authorized user** to **manage preferences** such as **Color Theme**, **Time Zone**, and **Language** from this page in the **BI Dashboard Web Application**.

This page will be **accessible when the user clicks on their profile image** located at the **top right-hand corner** of the screen. All **preference options on this page** will be **available to all users**.

Prototype:

→ The screen below illustrates the **options available for users** to select their own preferences, like **Theme**, **Time Zone**, and **Language**:



Business Rules:

- **Color Theme:**
 - **All users** will be able to select this option.
 - Users can choose from the following **color theme** options:
 - Light
 - Dark
 - The **UI** will be appeared based on the **selected theme** only.
 - By default, light theme will be selected.
- **Time Zone:**
 - The system will provide following two options for time zone configuration:
 - Synce with OS
 - Manual Setup
 - When user selects the **“Synce with OS”** option,

- The system will fetch the information from the OS & configure the time zone accordingly.
- ➔ When user selects the “**Manual Setup**” option,
 - The system will display the predefined list of time zones.
 - The user will be able to select a single time zone from the dropdown list.
- ➔ Authorized users will be able access this option.
- ➔ Based on the selected time zone, the system will display the date and time accordingly to the logged-in user in entire system.
- ➔ The system will also allow users to enable “**Daylight Savings**”. If selected, the system will automatically adjust the time for users located in the respective time zones **where Daylight Savings is applicable**.
- **Language:**
 - ➔ The system will display the predefined list of languages.
 - ➔ In this scope, the system will support & provide only single option. That is “**English**”.
- The system will display an appropriate “**Success**” or “**Error**” message when the user clicks the “**Apply**” button.

Data Dictionary:

userPreferences					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	Objectid	No	674daaf7d34a2b901c847454	Autogenerated id
2	userId	String	No	674daaf7d34a2b901c847454	reference id of the specific user from userMaster table
3	Theme	String	No	Light	user theme preference, by default - it will be Light theme only
4	timeZone	Objectid	No	674daaf7d34a2b901c847454	Reference id of timeZoneMaster table
4	IsDayLightSavings	Boolean	No	FALSE	Day light savings of the specific timezone
5	language	String	No	English	system will support english language only
9	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
10	createdBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
11	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
12	updatedBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
13	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.12.Header Section > Notifications

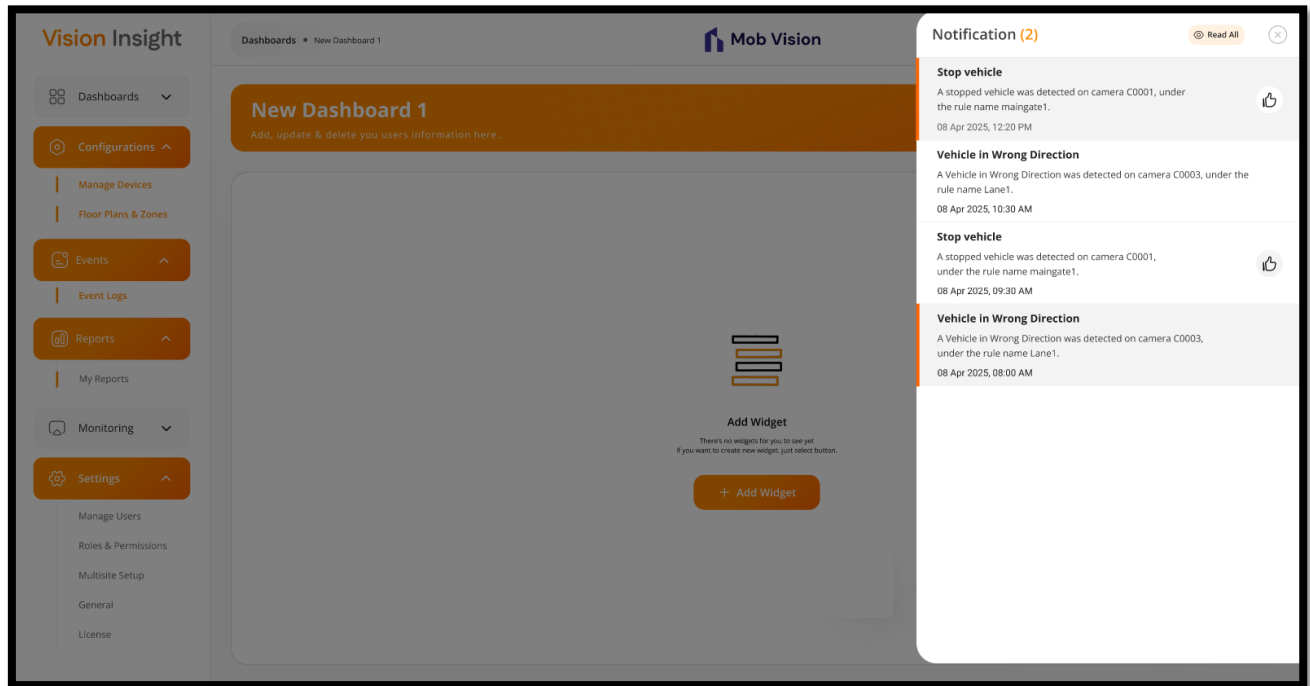
The system allows the **authorized users** to view all the notification received within the BI Dashboard Web Application.

This page will be **accessible when the user clicks the bell icon**, located at the **top-right corner of the screen**. **Only authorized users** will be able to **view notifications** under this section.

Prototype:

- ➔ The screen below illustrates the **notification section** which includes **counts of the total unread notifications**, **option to mark all notifications as read**, **acknowledge the event notification & click on any notification and mark it as read**. This will be appeared when **user clicks on the Bell Icon** of the

header section:



Business Rules:

- **Authorized users** will be able to **view the bell icon** in the **header section**. Upon clicking the icon, they can **access all system notifications**.
- Each notification will include the **title, description, and date & time** of the notification.
- The system will display the following **types of notifications only**:
 - ➔ **Event Notification** (*may include an option to **acknowledge** it*)
 - ➔ **Alert Notification** (*displayed in case of **emergencies** or where **immediate action is required***)
- The system will show a **visual indication in RED DOT** on the bell icon when **unread notifications** are present.
- The system will display the **count of unread notifications** next to the bell icon.
- **Authorized users** will have the option to **mark all unread notifications as read** by clicking the **“Read All”** button. This action will be **reflected for all authorized users**, as the system does **not generate or maintain user-specific notifications**.

Data Dictionary:

notification					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	title	String	No	Vehicle in wrong direction	Title of the notification
3	content	String	Yes	A stopped vehicle detected on cam12	Content of the notification
3.1	isRead	Boolean	No	FALSE	Notification is read or not
3.2	ActionName	String	Yes	Acknowledge	Action the name of the notification
3.3	ActionParameter	Object	Yes	{}	Action parameters for the notification
4	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
5	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
6	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
7	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

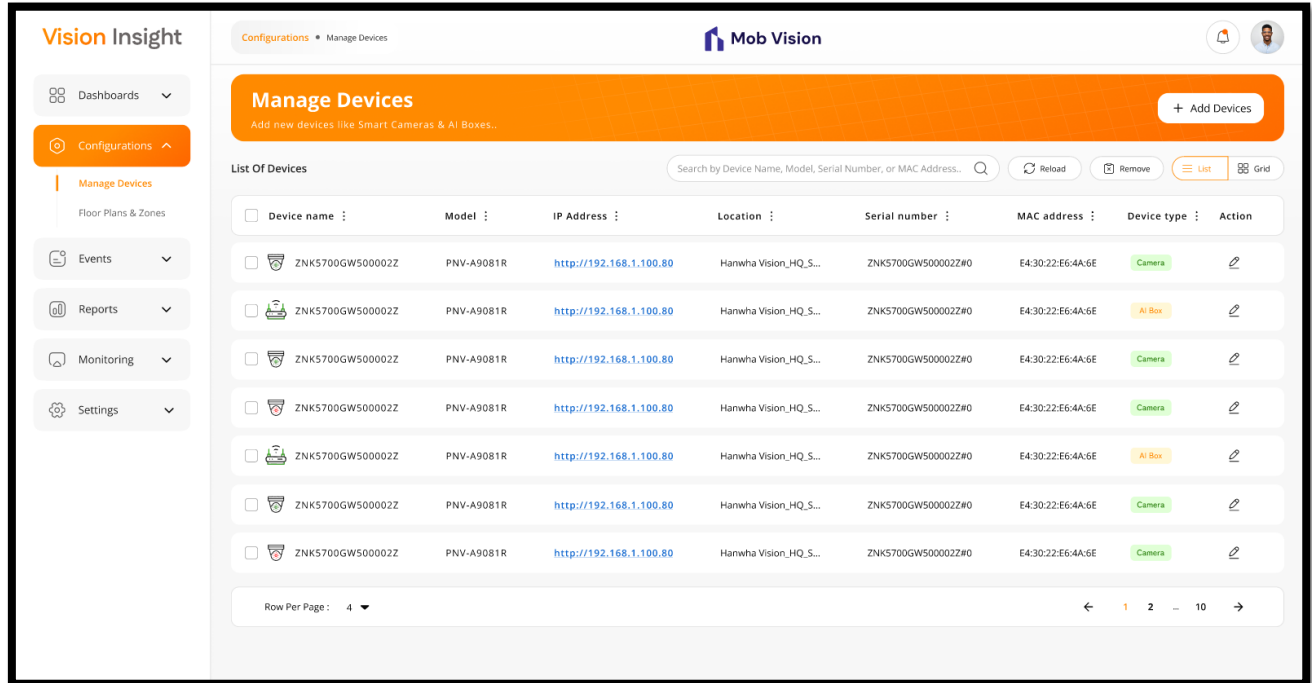
3.13.Configurations > Manage Devices Page

The system provides a feature for **Super Admin and authorized users** to **manage devices** (such as **Cameras** and **AI Boxes**) through the **BI Dashboard Web Application**.

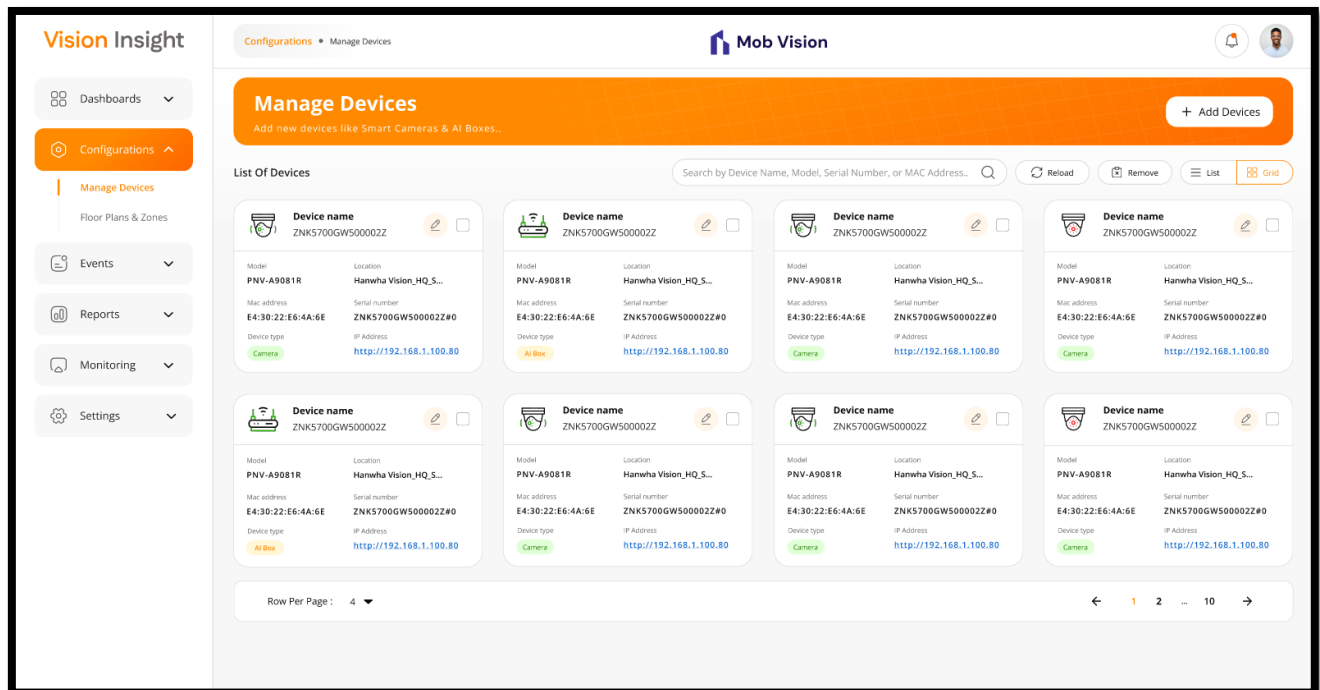
This page will be **accessible exclusively to Super Admin and authorized users**, with actions such as **adding, editing, and deleting devices** displayed based on the **permissions assigned to the logged-in user**.

Prototype:

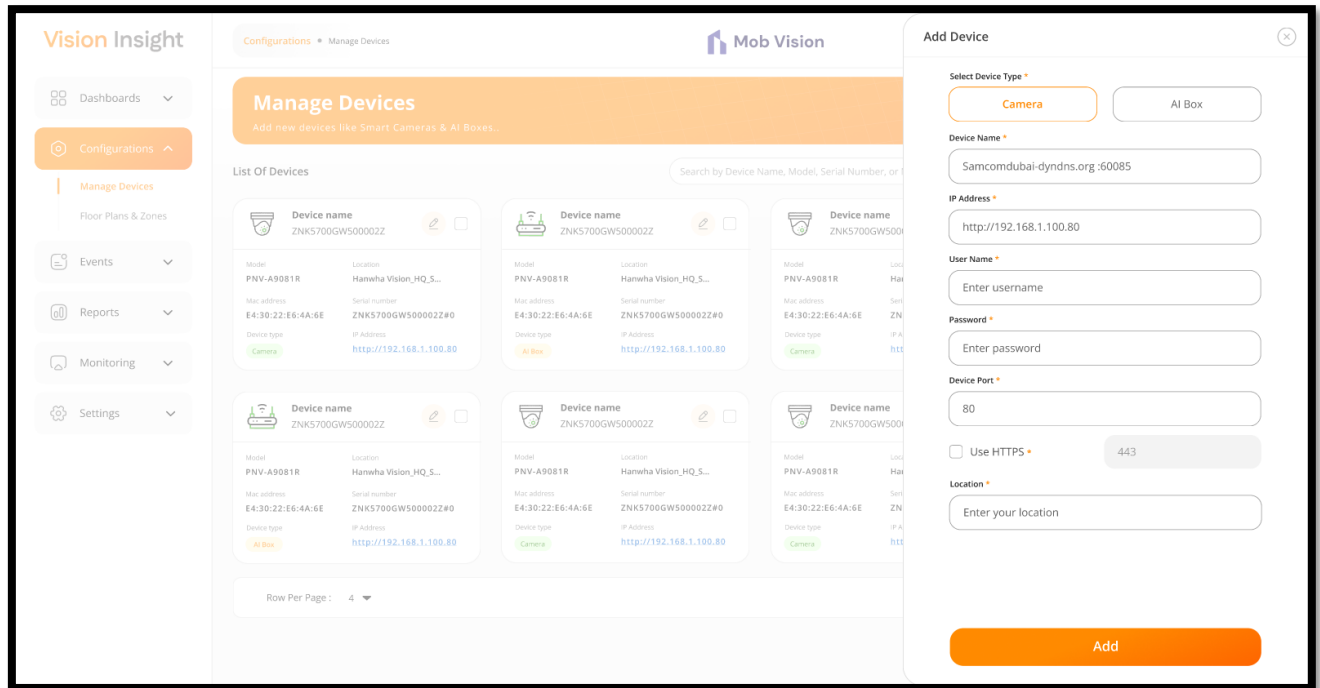
- ➔ The screen below illustrates the **list of all created devices** under **List view**, including options for **Search, Reload, Remove, Add, and Edit**:



➔ The screen below illustrates the **list of all created devices** under **Grid view**, including options for **Search, Reload, Remove, Add, and Edit**:

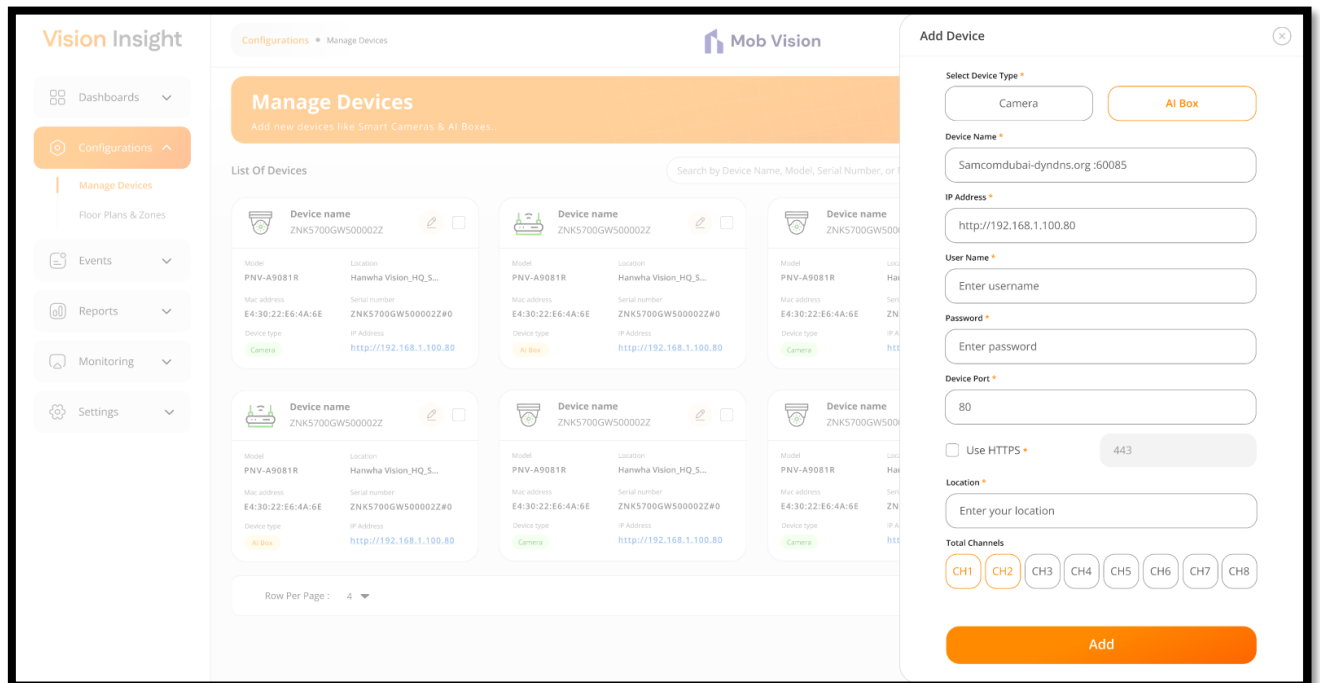


→ The screen below represents the form for **creating a new camera device**, allowing users to enter relevant details and submit the information:



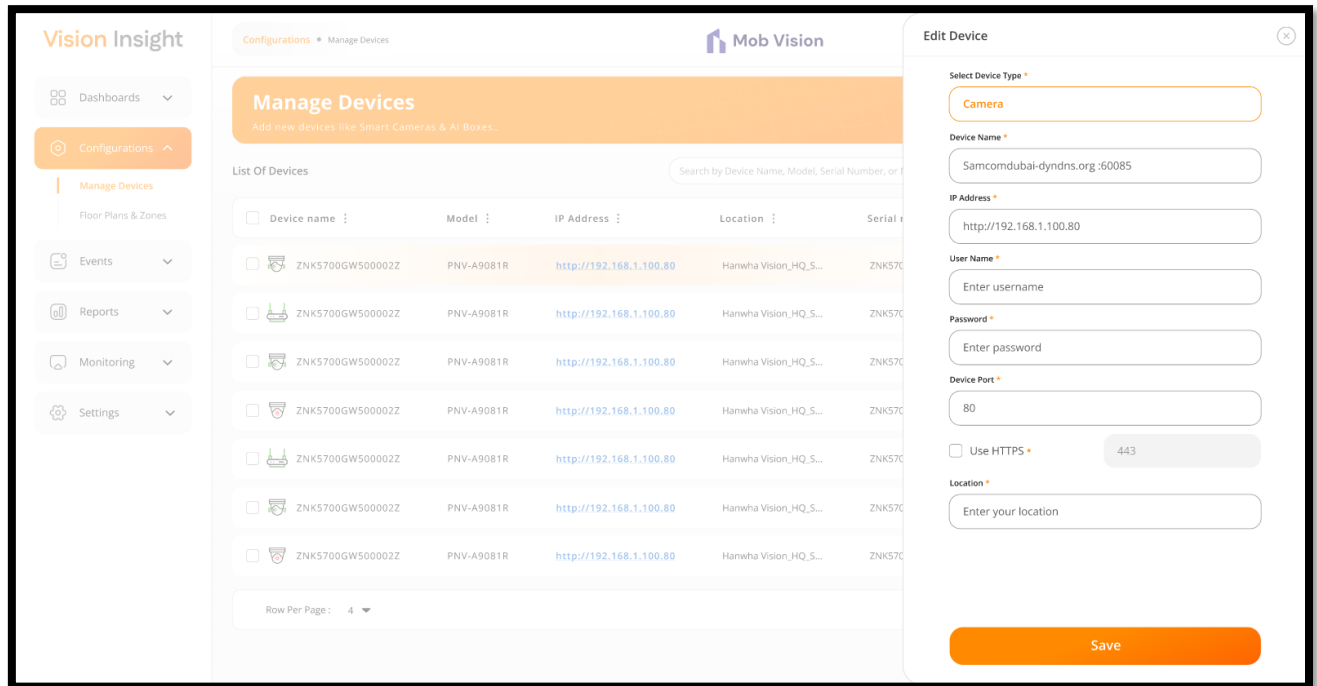
The screenshot shows the 'Add Device' form in the Vision Insight application. The form is titled 'Add Device' and has a close button (X). It features a 'Select Device Type' dropdown with 'Camera' selected. The form includes fields for 'Device Name' (Samcomdubai-dyndns.org :60085), 'IP Address' (http://192.168.1.100.80), 'User Name' (Enter username), 'Password' (Enter password), 'Device Port' (80), and 'Location' (Enter your location). There is a checkbox for 'Use HTTPS' and a '443' button. The 'Add' button is orange and located at the bottom right.

→ The screen below represents the form for **creating a new AI Box device**, allowing users to enter relevant details and submit the information:



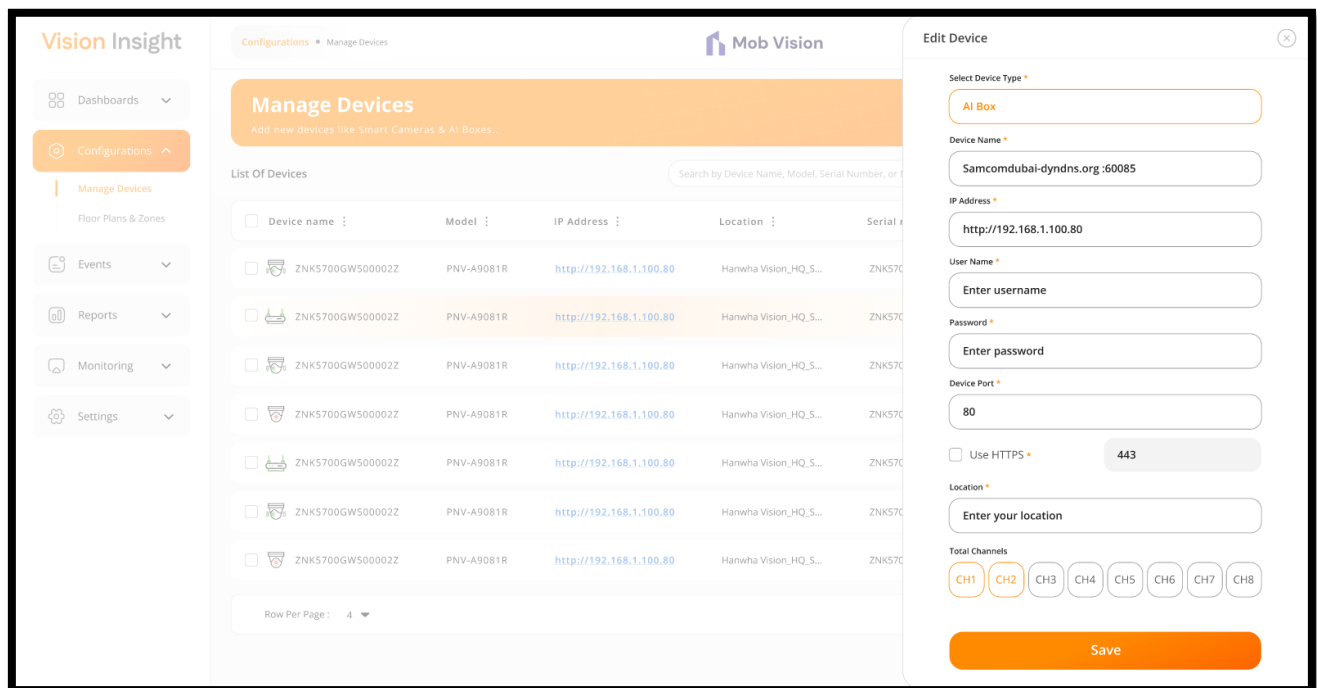
The screenshot shows the 'Add Device' form in the Vision Insight application. The form is titled 'Add Device' and has a close button (X). It features a 'Select Device Type' dropdown with 'AI Box' selected. The form includes fields for 'Device Name' (Samcomdubai-dyndns.org :60085), 'IP Address' (http://192.168.1.100.80), 'User Name' (Enter username), 'Password' (Enter password), 'Device Port' (80), and 'Location' (Enter your location). There is a checkbox for 'Use HTTPS' and a '443' button. The 'Add' button is orange and located at the bottom right.

→ The screen below displays the form for **editing an existing camera device**, allowing users to modify relevant details and update the information:



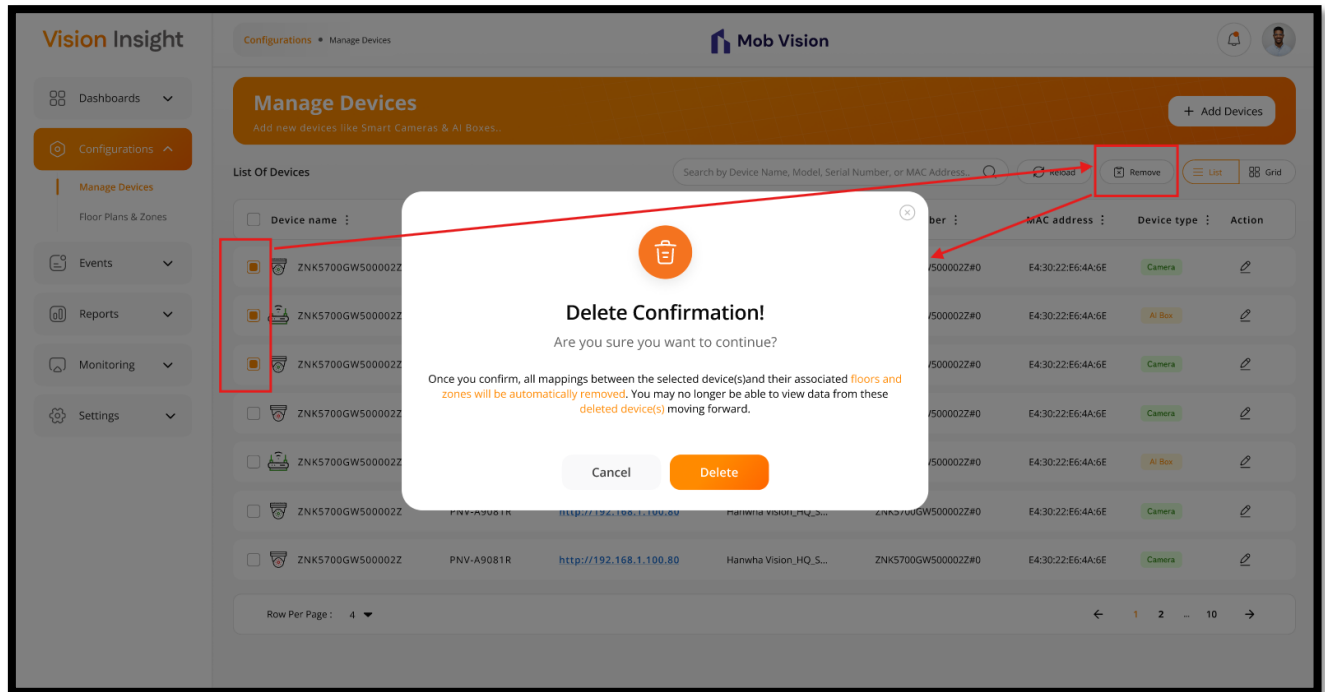
The screenshot shows the 'Edit Device' form in the Vision Insight application. The form is titled 'Edit Device' and includes a close button (X). The 'Select Device Type' dropdown is set to 'Camera'. The 'Device Name' field contains 'Samcomdubai-dyndns.org :60085'. The 'IP Address' field contains 'http://192.168.1.100.80'. The 'User Name' field is empty with the placeholder 'Enter username'. The 'Password' field is empty with the placeholder 'Enter password'. The 'Device Port' field contains '80'. The 'Use HTTPS' checkbox is checked, and the '443' port is shown in a greyed-out state. The 'Location' field is empty with the placeholder 'Enter your location'. A 'Save' button is at the bottom right.

→ The screen below displays the form for **editing an existing AI Box device**, allowing users to modify relevant details and update the information:

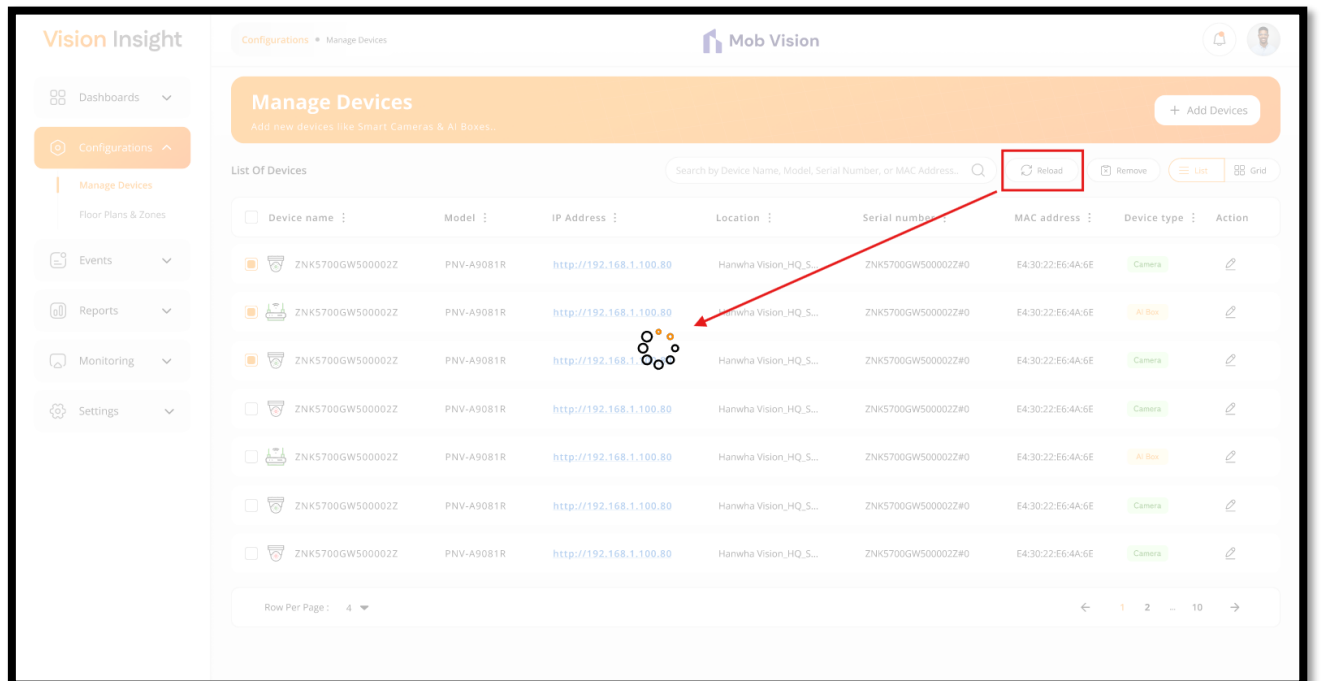


The screenshot shows the 'Edit Device' form in the Vision Insight application. The form is titled 'Edit Device' and includes a close button (X). The 'Select Device Type' dropdown is set to 'AI Box'. The 'Device Name' field contains 'Samcomdubai-dyndns.org :60085'. The 'IP Address' field contains 'http://192.168.1.100.80'. The 'User Name' field is empty with the placeholder 'Enter username'. The 'Password' field is empty with the placeholder 'Enter password'. The 'Device Port' field contains '80'. The 'Use HTTPS' checkbox is checked, and the '443' port is shown in a greyed-out state. The 'Location' field is empty with the placeholder 'Enter your location'. Below the location field, there are 'Total Channels' with buttons for CH1, CH2, CH3, CH4, CH5, CH6, CH7, and CH8. A 'Save' button is at the bottom right.

- The system will request **confirmation** along with an **appropriate message** when a logged-in user attempts to **delete any authorized user**:



- The screen below represents the **loader** on the page when the **authorized user clicks on the “Reload” button**:



Business Rules:

- **Authorized users** will have the ability to **add, update, and delete devices** within the system.
- The system will allow users to **select one or more devices** from the list and provide the ability to **remove the selected device(s)** from the records.
- When a user attempts to delete one or more devices, the system will display a **confirmation popup** with an **appropriate message**. Upon confirmation, the system will **automatically un-map the selected device(s)** from their associated **floors and zones**, and users will **no longer be able to view data** from these **deleted device(s)** moving forward.
- When a logged-in user attempts to delete device(s), the system will prompt for confirmation. Once confirmed, the system will **soft delete** the selected device(s), meaning the records will remain in the database but will no longer be visible in the web application.
- The system will prevent duplicate entries for devices by validating the **Device Name** and **IP Address**.
- The "Add Device" and "Edit Device" screens will include the following fields:
 - ➔ Select Device Type* - (Camera / AI Box)
 - ➔ Device Name*
 - ➔ IP Address*
 - ➔ Username*
 - ➔ Password*
 - ➔ Device Port*
 - ➔ Location*
 - ➔ Use HTTP (*Ability to specify HTTP port; default value will be 443*)
 - ➔ Total Channels (*This will be displayed only if the device type is AI Box*)
- The asterisk (*) indicates a mandatory field only.
- The system will provide authorized users the ability to view the list of added devices using two options: "Card View" and "List View".
- **Both views** will support **pagination** and allow users to **configure the number of rows per page**.
- The list will include the following fields along with pagination:
 - ➔ Checked / Unchecked (selection checkbox)
 - ➔ Device Online / Offline status (The device icon for Camera / AI Box will be displayed based on its status)
 - ➔ Device Name
 - ➔ Model
 - ➔ Location
 - ➔ Serial Number
 - ➔ MAC Address
 - ➔ Device Type
 - ➔ Action (Edit & View)
- The system will allow users to search for specific device(s) using **Device Name, Model, Serial Number, or MAC Address** only.
- The system will allow **authorized users** to **click on the value in the "IP Address" column**, and upon clicking, it will **open the corresponding URL in a new browser window**.
- The system will allow users to **"Reload"** the device list.
- The system will **make API calls** to retrieve the **current online/offline status of devices** when the user clicks on the **"Reload"** button. The **fetch status** will be **stored in the database** and will be **reflected in the web application** accordingly.

- Under “List View” only, the system will offer **filter options on specific columns** using the **MUI library**, which is **integrated throughout the web application**. These **filters will reset** when the user **reloads the page or navigates to another page**.
- The system will **retrieve device status data from a background job** at a **predefined time interval (default: 15 minutes)**. This interval can be **modified from the source code** and may require **restarting the application from the hosted environment**. The **retrieved status will be stored in the database** for all configured devices and will be **reflected in the web application**.

Data Dictionary:

deviceMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceType	String	Yes	Camera	Type of the device
3	deviceName	String	Yes	Netgear	Name of the device
4	ipAddress	String	Yes	10.10.10.1	device IP address
5	userName	String	Yes	admin	Username of the device
6	password	String	Yes	admin	Password of the device
7	devicePort	String	Yes	1234	Port number of the device
8	isHttps	Boolean	No	TRUE	Is device URL us secured
9	httpPort	String	Yes	1234	httpport number
10	location	String	Yes	Dubai	Location of the device
11	apiModel	String	Yes	SUNAPI / WiseAI	AI model installed in the camera
12	model	String	Yes	QNO-C8083R	Model of the camera
13	serialNumber	String	Yes	ZS4D6V4TC00012K	serial number of the camera
14	macAddress	String	Yes	E4:30:22:8D:20:FF	MAC address of the camera
15	channelEvent	Object[]	Yes	[{}]	Channel list supported by device
16	peopleCount	Object[]	Yes	[{}]	People count configuration data
17	objectCountingConfiguration	Object[]	Yes	[{}]	Configuration of all types of WiseAI counting rules
18	vehicleCount	Object[]	No	[{}]	Vehicle count configuration data
19	features	Object	Yes	{}	All the supported feature data
20	isDefaultZone	Boolean	Yes	FALSE	is device is mapped or not
21	isOnline	Boolean	Yes	TRUE	device is online or offline
22	isDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
23	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
24	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
25	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
26	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

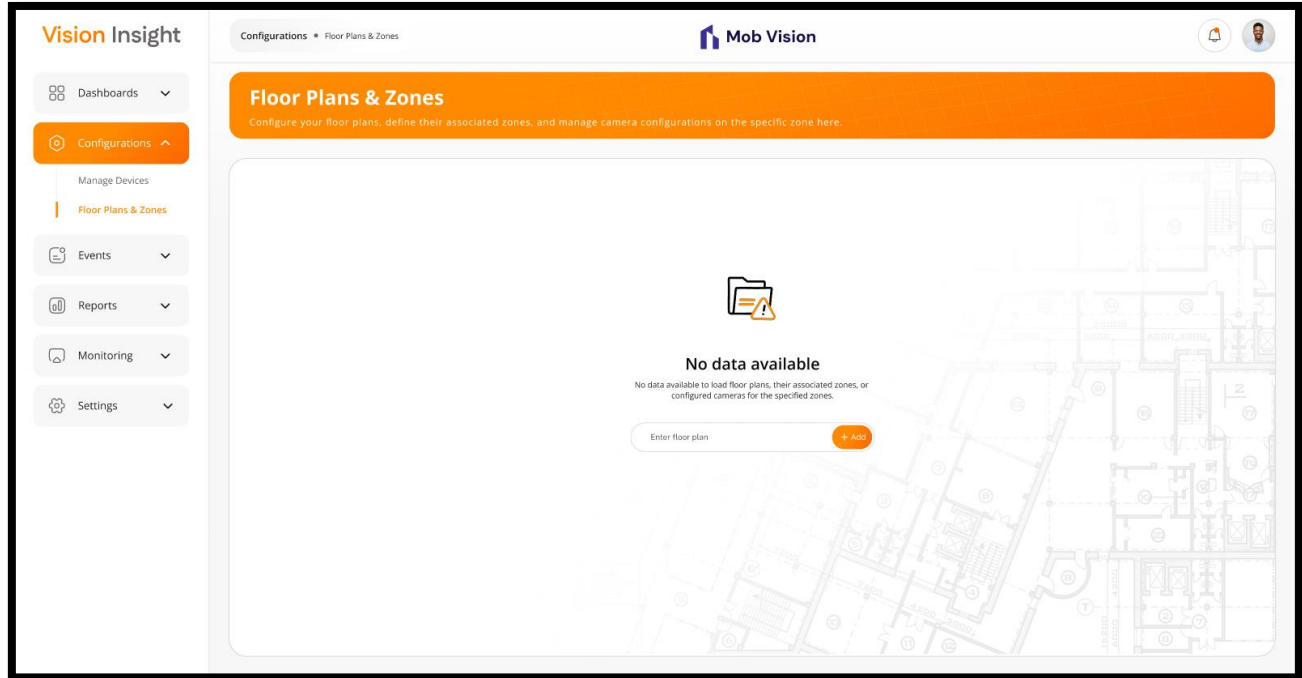
3.14.Configurations > Floor Plans & Zones Page

The system provides a feature for **authorized users** to **manage Floor Plans and Zones** through the **BI Dashboard Web Application**.

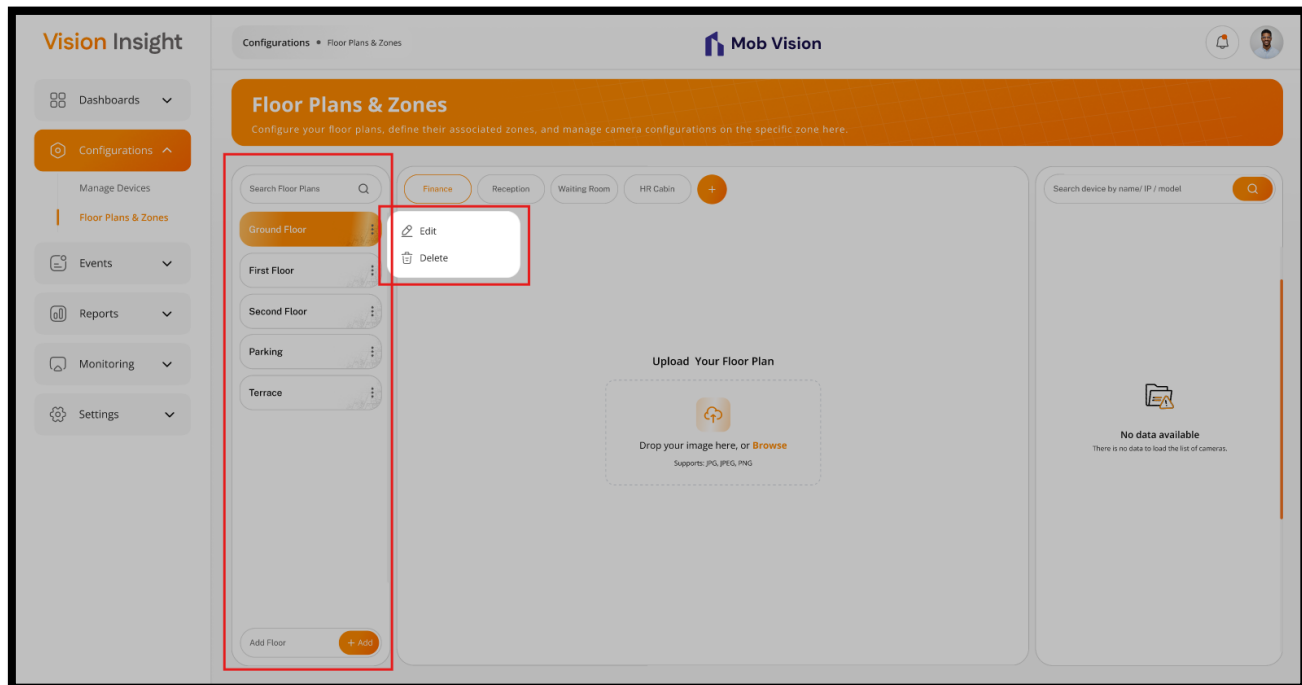
This page will be **accessible exclusively to authorized users**, with actions such as **adding, editing, and deleting floor plans and their associated zones, adding camera on the specific zone, etc.**

Prototype:

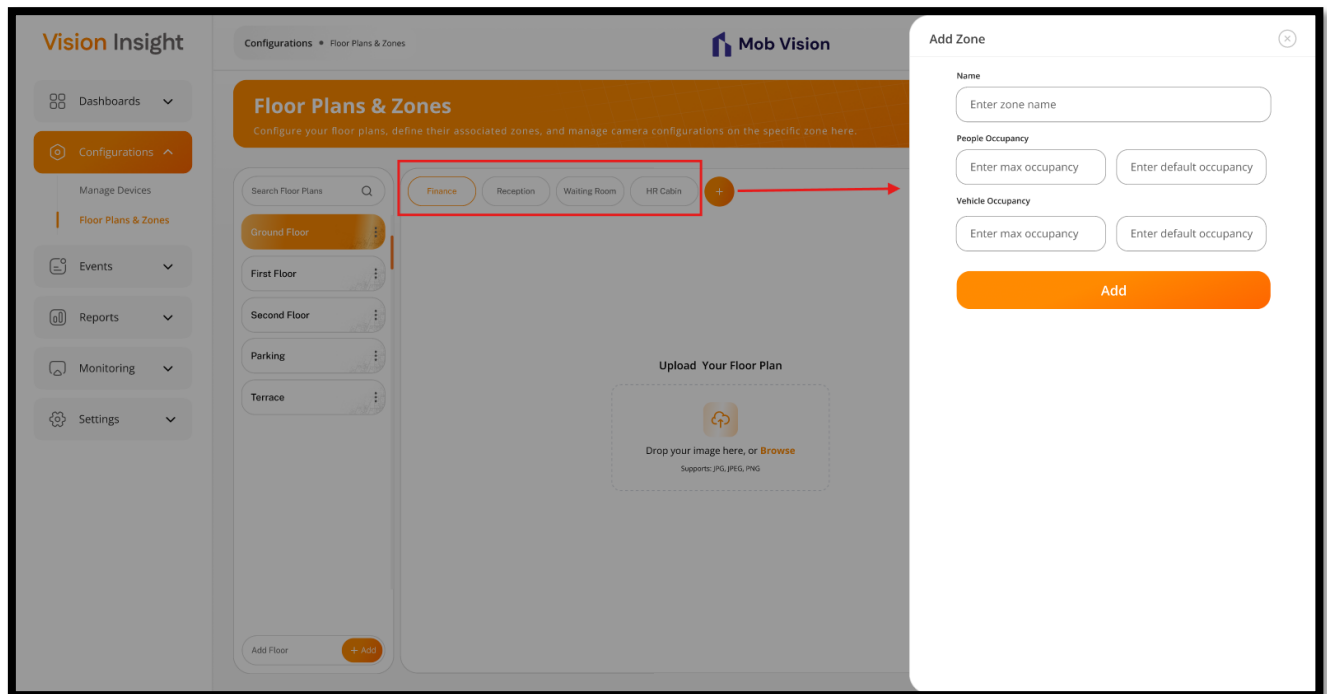
- The screen below illustrates **first landing page with appropriate message & action to add new floor plan** when there is **no any floor plan yet configured in the system**:



- The screen below illustrates the **list of added floor plans**, including options to **Search, Add, Update & Delete** the floor plan from:



- The screen below represents the **form for creating a new zone**, allowing users to **enter relevant details and submit** the information. Additionally, the **list of all the created zones**:

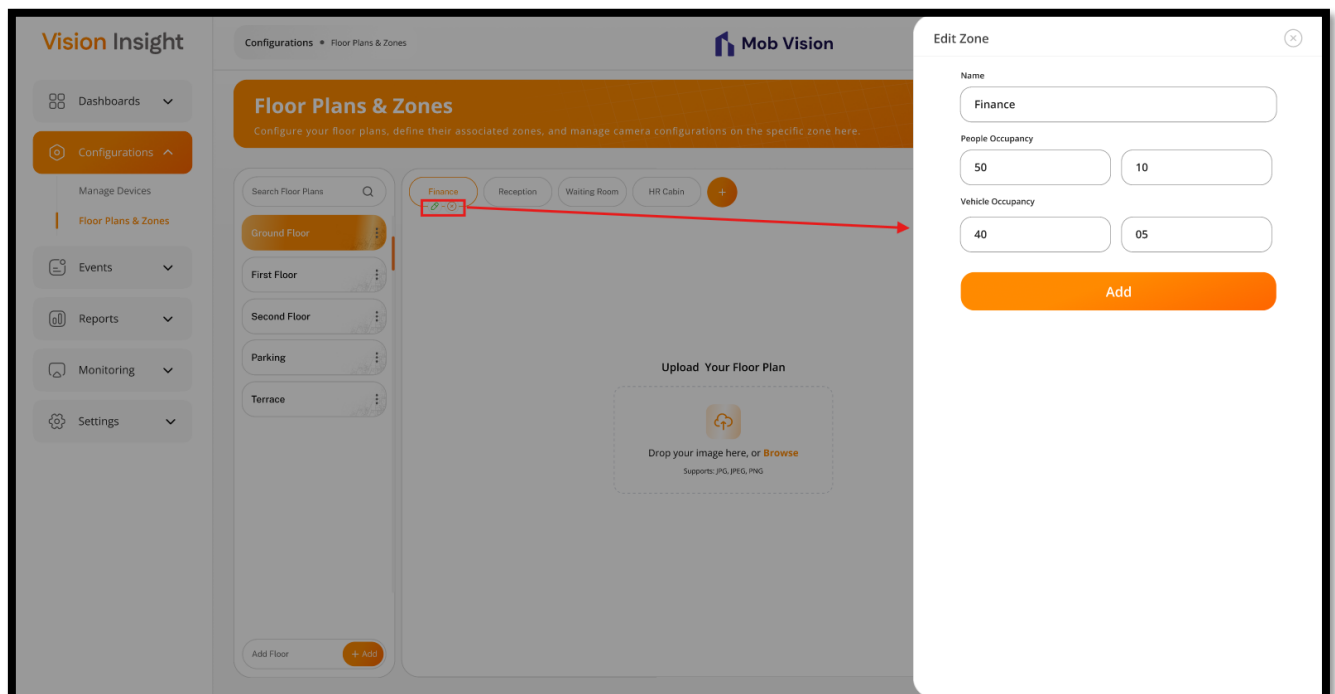


The screenshot shows the 'Add Zone' form in the Vision Insight application. The form is titled 'Add Zone' and includes the following fields:

- Name:** Enter zone name
- People Occupancy:** Enter max occupancy, Enter default occupancy
- Vehicle Occupancy:** Enter max occupancy, Enter default occupancy

An orange 'Add' button is located at the bottom of the form. A red box highlights the 'Finance' button in the 'Floor Plans & Zones' section of the main interface, with an arrow pointing to the form.

- The screen below displays the **form for editing an existing zone**, allowing users to **modify relevant details and update** the information:

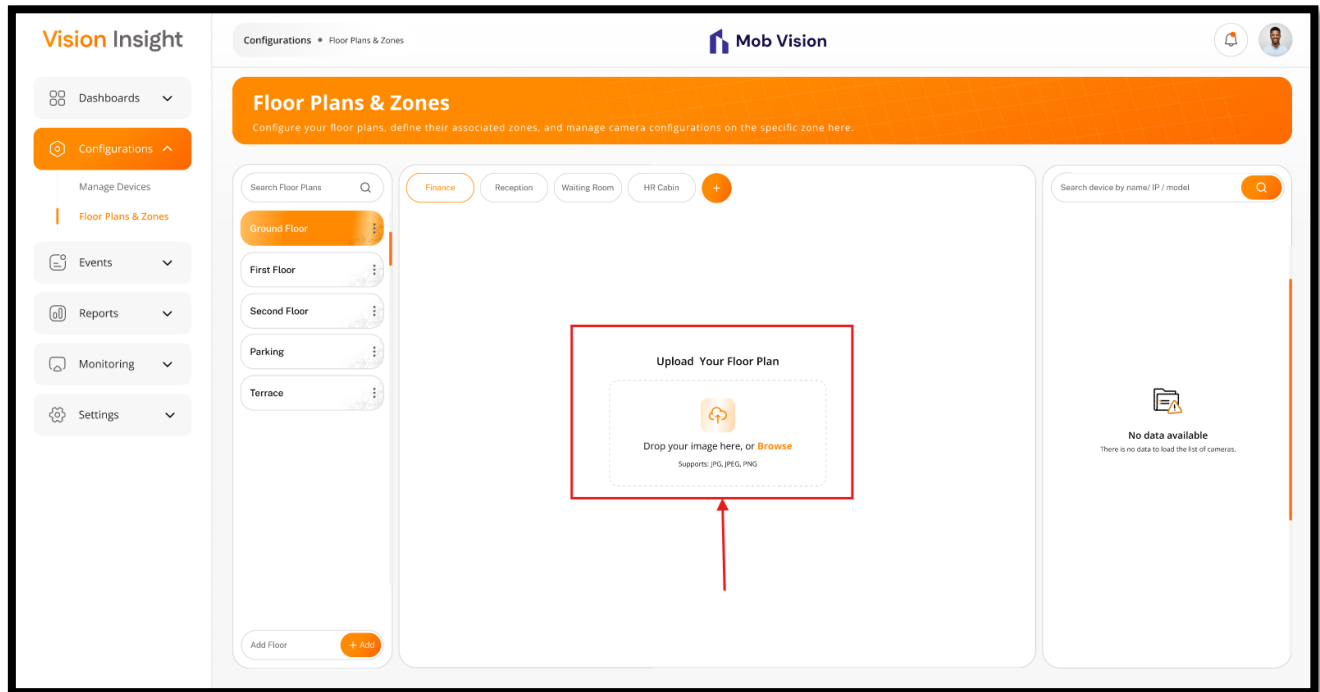


The screenshot shows the 'Edit Zone' form in the Vision Insight application. The form is titled 'Edit Zone' and includes the following fields:

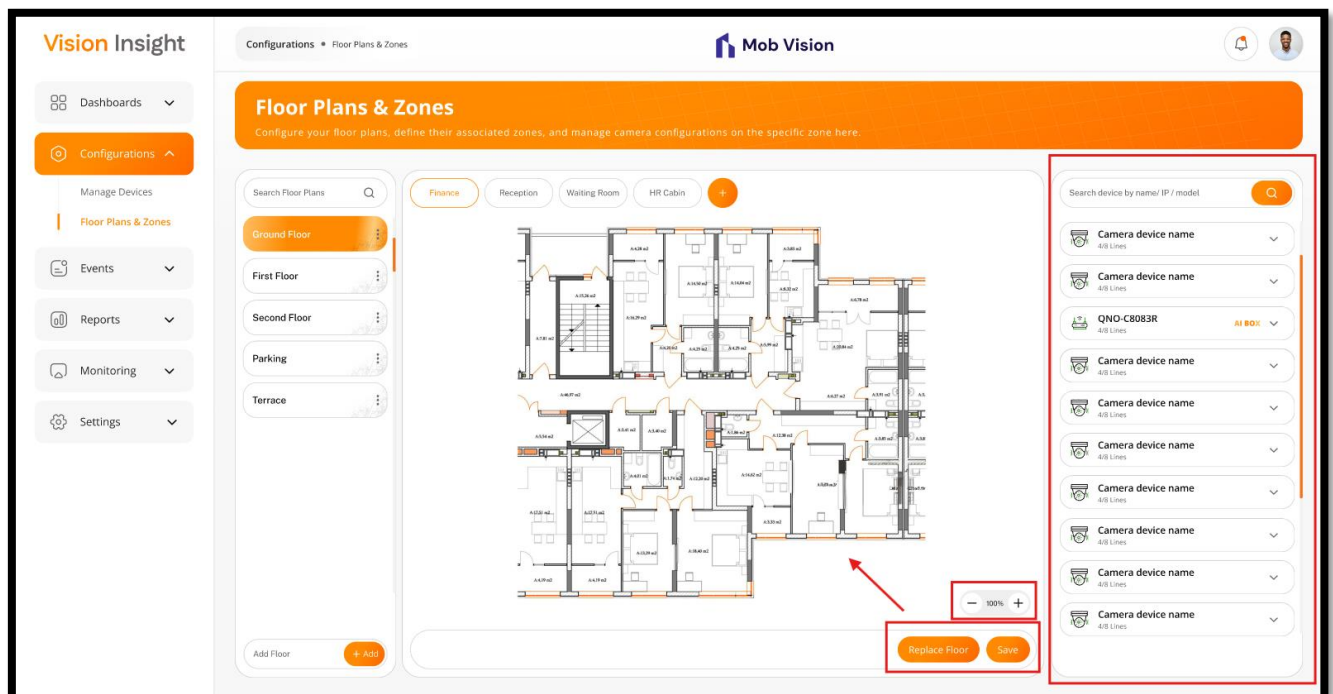
- Name:** Finance
- People Occupancy:** 50, 10
- Vehicle Occupancy:** 40, 05

An orange 'Add' button is located at the bottom of the form. A red box highlights the 'Finance' button in the 'Floor Plans & Zones' section of the main interface, with an arrow pointing to the form.

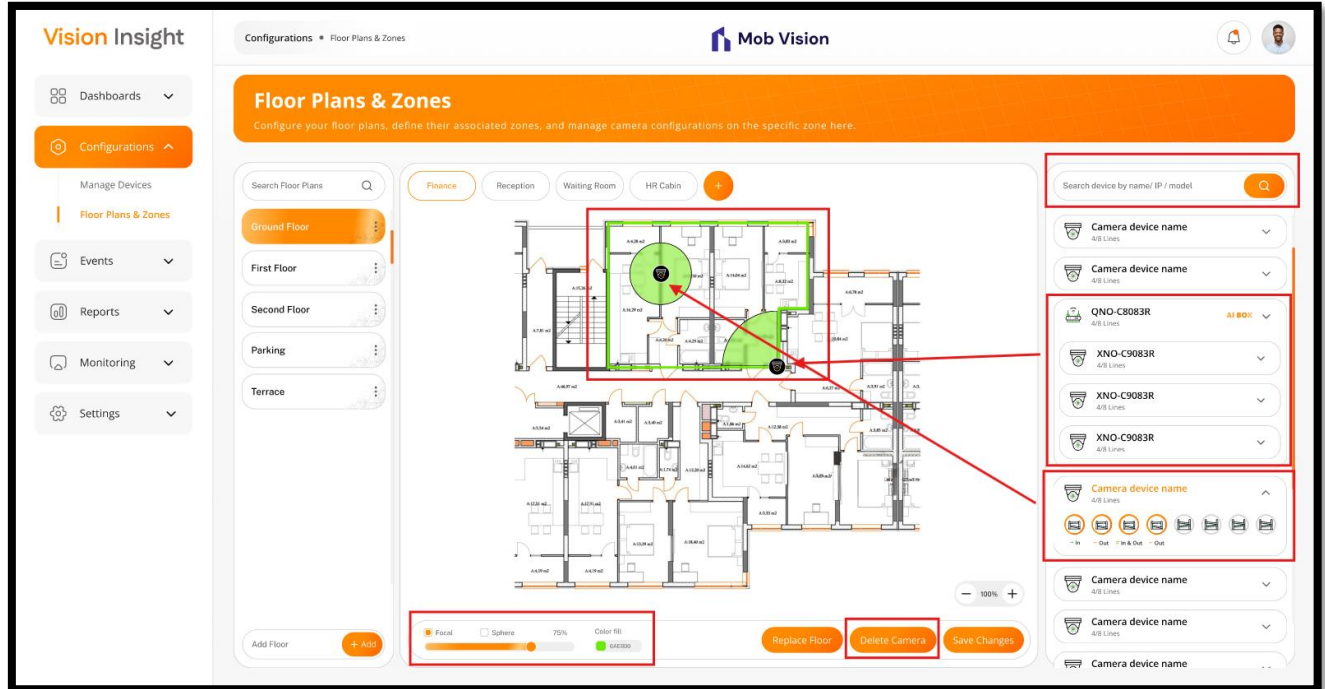
→ The screen below displays the **option to select & upload floor plan image**:



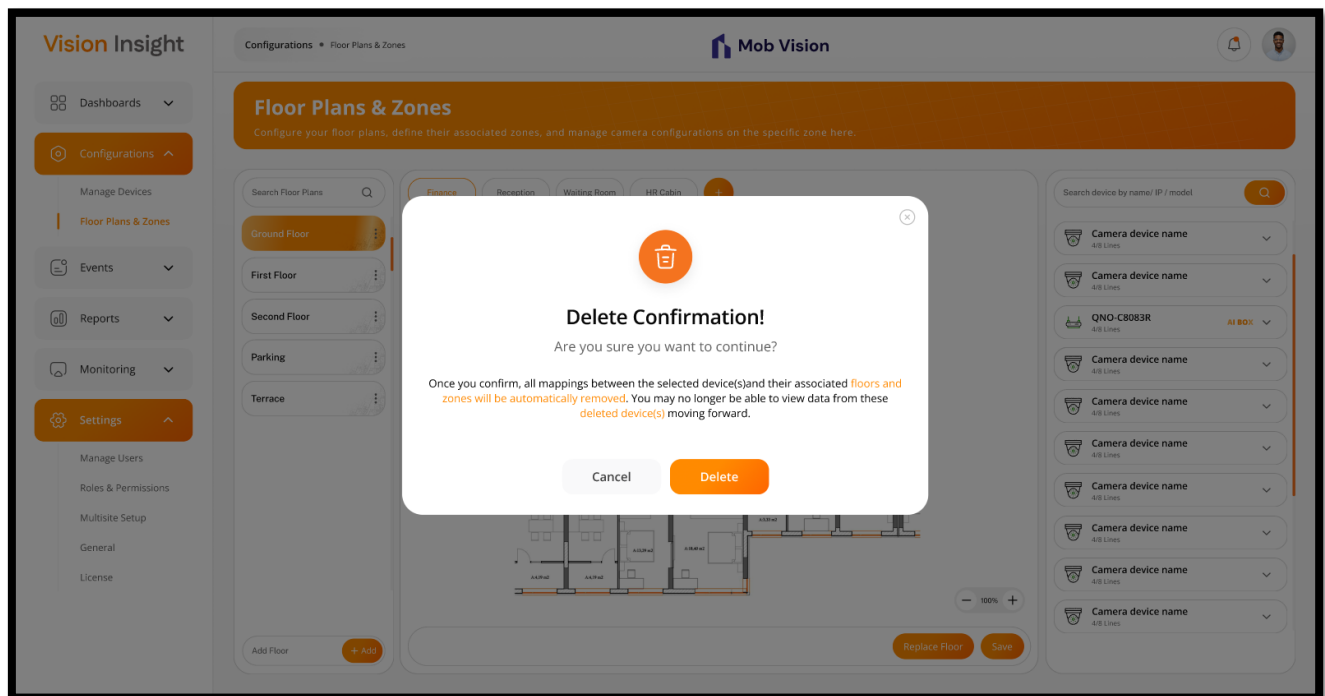
→ The screen below displays the **uploaded floor plan image** along with the **options to replace floor plan image, save entire changes, zoom-in & zoom-out floor plan image, & list of all online cameras which are yet pending to configure on the zone**:



- The screen below displays the **camera-related options** including **search the camera, expand, select & drag-and-drop the camera on the floor plan image under selected zone, adjust the sphere & focal view, adjust the color of sphere & focal view, rotate the camera, selection of line(s) while configuring the camera, and delete the camera from the specific area:**



- The system will request **confirmation** with an **appropriate message** when a logged-in user attempts to **delete any authorized zone**:



Business Rules:

- **Authorized users** will have the ability to **add, update, and delete specific floor plans** within the system.
- The system will display a **confirmation popup with an appropriate message** when a user attempts to **delete a floor plan** from the list:
 - ➔ Once the user **confirms the prompt**, the system will **automatically delete all associated zones, unmapped their devices**, and the user **may no longer be able to view data** from these deleted **floor plans, zones, and mapped devices** moving forward.
- **Authorized users** will also have the ability to **add, update, and delete specific zones** within the system.
- The system will display a **confirmation popup with an appropriate message** when a user attempts to **delete a zone** from the list:
 - ➔ Once confirmed, the system will **automatically unmapped all configured devices**, and the user **may no longer be able to view data** from these deleted **zones and mapped devices**.
- When a authorized users attempts to delete **floor plans or zones** along with their configured devices, the system will **prompt for confirmation**:
 - ➔ Once confirmed, the system will perform a **soft delete**, meaning the records will **remain in the database** but will **no longer be visible** in the web application.
- The system will **prevent duplicate entries** for floor plans and zones by **validating their names**.
- The “Add Floor Plan” and “Edit Floor Plan” features will include the following fields:
 - ➔ Floor Plan Name*
- The “Add Zone” and “Edit Zone” features will include the following fields:
 - ➔ Name*
 - ➔ People Occupancy*
 - Max Occupancy
 - Default Occupancy
 - ➔ Vehicle Occupancy*
 - Max Occupancy
 - Default Occupancy
- **NOTE:** Either “**People Occupancy**” or “**Vehicle Occupancy**” is a **mandatory field** - at least one must be provided.
- The asterisk (*) indicates a mandatory field only.
- **NOTE:** The system will automatically create a **default Floor Plan and Zone** when a user adds a **single device for the first time** or if the device is **not yet assigned to any zone**. By default, such devices will be **assigned to this default Floor Plan and Zone**.
- The system will provide authorized users the ability to view the list of added floor plans & their zones.
- The system will allow users to search for specific floor plan using **Floor Plan Name** only.
- The system will allow users to upload floor plan image with a **maximum size of 10 MB**, supporting only the following formats: **.pdf, .jpg, .png, .jpeg, and .svg**.
- **The system should allow users to upload only single-page PDF files**. If a file contains multiple pages, the system must reject it and display an appropriate error message.
- Authorized users will be able to Zoom-In and Zoom-Out on the uploaded floor plan image.
- **Authorized users will have the option to replace the existing floor plan image** by clicking on the “**Replace Floor Plan**” button.
- The system will display a list of online Camera / AI Box devices that are not yet configured in any zone.
- The camera / AI Box list will support data filtration for the following columns:
 - ➔ Camera Name
 - ➔ IP Address

➔ Model Name

- **Users will be allowed to configure an unlimited number of cameras within a specific zone** (no limitation).
- Each configured camera will be exclusively associated with one specific zone.
- During real implementation, if the system fails to retrieve the model's name of a camera connected to an AI Box, it should default to generic names like "Camera-1", "Camera-2", etc.
- **Users will be able to drag a camera device from the device list and drop it onto a specific zone** on the floor plan image.
- After dragging, users can move and position the camera within the zone as needed.
- Users can **select a specific camera lane** and **map it to different zones** within the same floor. However, when **camera lanes are divided across multiple zones**, the **data counts will not be automatically synchronized across all zones**, as discussed with Bala.
- Once a camera is placed, users will be able to select it and perform the following actions.
 - ➔ Choose between "Focal" or "Sphere" view and adjust its size up to 100%.
 - ➔ Select a color for the Focal or Sphere area.
 - ➔ Delete the selected camera from the zone.
 - ➔ Rotate the camera
 - ➔ Save all changes to the database by clicking the "Save Changes" button.]

Floor Plans and Zones Permissions:

- The system will maintain permissions at the Floor Plan and Zone levels only.
- It will be the responsibility of the admin user (who configures the application, users, and user permissions) to assign appropriate permissions for specific floor plans and zones to each user role, especially in cases where a single camera is configured in multiple zones.
- As per Bala's request, the system will not handle this scenario automatically.
- The system will display information strictly based on the assigned permissions at the Zone and Floor plan levels.

Data Dictionary:

floorPlanMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	floorPlanName	String	No	Ground floor	Name of the floor plan
3	floorImage	String	Yes	data:image/png;base64,/9j	Image of the floor plan
4	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
5	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
6	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
7	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

zoneMaster					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	floorId	String	No	674daaf7d34a2b901c847454	The zone is mapped with floor
3	zoneName	String		674daaf7d34a2b901c847454	Name of the zone
4	peopleOccupancy	Int32		10	People Occupancy
5	vehicleOccupancy	Int32		10	Vehicle Occupancy
6	peopleDefaultOccupancy	Int32		5	People Default Occupancy
7	vehicleDefaultOccupancy	Int32		5	Vehicle Default Occupancy
8	resetAt	Int32		870	Reset time for zones camera in number format
9	zoneArea	Object[]		[{"x": 50, "y": 50}]	Position array of zone on floor plan
9.1	> x	Int32		50	X position of zone area
9.2	> y	Int32		50	Y position of zone area
10	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
11	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
12	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
13	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
14	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

zoneCamera					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	floorId	String	No	674daaf7d34a2b901c847454	Device belongs to which floor
3	zoneId	String	No	674daaf7d34a2b901c847454	Device belongs to which zone
4	deviceId	String	No	674daaf7d34a2b901c847454	Device id for create mappings
5	position	Object	Yes	{x:50,y:50,angle:30}	Position of camera over zone
5.1	> x	Int32	No	50	x position over zone
5.2	> y	Int32	No	50	y position over zone
5.3	> angle	Int32	No	30	Angle of the camera
6	fovLength	Int32	No	45	Camera focal length
7	fovColor	String	No	#####	camera focal color
8	peopleLineIndex	int32[]	No	[1]	Mapped people line index
9	vehicleLineIndex	int32[]	No	[1]	Mapped vehicle line index
10	isSphere	Bool	No	FALSE	camera capture type, is sphere or not
11	channel	int	No	0	Mapped channel number
12	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
13	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
14	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
15	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
16	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

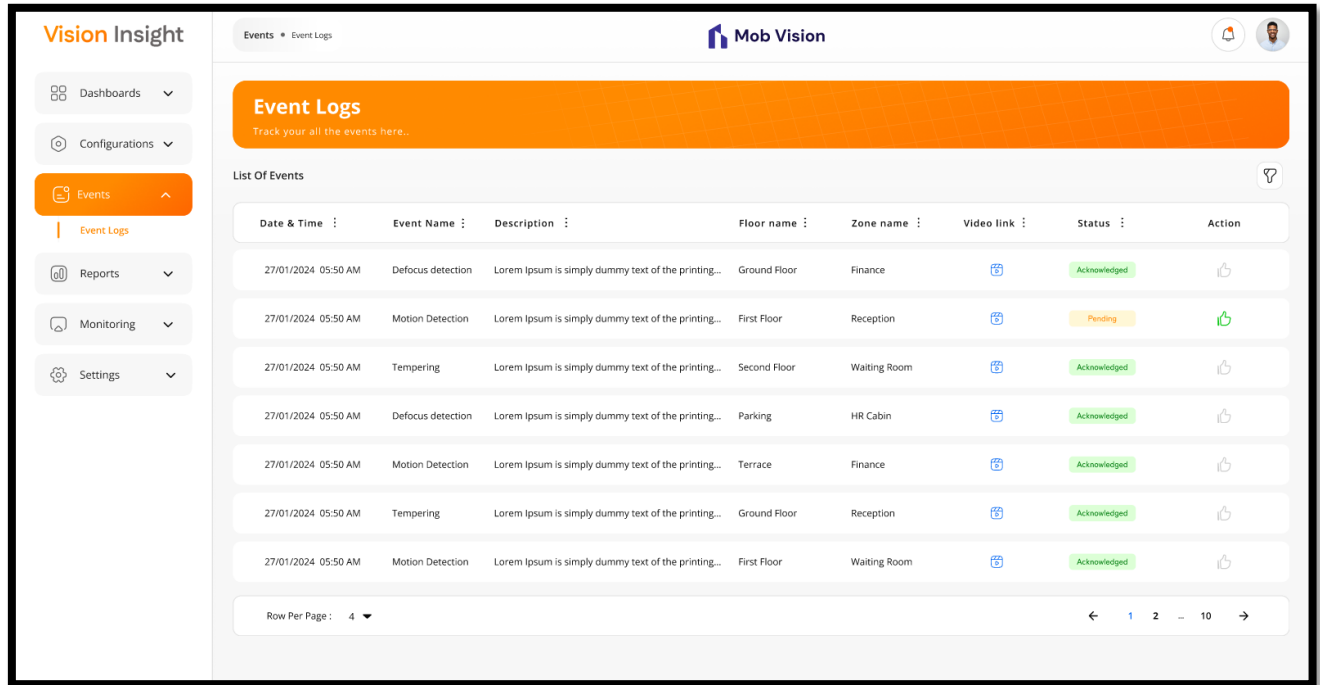
3.15.Events > Event Logs Page

The system provides a feature for **Super Admin** and **authorized users** to **view the details of all detected events** through the **BI Dashboard Web Application**.

This page will be **accessible only to Super Admin and authorized users**, with actions such as **acknowledging events** and **viewing pre- and post-5-second videos**, based on the **permissions assigned to the logged-in user**.

Prototype:

→ The screen below illustrates the list of all detected events, including options for searching using filters, acknowledging specific events, and an icon to view the pre- and post-5-second video clips:



Vision Insight | Events • Event Logs | Mob Vision

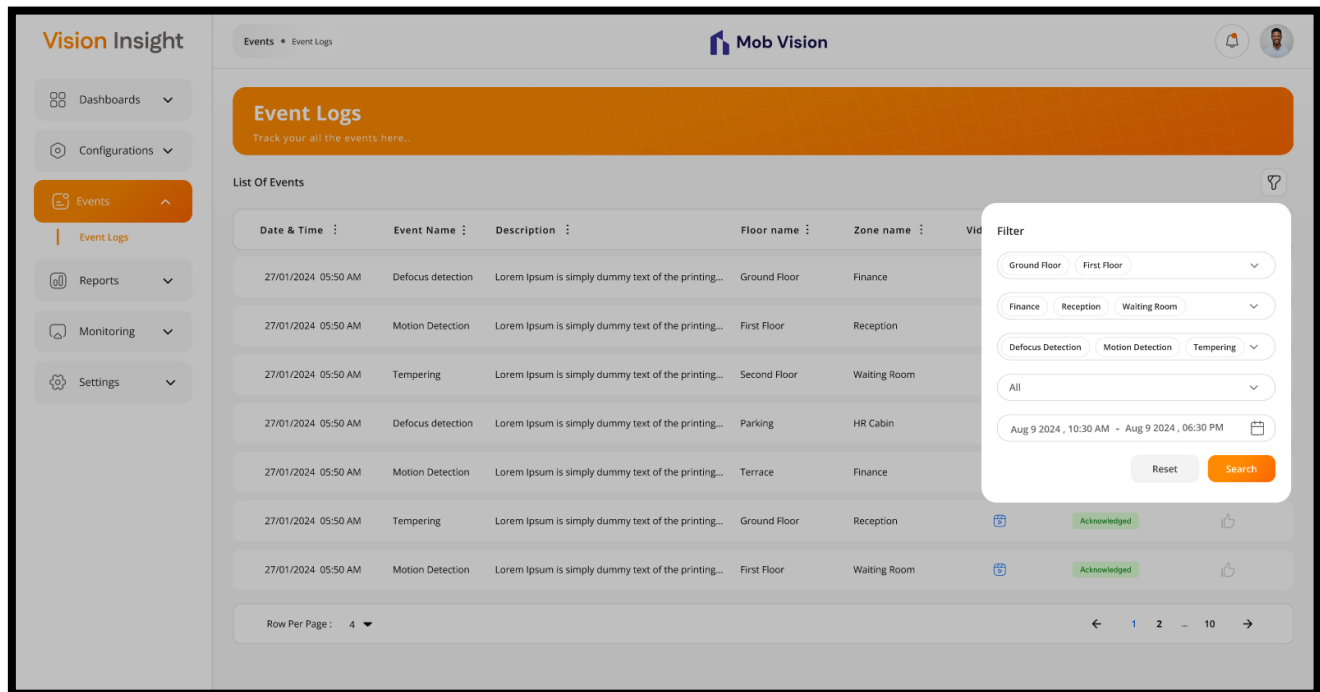
Event Logs
Track your all the events here..

List Of Events

Date & Time	Event Name	Description	Floor name	Zone name	Video link	Status	Action
27/01/2024 05:50 AM	Defocus detection	Lorem Ipsum is simply dummy text of the printing...	Ground Floor	Finance		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	First Floor	Reception		Pending	
27/01/2024 05:50 AM	Tempering	Lorem Ipsum is simply dummy text of the printing...	Second Floor	Waiting Room		Acknowledged	
27/01/2024 05:50 AM	Defocus detection	Lorem Ipsum is simply dummy text of the printing...	Parking	HR Cabin		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	Terrace	Finance		Acknowledged	
27/01/2024 05:50 AM	Tempering	Lorem Ipsum is simply dummy text of the printing...	Ground Floor	Reception		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	First Floor	Waiting Room		Acknowledged	

Row Per Page: 4

→ The screen below illustrates the filter options:



Vision Insight | Events • Event Logs | Mob Vision

Event Logs
Track your all the events here..

List Of Events

Date & Time	Event Name	Description	Floor name	Zone name	Video link	Status	Action
27/01/2024 05:50 AM	Defocus detection	Lorem Ipsum is simply dummy text of the printing...	Ground Floor	Finance		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	First Floor	Reception		Acknowledged	
27/01/2024 05:50 AM	Tempering	Lorem Ipsum is simply dummy text of the printing...	Second Floor	Waiting Room		Acknowledged	
27/01/2024 05:50 AM	Defocus detection	Lorem Ipsum is simply dummy text of the printing...	Parking	HR Cabin		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	Terrace	Finance		Acknowledged	
27/01/2024 05:50 AM	Tempering	Lorem Ipsum is simply dummy text of the printing...	Ground Floor	Reception		Acknowledged	
27/01/2024 05:50 AM	Motion Detection	Lorem Ipsum is simply dummy text of the printing...	First Floor	Waiting Room		Acknowledged	

Row Per Page: 4

Filter

Floor name: Ground Floor | First Floor

Zone name: Finance | Reception | Waiting Room

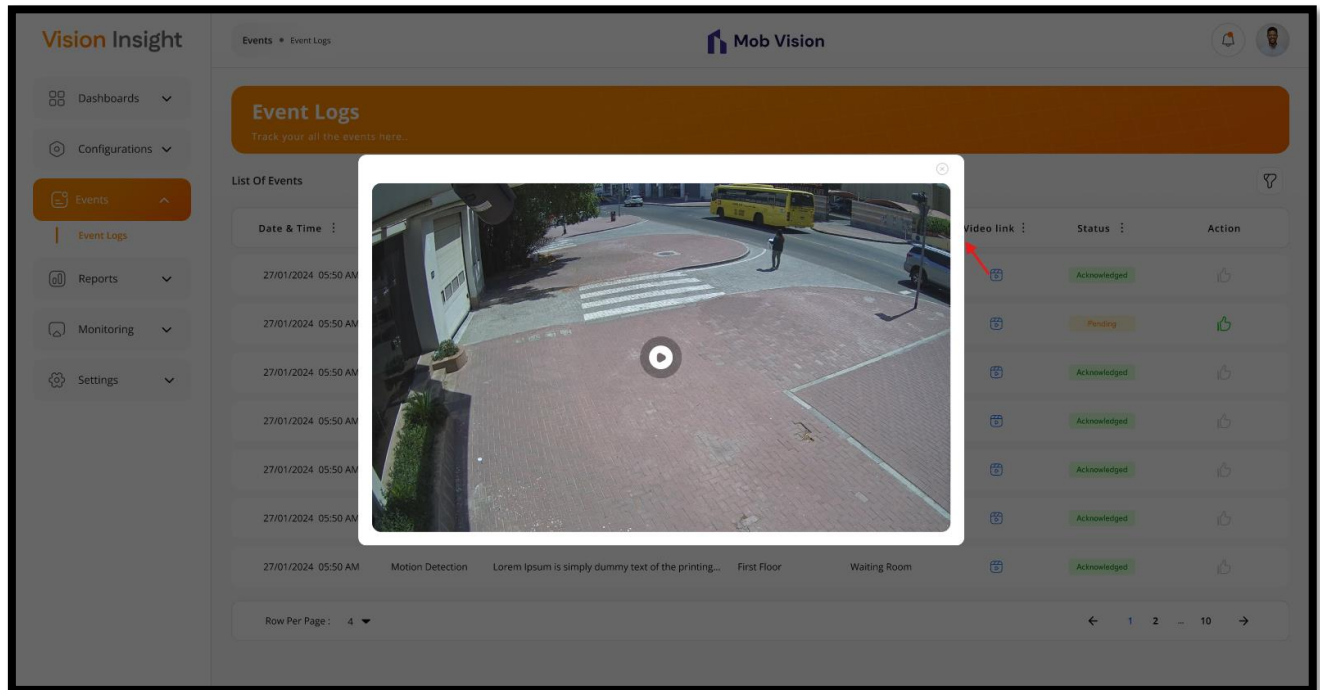
Event Name: Defocus Detection | Motion Detection | Tempering

All

Aug 9 2024, 10:30 AM - Aug 9 2024, 06:30 PM

Reset Search

→ The screen below illustrates the **view for the pre- and post-5-second video clips:**



Business Rules:

- **Authorized users** will be able to view all the **detected events** on this page.
- **Events** will be displayed in **descending order** by default based on the **created time** in the database.
- The list will be displayed with the following **columns**:
 - Date & Time
 - Event Name
 - Description
 - Floor Name
 - Zone Name
 - Video Link
 - Status (*Acknowledged / Pending*)
 - Action (*to acknowledge the event, if it is still pending to acknowledge*)
- The system will offer filter options (**like Sort By ASC, Sort by DESC, Filter, Hide Column, Manage Columns**) on specific columns using the MUI library, which is integrated into the entire web application. These filters will reset when the user reloads the page or navigates to another page.
- **Pagination** will be applied to the list so that users can navigate to any page and check specific events.
- Authorized users will be able to **filter the data** by the following:
 - Event Name
 - From Date – To Date (*Date Range*)
 - Floor Name (*Dropdown List*)
 - Zone Name (*Dropdown List*)
 - Status (*Acknowledged / Pending*) (*Dropdown List*)
- The system will allow authorized users to **acknowledge events** that are still pending acknowledgment.

- The system will display an **appropriate message** to the user when an event is acknowledged.
- Authorized users will be able to **view pre- and post-5-second videos** by clicking on the **video icon**.
- The **camera must store all event videos exclusively on the FTP server**, with no local or alternative storage as BI Dashboard Web Application will always stream the video from the configured FTP server only.
- The system will open a **popup** when users click on the above option to view the video.

Data Dictionary:

deviceEvents					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	eventName	String	No	OpenSDK.WiseAI.VehicleQue	Name of the event
3	deviceId	String	No	674daaf7d34a2b901c847454	DeviceId id from where event is fire
4	channelNo	String	No	0	ChannelNo from where event is fire
5	ruleIndex	ObjectId	No	1	Rule index from where event is fire
6	ruleName	DateTime	No	rule1	Rule name from where event is fire
7	videoSourceToken	ObjectId	No	VideoSourceToken-0	Video source token of the event
8	vehicleQueue	Object	Yes	{state:false}	Data of the vehicle events
8.1	> state	Boolean	No	FALSE	State of the vehicle events
9	peopleQueue	Object	Yes	{state:false}	Data of the people events
9.1	> state	Boolean	No	FALSE	State of the people events
10	ShoppingCartQueue	Object	Yes	{state:false}	Data of the shopping cart events
10.1	> state	Object	No	FALSE	State of the shopping cart events
11	forkLiftQueue	Object	Yes	{state:false}	Data of the fork lift events
11.1	> state	Object	No	FALSE	State of the fork lift events
12	slipFallDetection	Object	Yes	{state:false}	Data of the Slip & Fall events
12.1	> state	Object	No	FALSE	State of the Slip & Fall events
13	wrongWayDetection	Object	Yes	{state:false}	Data of the Wrong Way Detection events
13.1	> state	Object	No	FALSE	State of the Wrong Way Detection events
14	pedestrianDetection	Object	Yes	{state:false}	Data of the Pedestrian Detection events
14.1	> state	Object	No	FALSE	State of the Pedestrian Detection events
15	blockedExitDetection	Object	Yes	{state:false}	Data of the Blocked Exit Detection events
15.1	> state	Object	No	FALSE	State of the Blocked Exit Detection events
16	vehicleSpeedDetection	Object	Yes	{state:false}	Data of the Vehicle Speed Detection events
16.1	> state	Object	No	FALSE	State of the Vehicle Speed Detection events
17	trafficJamDetection	Object	Yes	{state:false}	Data of the Traffic Jam Detection events
17.1	> state	Object	No	FALSE	State of the Traffic Jam Detection events
18	stoppedVehicleByType	Object	Yes	{state:false}	Data of the Stopped Vehicle By Type events
18.1	> state	Object	No	FALSE	State of the Stopped Vehicle By Type events
19	proximityDetection	Object	Yes	{state:false}	Data of the Proximity Detection events
19.1	> state	Object	No	FALSE	State of the Proximity Detection events
20	IsAcknowledged	Object	No	FALSE	Event is Acknowledge or not
21	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
22	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
23	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
24	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
25	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

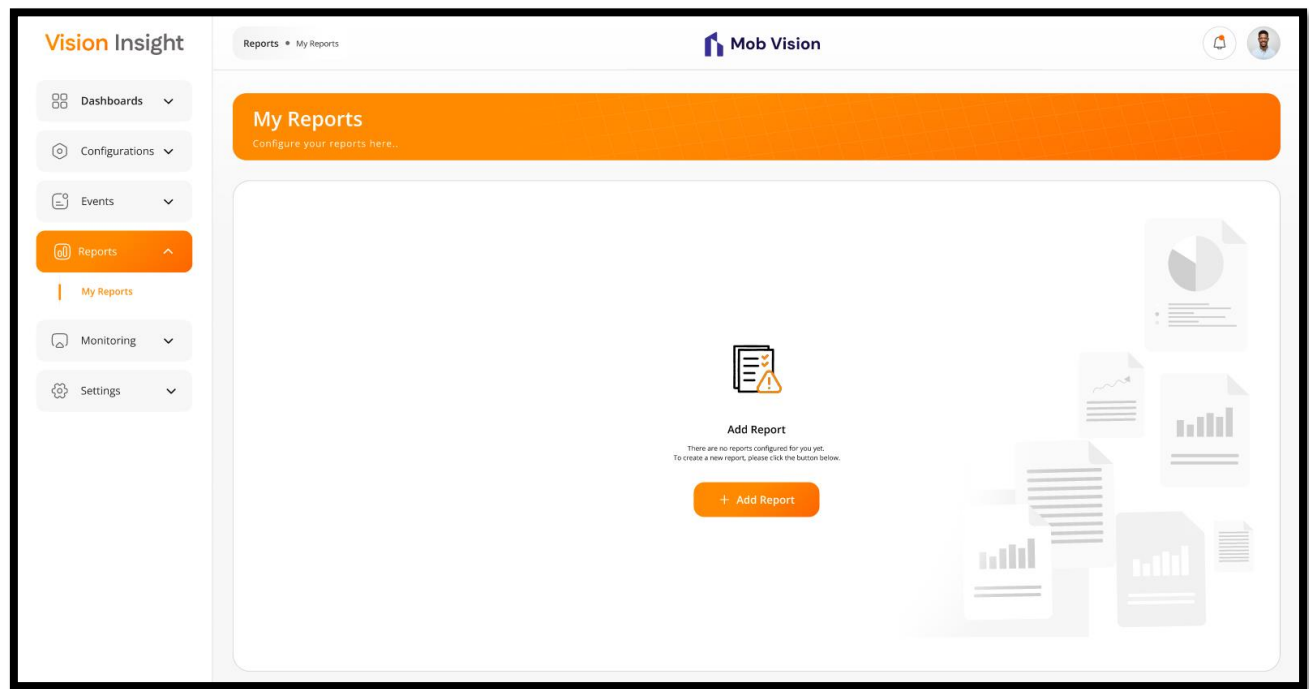
3.16.Reports > My Reports Page

The system provides a feature for **Super Admin and authorized users** to **manage reports** by selecting from **predefined report templates** while adding them to the list through the **BI Dashboard Web Application**.

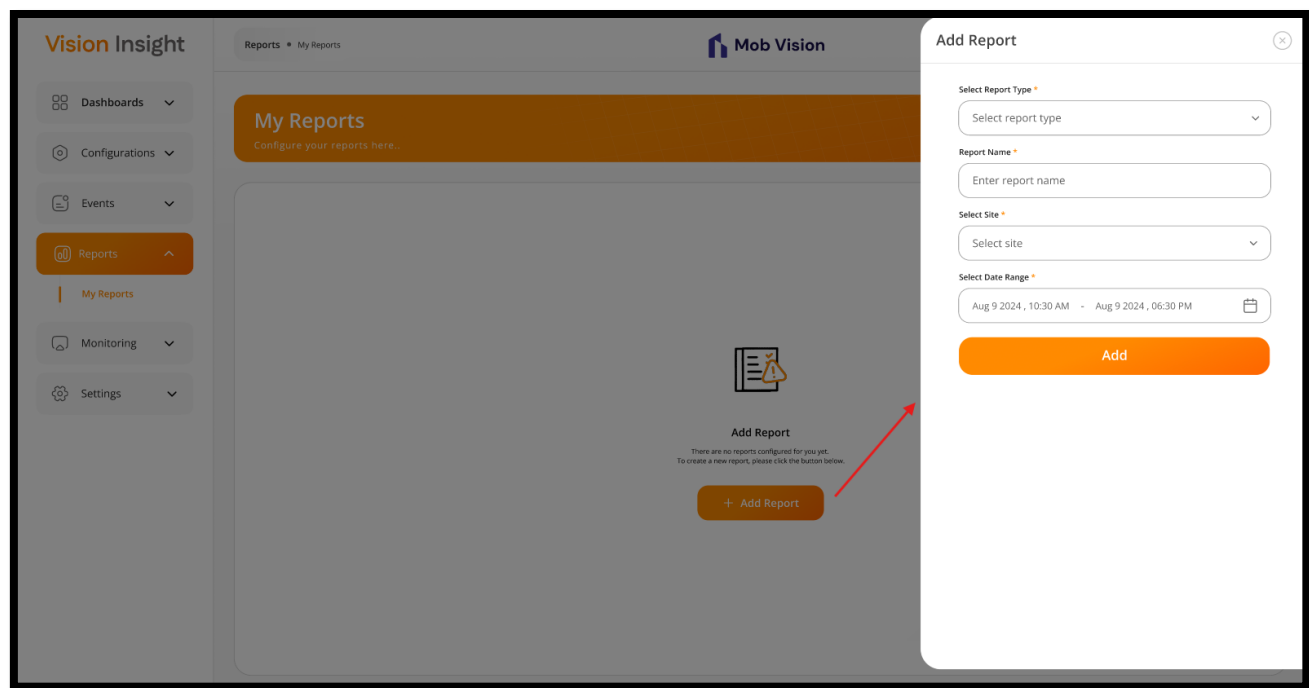
This page will be **accessible only to Super Admin and authorized users**, with actions such as **add, edit, delete**, and **export reports as PDF**, based on the **permissions assigned to the logged-in user**.

Prototype:

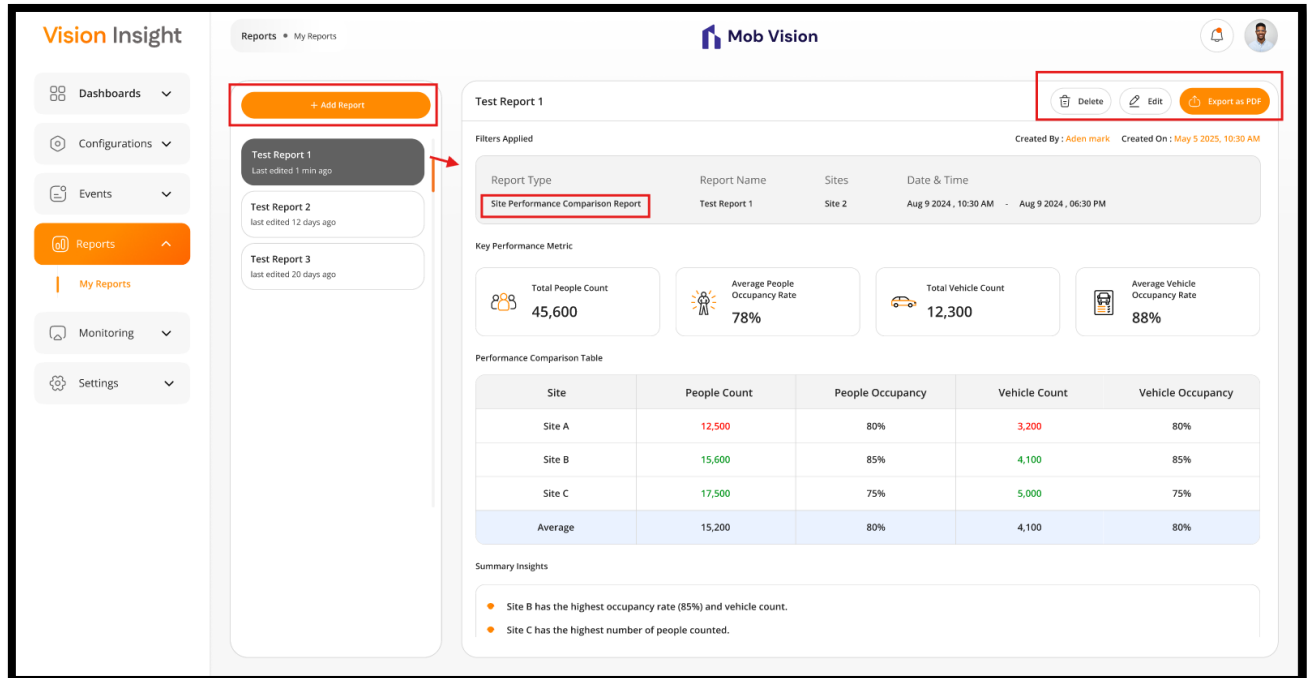
➔ The screen below illustrates **first landing page with appropriate message & action to add new report** when there is **no any report yet configured in the system**:



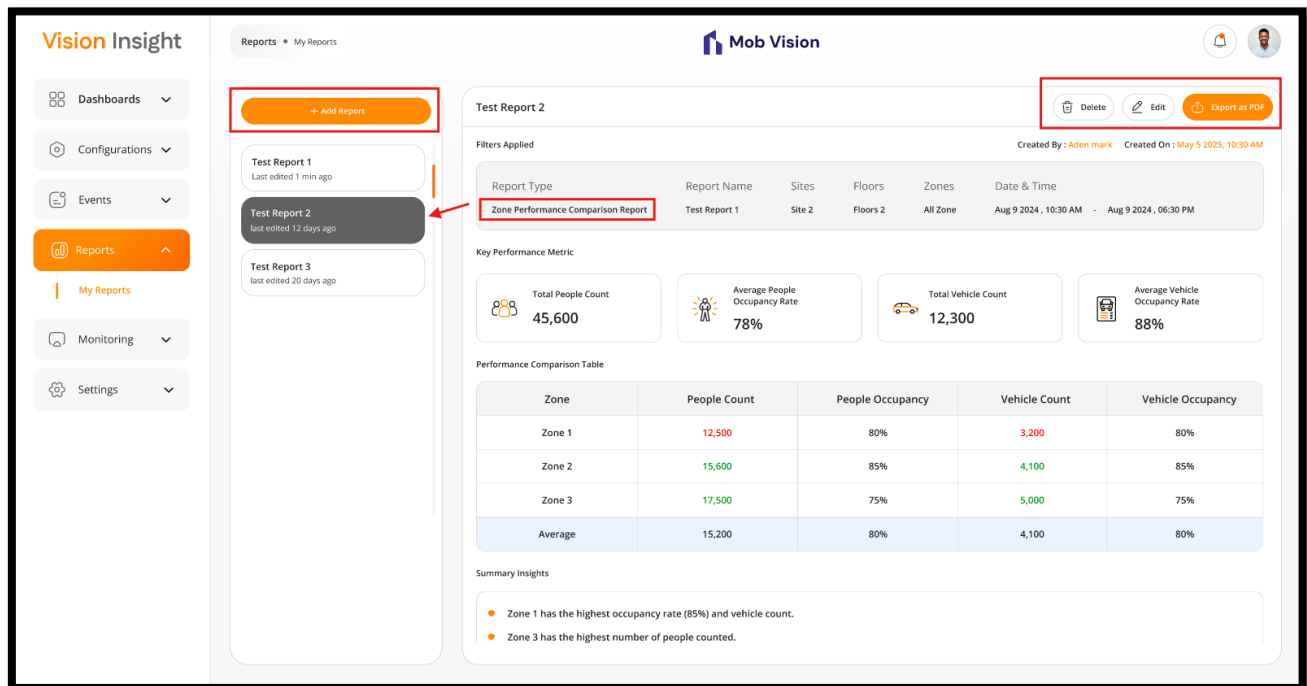
➔ The screen below represents the **report creation form**, allowing users to **enter relevant details** and **submit the information** to create a new report:



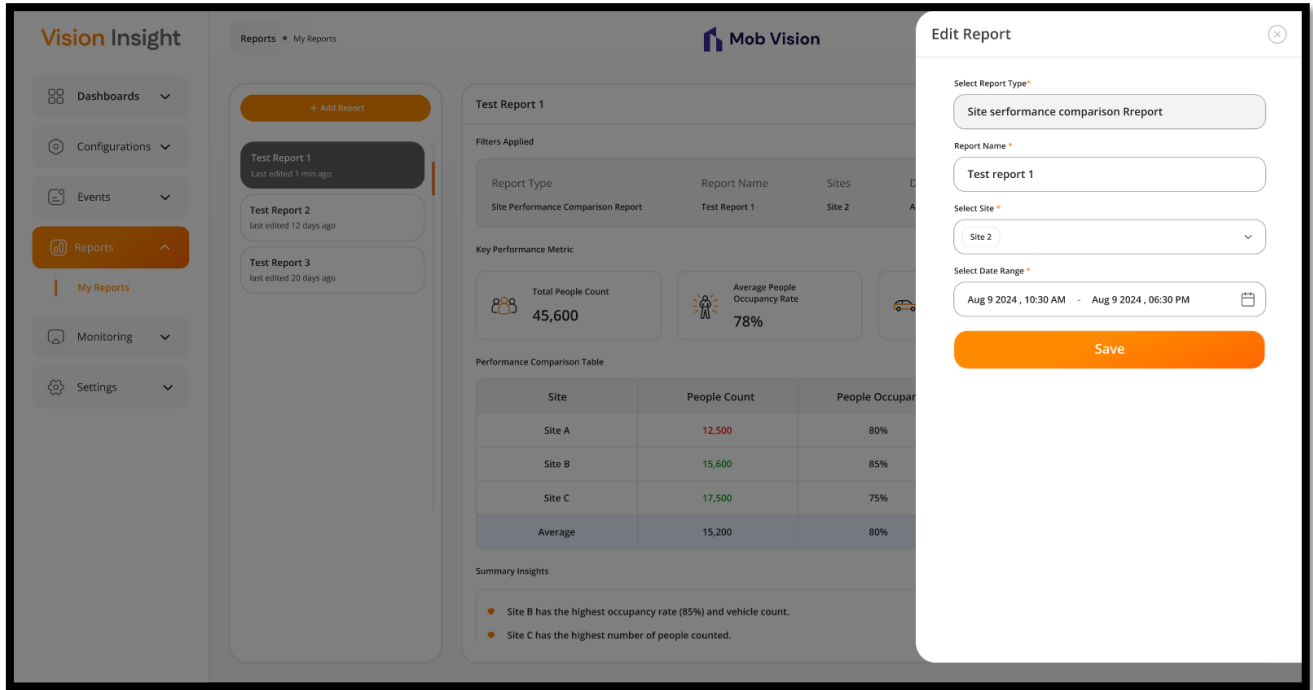
- The screen below displays the **"Site Performance Comparison Report"** page, featuring options to **Edit**, **Delete**, and **Export as PDF**. Users can **scroll through the page** to view the **entire report**, including predefined charts:



- The screen below displays the **"Zone Performance Comparison Report"** page, featuring options to **Edit**, **Delete**, and **Export as PDF**. Users can **scroll through the page** to view the **entire report**, including predefined charts:



→ The screen below displays the **form for editing an existing report**, allowing users to **modify relevant details** and **update the information**:



The screenshot shows the 'Vision Insight' dashboard with a sidebar containing 'Dashboards', 'Configurations', 'Events', 'Reports', 'Monitoring', and 'Settings'. The 'Reports' section is active, showing a list of reports: 'Test Report 1' (last edited 1 min ago), 'Test Report 2' (last edited 12 days ago), and 'Test Report 3' (last edited 20 days ago). The 'Test Report 1' is selected, and the 'Edit Report' modal is open. The modal contains the following fields:

- Select Report Type* (Site performance comparison Rreport)
- Report Name* (Test report 1)
- Select Site* (Site 2)
- Select Date Range* (Aug 9 2024 , 10:30 AM - Aug 9 2024 , 06:30 PM)
- Save button

The background dashboard shows a 'Mob Vision' header and a 'Test Report 1' section with filters applied. It displays key performance metrics: Total People Count (45,600) and Average People Occupancy Rate (78%). Below this is a 'Performance Comparison Table' with columns for Site, People Count, and People Occupancy Rate.

Site	People Count	People Occupancy Rate
Site A	12,500	80%
Site B	15,600	85%
Site C	17,500	75%
Average	15,200	80%

Summary Insights:

- Site B has the highest occupancy rate (85%) and vehicle count.
- Site C has the highest number of people counted.

Business Rules:

- **Authorized users** will be able to **add**, **edit**, and **delete** specific configured reports from the list.
- The system will display a **blank screen with the following message** when no reports are configured yet:
"There are no reports configured for you yet. To create a new report, please click the button below"
- Users will have the option to **add a new report** by clicking on the **"Add Report"** button on the above screen.
- The authorized user can **configure a new "Site Performance Comparison Report"** by providing the following required information:
 - Select Report Type* (e.g., Site Performance Comparison)
 - Report Name* (custom input by user)
 - Select Site* (sites to be compared)
 - Select Date Range*
- The authorized user can **configure a new "Zone Performance Comparison Report"** by providing the following required information:
 - Select Report Type* (e.g., Zone Performance Comparison Report)
 - Report Name* (custom input by user)
 - Select Site* (sites to be compared, single selection)
 - Select Floor*
 - Select Zone*
 - Select Date Range*
- The asterisk (*) indicates a mandatory field only.
- **Authorized users** will be able to **view all configured reports** in the list.
- Each report will allow users to **Edit**, **Delete**, and **Export as PDF**.

- When an authorized user attempts to **delete** a report, the system will **prompt for confirmation**:
 - ➔ Upon confirmation, the system will perform a **soft delete** (the record remains in the database but is hidden in the application).
- When an authorized user selects **“Export as PDF”**, the system will **download the report in PDF format**.
- The **“Site Performance Comparison Report”** will contain the following information:
 - ➔ Title (user-defined report name)
 - ➔ Created By & Created On
 - ➔ Filters Applied:
 - Report Type
 - Report Name
 - Sites
 - Date & Time
 - ➔ Key Performance Matric
 - Total People Count
 - Average People Occupancy Rate
 - Total Vehicle Count
 - Average Vehicle Occupancy Rate
 - ➔ Performance Comparison Table
 - Site (name of the sites)
 - People Count
 - People Occupancy
 - Vehicle Count
 - Vehicle Occupancy
 - Average of “People Count, People Occupancy, Vehicle Count, Vehicle Occupancy”
 - ➔ Summary Insights
 - **Bullet-point summary** to provide clear comparison highlights.
 - ➔ Visualizations
 - It will display following Charts:
 - People & Vehicle Counts by Site
 - Occupancy Rate Comparison
 - Traffic Composition (Peoples vs Vehicles)
- The **“Zone Performance Comparison Report”** will contain the following information:
 - ➔ Title (user-defined report name)
 - ➔ Created By & Created On
 - ➔ Filters Applied
 - Report Type
 - Report Name
 - Sites
 - Floors
 - Zones
 - Date & Time
 - ➔ Key Performance Matric
 - Total People Count
 - Average People Occupancy Rate
 - Total Vehicle Count
 - Average Vehicle Occupancy Rate

- ➔ Performance Comparison Table
 - Zone (name of the zones)
 - People Count
 - People Occupancy
 - Vehicle Count
 - Vehicle Occupancy
 - Average of “People Count, People Occupancy, Vehicle Count, Vehicle Occupancy”
- ➔ Summary Insights
 - **Bullet-point summary** to provide clear comparison highlights.
- ➔ Visualizations
 - It will display following Charts:
 - People & Vehicle Counts by Site
 - Occupancy Rate Comparison
 - Traffic Composition (Peoples vs Vehicles)
- **Authorized users can add any number of reports** using the same report type.
- The system will **prevent duplicate entries** using the same report name.

Data Dictionary:

reports					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	reportType	String	No	Site comparison report	type of the report
3	siteComparisionReport	Object	Yes	{}	Configuration for site comparison report
3.1	> reportName	String	No	my sites compare report	name of the report
3.2	> sitesIds	String[]	No	[]	sites configured for the report
3.3	> startDate	DateTime	No	23-03-2025	Start date of the report
3.4	> endDate	DateTime	No	23-03-2025	End date of the report
4	zoneComparisionReport	object	Yes	{}	Configuration of the zone comparison report
4.1	> reportName	String	No	my zone compare report	Name of the report
4.2	> siteId	Object	No	674daaf7d34a2b901c847454	sites configured for the report
4.3	> floorIds	Object[]	No	[674daaf7d34a2b901c847454]	floor configured for the report
4.4	> zoneIds	Object[]	No	[674daaf7d34a2b901c847454]	Zones configured for the report
4.5	> startDate	DateTime	No	23-03-2025	Start date of the report
4.6	> endDate	DateTime	No	23-03-2025	End date of the report
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.17.Monitoring > Monitoring Page

The system provides a feature for users to **manage and configure one or multiple monitoring setups** through the **BI Dashboard Web Application**.

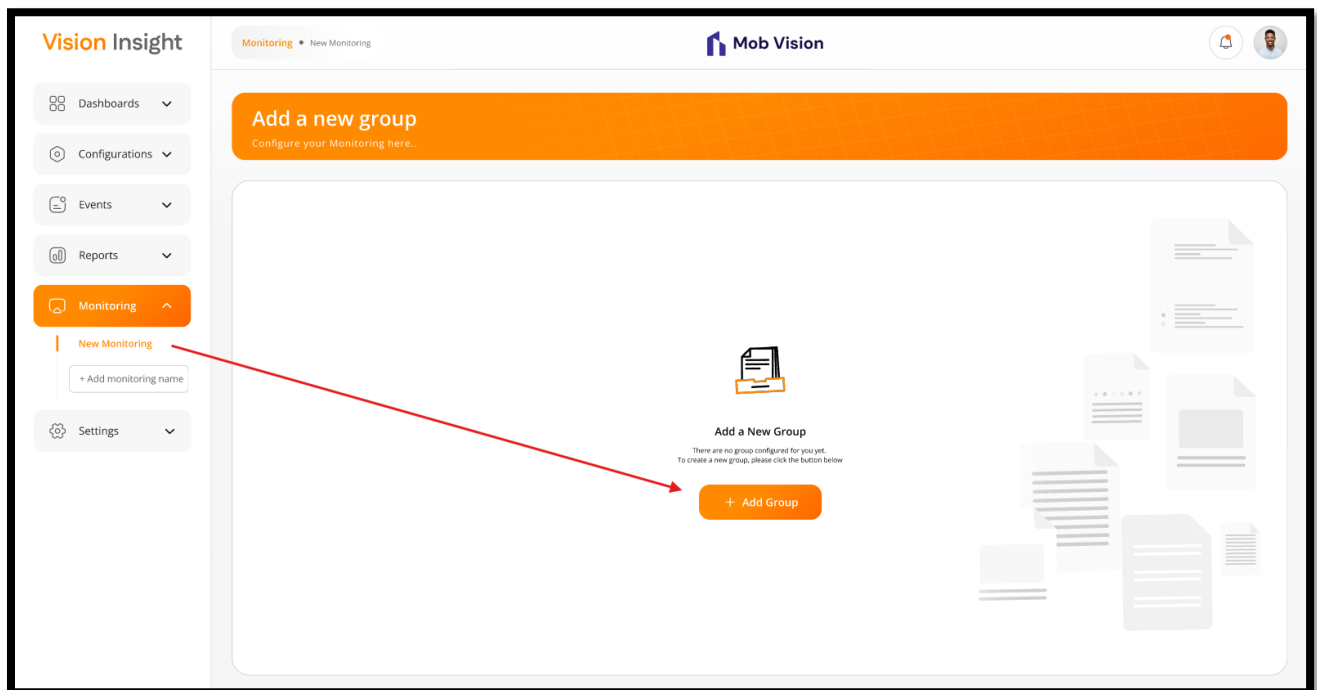
This page will be **accessible only to Super Admin and authorized users**, with actions such as **add, update, and delete** for:

- Monitoring
- Groups
- URL Previews (Poles)

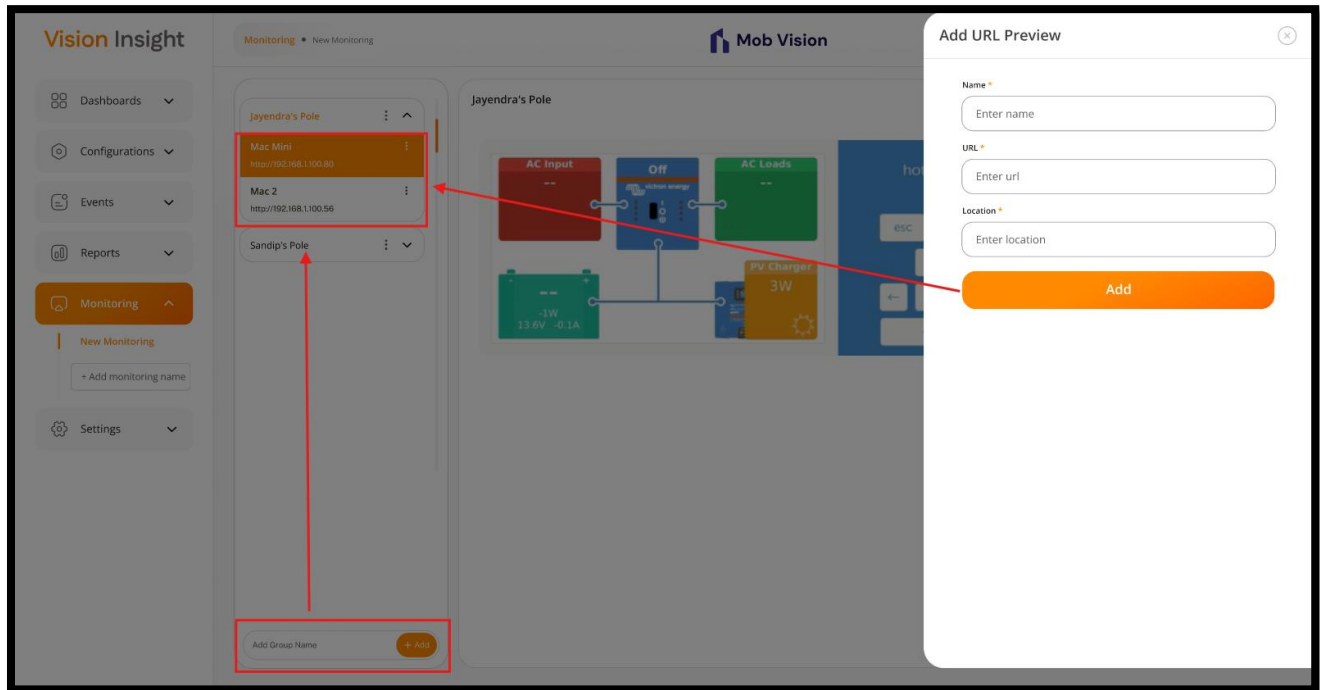
Authorized users can configure **any number of monitoring entries** based on their operational needs and the **permissions assigned** to their user account.

Prototype:

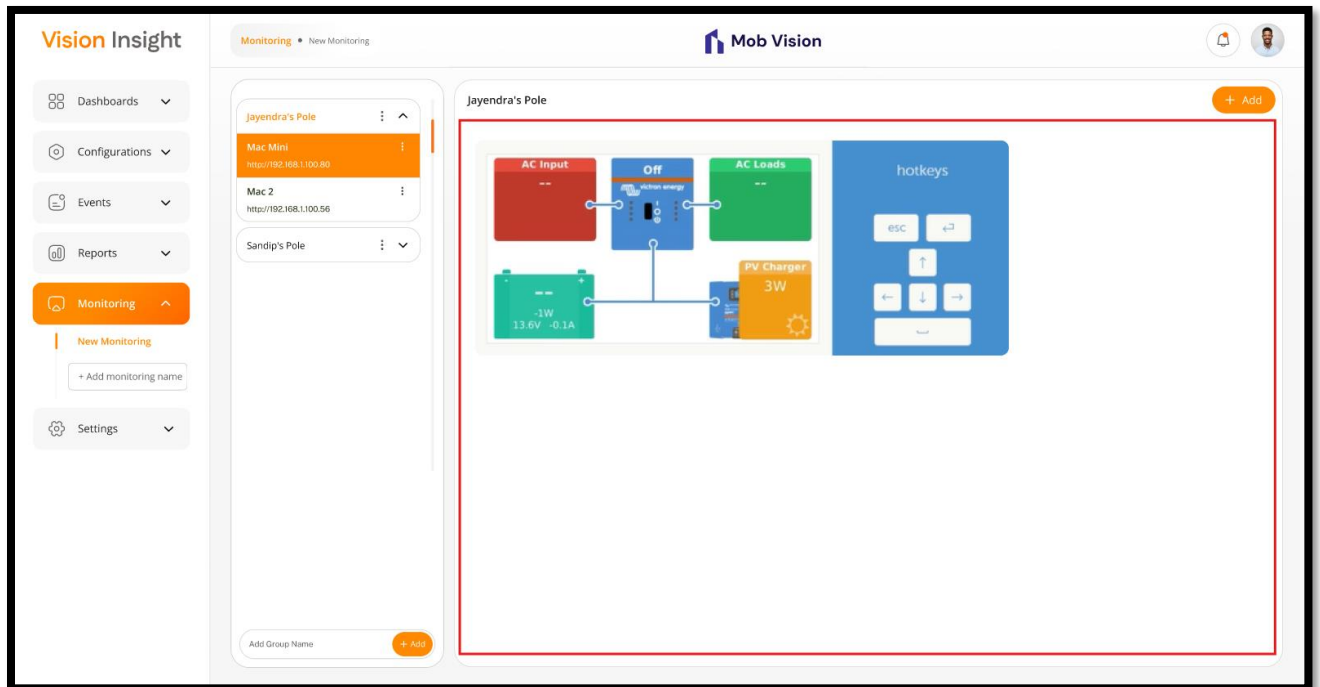
➔ The screen below illustrates **first landing page with appropriate message & action to add new group for the specific monitoring** when there is **no any group yet configured in the specific monitoring**:



→ The screen below illustrates the **list of created groups** along with their **URL previews (Poles)**, including options to **Add, Edit, and Delete** them:



→ The added URL preview will be displayed only within the highlighted area shown below:



Business Rules:

- The system will display a **blank screen with the following message** when no groups are configured yet:
"There are no group configured for you yet. To create a new group, please click the button below"
- Users will have the option to **add a new group** by clicking on the **"Add Group"** button on the above screen.
- Authorized users** will be able to **add, edit, and delete** any number of **monitoring** from the **main side menu** of the application by providing the **name of the monitoring**.
- Inside the above-added monitoring, users will be able to **add, edit, and delete** any number of **groups** by providing the **name of the group** only.
- Under each group, users will be able to **add, edit, and delete** any number of **URL Preview (Pole)** by providing the following fields:
 - Name of the URL preview*
 - URL* (the URL of the web page, which will get embedded into an iframe)
 - Location*
- The asterisk (*) indicates a mandatory field only.
- The system will **prevent duplicate entries** for the **"URL Preview"** using the same **Name & URL** combination.
- The provided URL while adding or updating **"URL Preview (Pole)"** will be directly **embedded into an iframe**, and whatever **user interface** is available at the added URL will be **displayed on the screen** directly.
- The **added/provided URL** will always be **accessible** from the **BI Dashboard Web Application**. If it is not accessible, the system **will not handle** such scenarios.
- The system will **prevent duplicate entries** using the same **Monitoring and Group name**.

Data Dictionary:

monitoring					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	monitoringName	String	Yes	Soudi area	Name of the monitoring
3	MonitoringGroup	Object[]	Yes	[{}]	Collection of the monitor groups
3.1	> GroupId	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated group id
3.2	> GroupName	String	No	West zone area	Name of the group
3.3	> GroupItem	Object[]	No	[{}]	Collection of group items
3.3.1	>> GroupItemId	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated group item id
3.3.2	>> Name	String	No	Pole1	Name of the monitor items
3.3.3	>> Url	String	No	http://1.1.1.1	Url of the monitor item
3.3.4	>> Location	String	Yes	Pole1	Location of the monitor item
4	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
5	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
6	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
7	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.18.Dashboards > Dashboard Creation

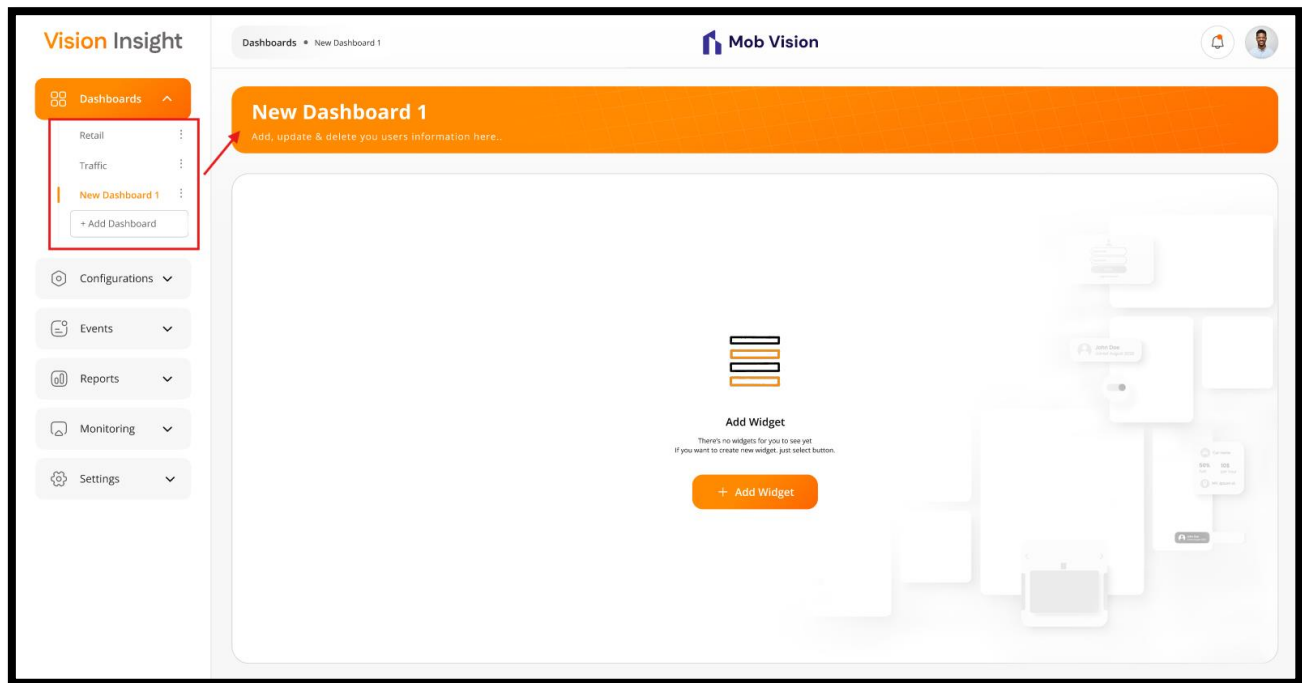
The system provides a feature for **authorized users** to **manage and configure one or multiple dashboards** through the **BI Dashboard Web Application**.

This page will be **accessible only to authorized users**, with actions such as **add, update, and delete** dashboards, based on the **permissions assigned on the widget** to the logged-in user.

Authorized users can **configure any number of dashboards** based on their requirements and the permissions assigned to them on the widgets.

Prototype:

➔ The screen below illustrates **first landing page with appropriate message & action to add new widget for the specific dashboard** when there is **no any widget yet configured in the specific dashboard**:



Business Rules:

- **Authorized users will be able to add, update, and delete any number of dashboards by providing name of the dashboard.**
- The system will **prevent duplicate entries** for the added dashboards using the **same Name**.
- The system will **allow users to enter characters only** in the **dashboard name field**. It is a **mandatory field** while **adding or updating** the dashboard name.

Data Dictionary:

dashboardPreference					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	userId	String	No	674daaf7d34a2b901c847454	Dashboard is mapped with this user
3	dashboardPreferenceDesign	String	Yes	[[]]	Collection if widgets for this dashboard
4	dashboardName	String	Yes	Retail	General description of the email template
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

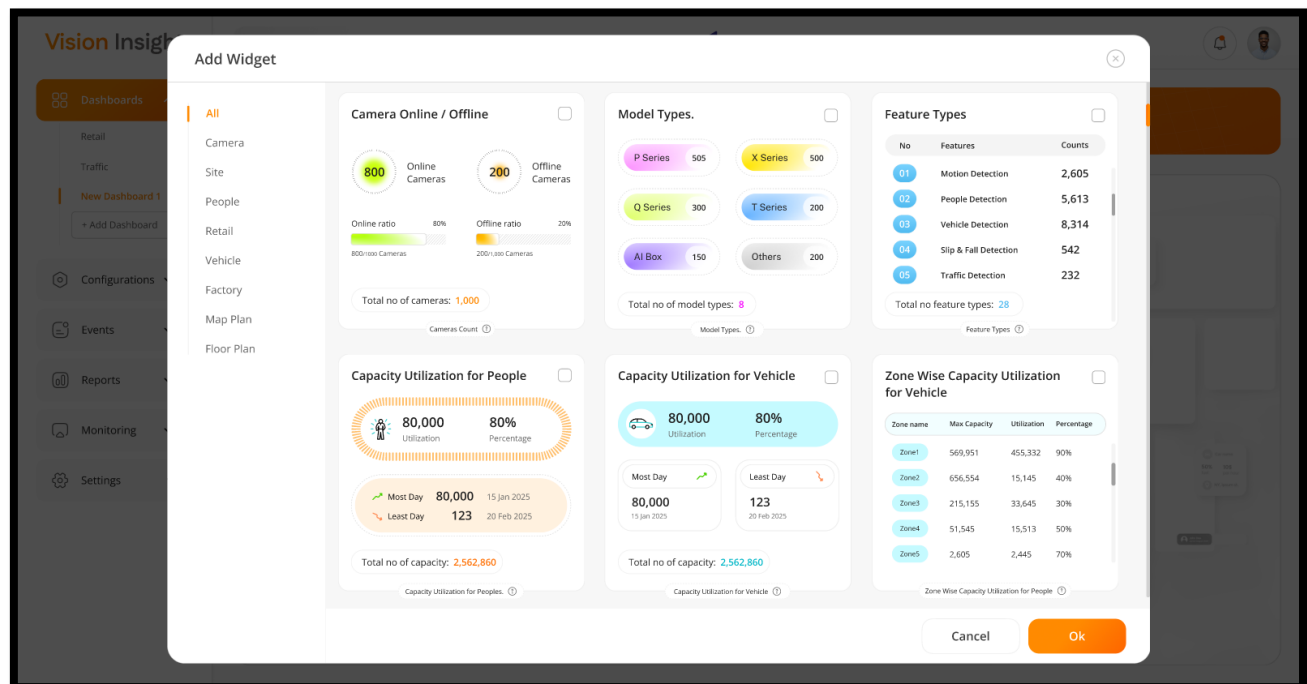
3.18.1. Dashboards > Dashboard Page > Add Widget Screen

The system provides a feature for **authorized users to manage and add one or multiple widgets under the specific dashboard** through the **BI Dashboard Web Application**.

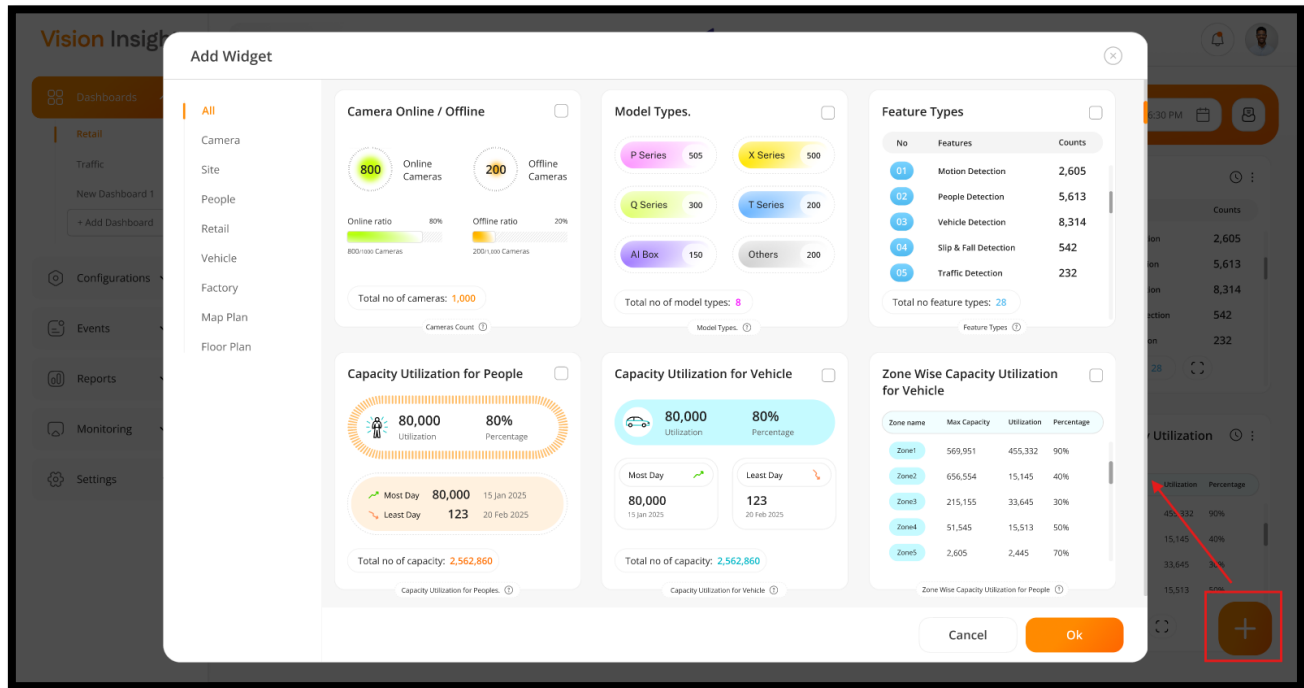
Authorized users can **select & add any number of widgets from the “Add Widget” screen** based on their requirements and the permissions assigned to them.

Prototype:

➔ The screen below illustrates the **flow of adding new widget(s) to a specific dashboard**. Users can **select any category** and choose **any number of widgets to add**:



→ The screen below illustrates the **second option** to add a new widget(s) to a specific dashboard:



Business Rules:

- **Authorized users** will be able to **add any number of widgets** from the **“Add Widget”** popup.
- The system will **prevent duplicate entries** for the added widget under any **specific dashboard**. It means, users will be able to **select & add any widgets** at a single time under the **specific dashboard only**, because the **added widget will not appear** under the **“Add Widget”** screen again for that dashboard.
- The system will allow users to **add the same widget(s)** under **other dashboards**.
- The system will provide this option at **two places**.
 - From the **blank screen** when there are **no widgets configured** yet.
 - From the **“+”** button at the **bottom-right corner**, when **any widget is already configured** on the specific dashboard.
- When there are **no widgets configured/added** on the dashboard, the system will display a **blank screen** with the following message:

“There are no widgets for you to see yet. If you want to create new widget, just select button”
- When user clicks on **“Add Widget”** button, the system will open a **popup**, and users will be able to see the **list of different categories** as a **left side menu**.
- Under this **left side menu**, users will be able to **click on any category**, and each category will **display their associated widgets** on the **right-hand side**.
- By default, **“All”** category will be selected under the left side category menu, and it will display **all widgets available in the system**, regardless of their category.
- The following **categories & their associated widgets** will be displayed:
 - All
 - All widgets will be displayed under this category, regardless of their actual category.
 - Camera

- Camera Online/ Offline
- Model Types
- Feature Types
- ➔ Site
 - Capacity Utilization for People
 - Capacity Utilization for Vehicle
 - Zone Wise Capacity Utilization for Vehicle
 - Zone Wise Capacity Utilization for People
- ➔ People
 - Slip & Fall Detection
 - People In & Out
 - Average People Counting
 - People Count by Gender
 - Cumulative People Count
 - New Vs Total Visitors
 - Safety Measures
 - Zone Wise People Counting
- ➔ Retail
 - Shopping Cart Counting
 - Queue Events for Shopping Cart
 - Queue Events for People
 - Shopping Cart Heatmap
 - Block Exit Detection
- ➔ Vehicle
 - Vehicle Count by Type
 - Vehicle in Wrong Direction
 - Vehicle U Turn Detection
 - Pedestrian Detection
 - Vehicle In & Out
 - Average Vehicle Counting
 - Vehicle Queue Analysis
 - Stopped Vehicle Count Time
 - Vehicle Turning Movement Counts
 - Vehicle Detection Heatmap
 - Speed Violation by Vehicle
 - Traffic Jam by Day
- ➔ Factory
 - Counting for Forklift
 - Queue Events for Forklift
 - Detected Forklifts
 - Forklift Heatmap
 - Block Exit Detection
 - Forklift Speed Detection
- ➔ Map Plan
 - Map Plan
- ➔ Floor Plan
 - Floor Plan

- The system will allow **authorized users** to **select/unselect any number of widgets** from this popup and under any category.
- All the **selected widgets** will be displayed on the **specific dashboard** from where the user tried to add new widgets.
- The system will provide a **“Delete” option** under each widget configured on the specific dashboard.
- Once user clicks on the **“Delete” option**, the system will display a **confirmation popup**, and once confirmed, that widget will be **permanently deleted** from the specific dashboard.

Data Dictionary:

→ The system will utilize below highlighted column in the **dashboardPreference** table:

dashboardPreference					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	userId	String	No	674daaf7d34a2b901c847454	Dashboard is mapped with this user
3	dashboardPreferenceDesign	String	Yes	{}	Collection if widgets for this dashboard
4	dashboardName	String	Yes	Retail	General description of the email template
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

3.18.2. Dashboards > Dashboard Page > Filters & Export Option

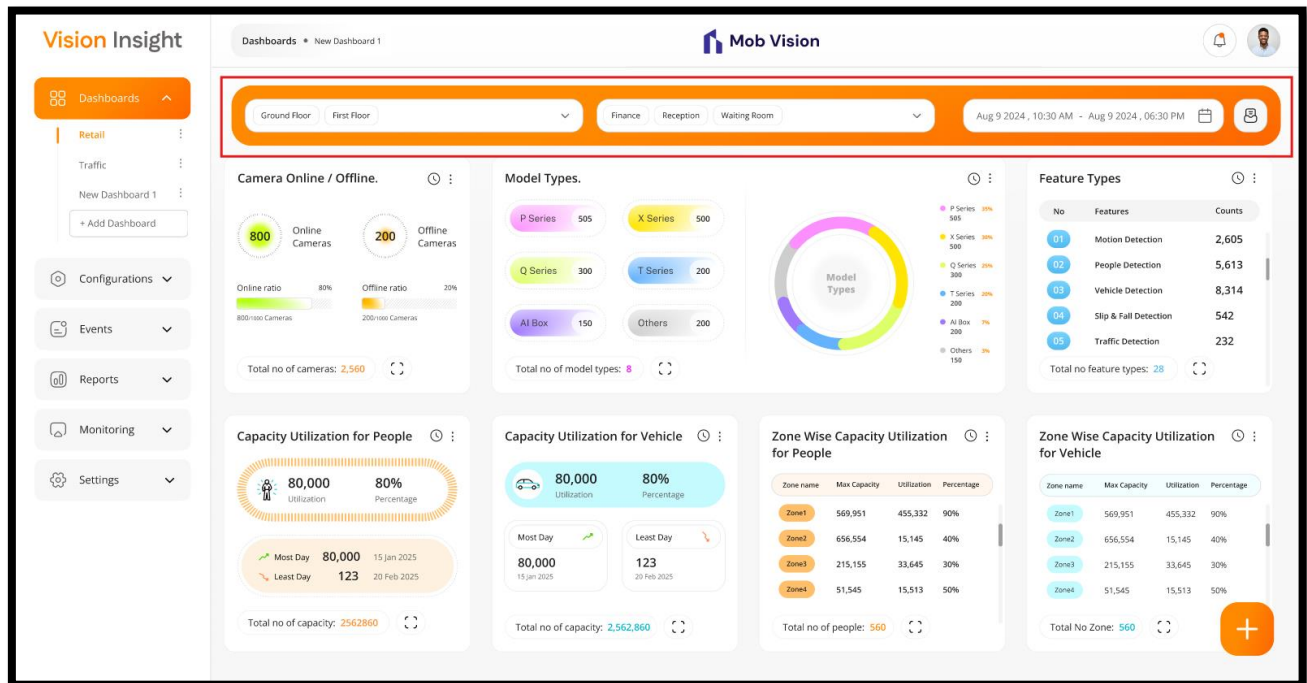
The system provides a feature for **authorized users** to **manage or apply the Floors, Zones, and Date & Time Range filters** under the **specific dashboard** through the **BI Dashboard Web Application**.

The system will **retrieve and display data** in **all the configured widgets** based on the **applied filters only**.

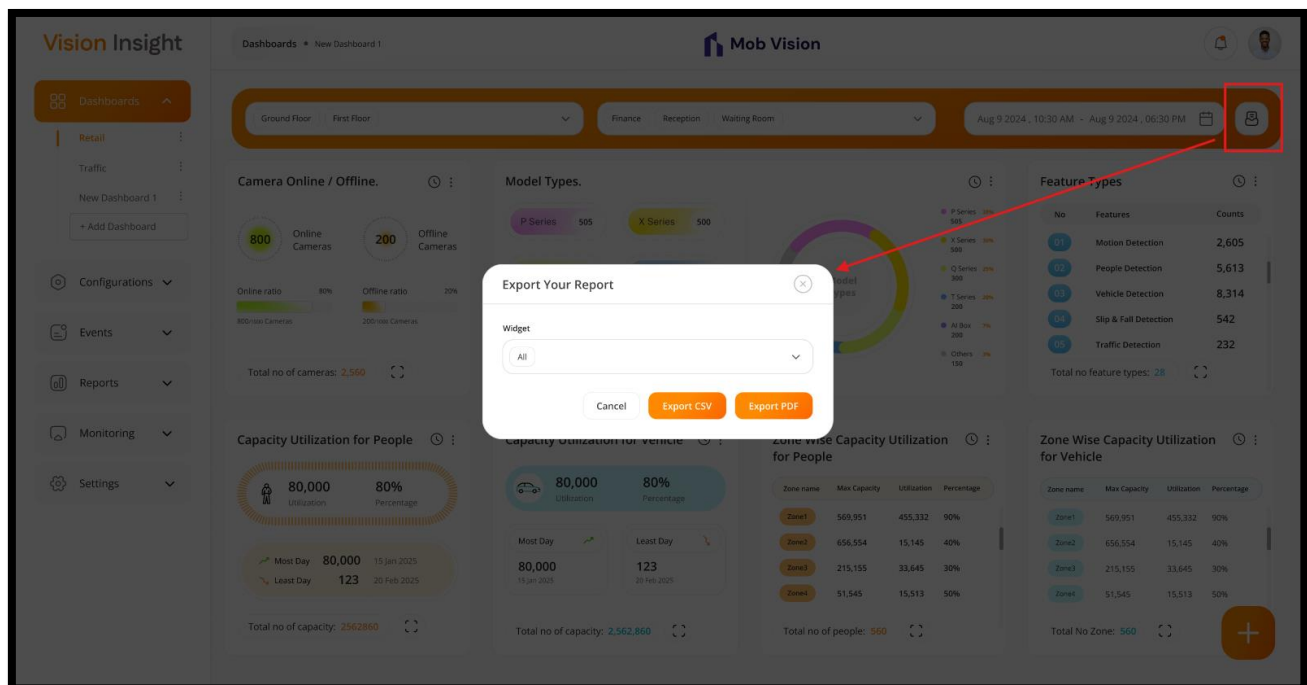
The system will also provide an option to **export your report** in the **.pdf** and **.csv format**.

Prototype:

→ The screen below illustrates the **options for Floors, Zones, Date & Time range filters**, along with the ability to **export your report**. These options become **available on each newly created dashboard** and will be **visible once the user adds at least one widget** to the specific dashboard:



➔ The screen below illustrates the **export report** option, allowing users to **select a specific widget** configured on the dashboard or choose **"All"** to export a report for **all configured widgets**. Reports can be exported in **PDF** or **CSV** format:



Business Rules:

- The system will provide **following filter options** on each dashboard once the users will add any single widget on the dashboard:
 - ➔ Floors
 - **Multi select dropdown list**, The users can select **one or more floors** from this dropdown list.
 - **By default**, default floor will be selected.
 - ➔ Zones
 - **Multi select dropdown list**, The users can select **one or more floors** from this dropdown list.
 - ➔ Date & Time Range
 - Allow users to select **any range** from date & time picker, along with following options:
 - Today, Yesterday, Two day ago, 7 Days, 1 Month, 3 Months, 6 Months, 1 Year.
- **Authorized users** will be able to **apply any of the above filters** & the system will **render the data accordingly** for all the **configured / added widgets** on that dashboard.
- The system will select **following filters by default** when user has not applied any filters, reloaded the page or redirected to the page:
 - ➔ **Default Floor** will be selected
 - ➔ **Default Zone** will be selected
 - ➔ **Today's date & time** will be selected
- The system will **not store applied filters in the database**. It will store those **applied filters in the local storage** of the browser page.
- The data for the **local storage will be removed** once the user **reload the page manually or close the browser window**

Export Your Report Option:

- The system will allow **authorized users** to access this option.
- Once user clicks on the **export icon** located at **top right-hand side** of the specific dashboard, the system will open **popup screen**.
- The above-mentioned popup screen will provide following options:
 - ➔ **Selection of Widgets** (*which are added on the specific dashboard only, non-mandatory field*)
 - ➔ Export CSV
 - ➔ Export PDF
- The users will be able to **select specific widget**, and the system will **export the report as CSV / PDF** for the **selected widgets only**.
- If users have **not selected any widget** from the dropdown list, the system will **export report as CSV / PDF** for **all the widgets** which are added on the specific dashboard.
- The **CSV** will include following information in the exported report:
 - ➔ Exported By
 - ➔ Exported On (Date & Time)
 - ➔ Selected Date Range & Time
 - ➔ **Data of all the selected widgets** with **whatever filters** they have applied on the specific widget & showcased on the specific dashboard
- The **PDF** will include following information in the exported report:

- ➔ Logo of the Vision Insight / company logo
- ➔ Exported By
- ➔ Exported On (Date & Time)
- ➔ Selected Date Range & Time
- ➔ **Images of all the selected widgets with whatever filters** they have applied on the specific widget & showcased on the specific dashboard.

Data Dictionary:

- ➔ The system utilizes the **floorPlanMaster** and **zoneMaster** tables to display the list of **floors** and **zones** in the filters mentioned above. These represent the **user-defined floor plans and zones** created manually within the system.

3.18.3. Dashboards > Dashboard Page > Widgets

The system provides a feature for **authorized users** to **manage up to 39 widgets** on the **specific dashboard** through the **BI Dashboard Web Application**.

The system will **retrieve and display data** in all the configured widgets based on the **applied filters only**.

The system will allow users to **configure only those widgets which are assigned to them** & it will only be **displayed based on the given permission** from the “**Roles & Permissions**” screen.

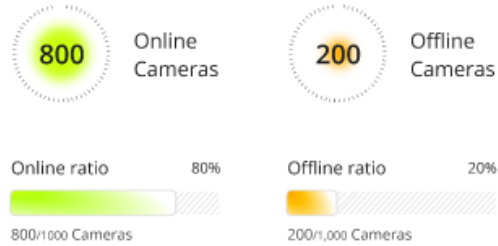
Prototype:

- ➔ The system will provide following 39 widgets:

1. Camera Online/ Offline Widget:

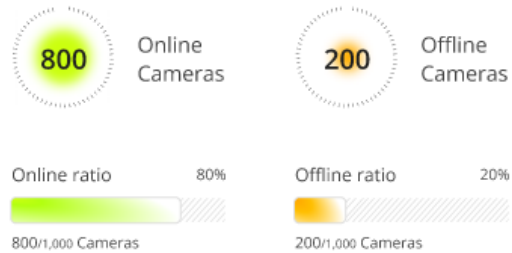
- This report will display counts & their ratio for total camera online & offline.
- It will be generated from the configured camera in the system only.
- It will provide only two charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Camera Online / Offline



Total no of cameras: **1,000**

Camera Online / Offline

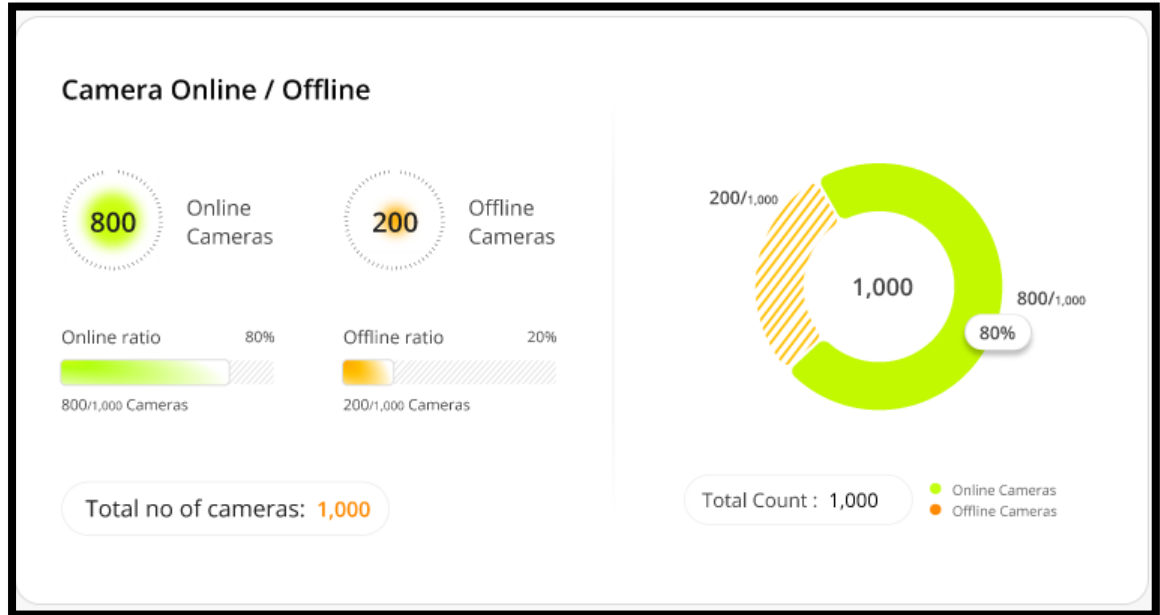


Total no of cameras: **1,000**



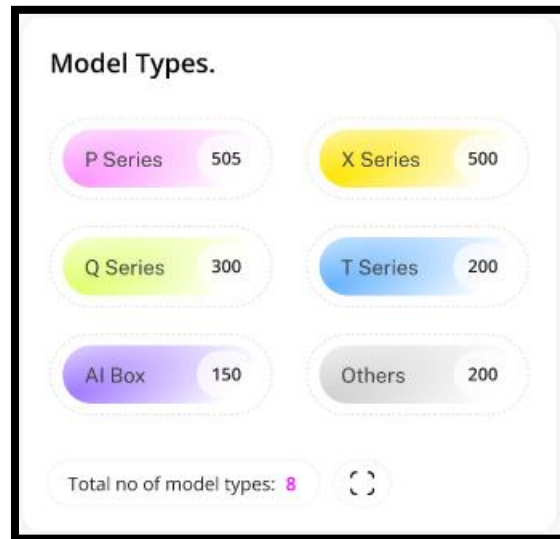
Total Count : **1,000**

● Online Cameras
● Offline Cameras

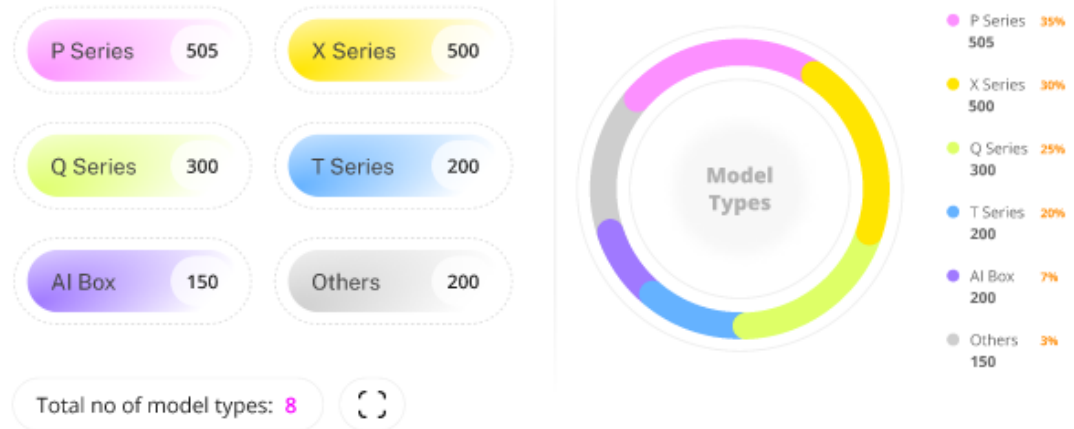


2. Model Types:

- This report will display counts configured cameras in the specific camera series.
- It will be generated from the configured camera in the system only.
- It will provide only two charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Model Types.



Model Types.



3. Feature Types:

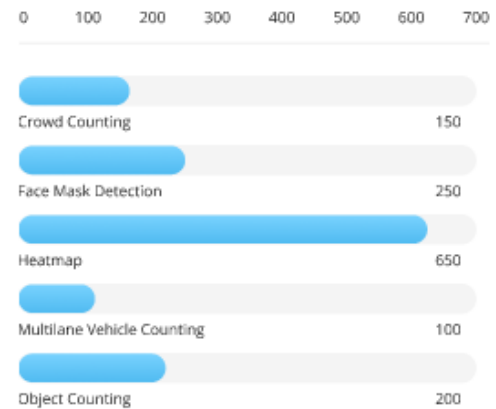
- This report will display counts configured cameras with specific camera features.
- It will be generated from the configured camera in the system only.
- It will provide only two charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

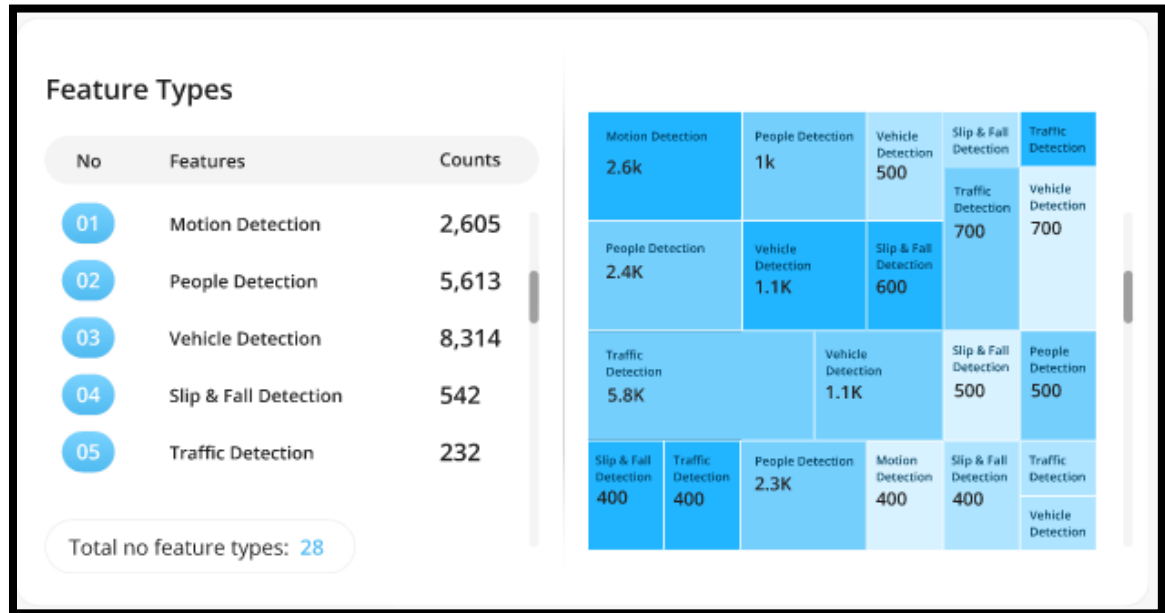
Feature Types

No	Features	Counts
01	Motion Detection	2,605
02	People Detection	5,613
03	Vehicle Detection	8,314
04	Slip & Fall Detection	542
05	Traffic Detection	232
Total no feature types: 28		

Feature Types

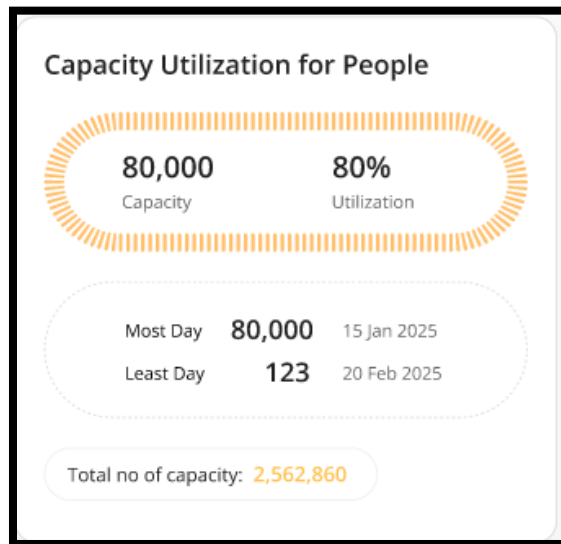
No	Features	Counts
01	Motion Detection	2,605
02	People Detection	5,613
03	Vehicle Detection	8,314
04	Slip & Fall Detection	542
05	Traffic Detection	232
Total no feature types: 28		



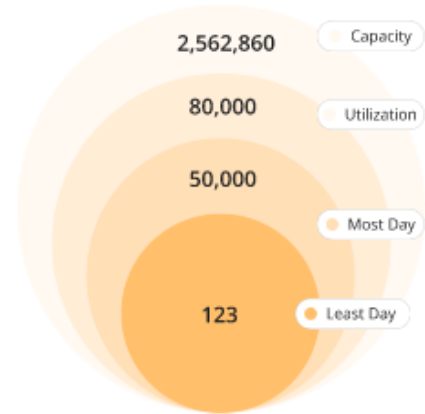
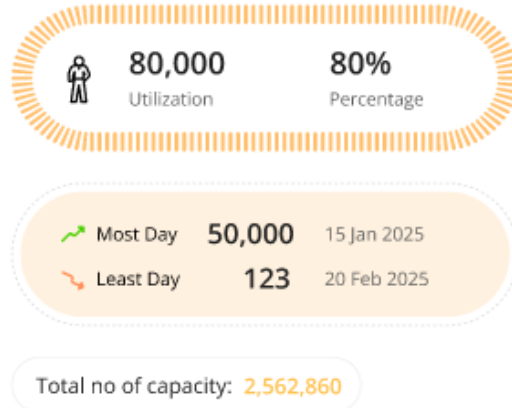


4. Capacity Utilization for People:

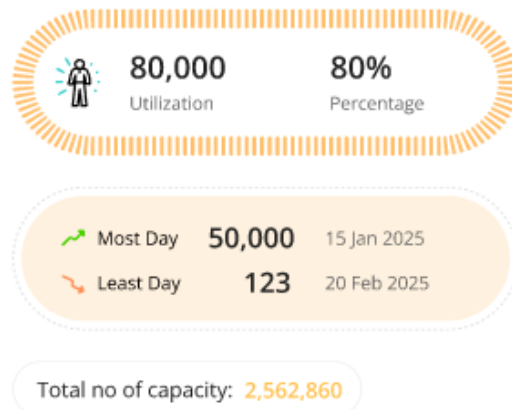
- This report will display the total capacity, percentage of total utilization, as well as the most and least utilized day along with the corresponding people counts.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

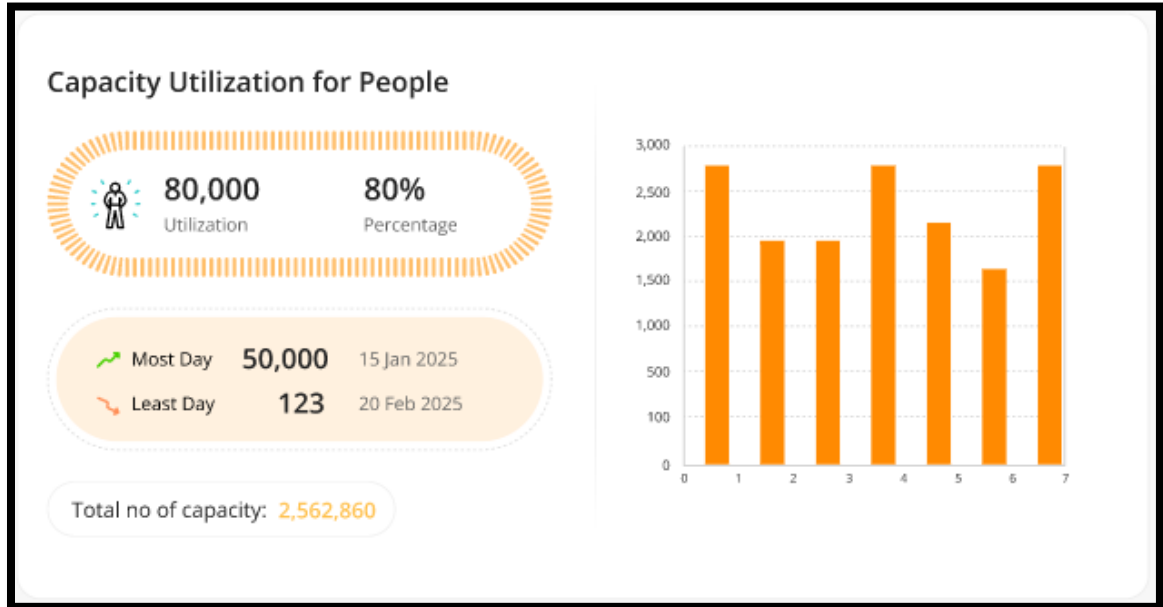


Capacity Utilization for People



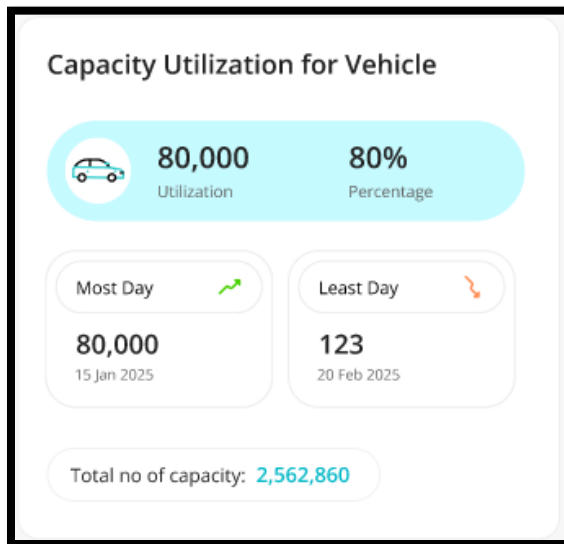
Capacity Utilization for People



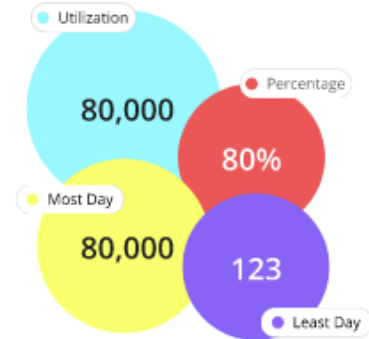
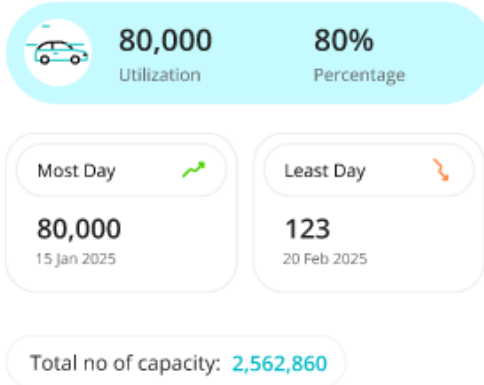


5. Capacity Utilization for Vehicle:

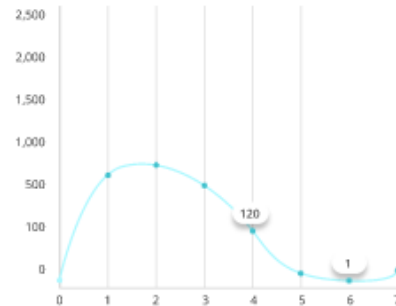
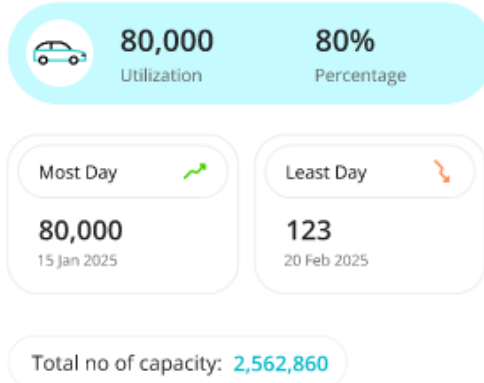
- This report will display the total capacity, percentage of total utilization, as well as the most and least utilized day along with the corresponding vehicle counts.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

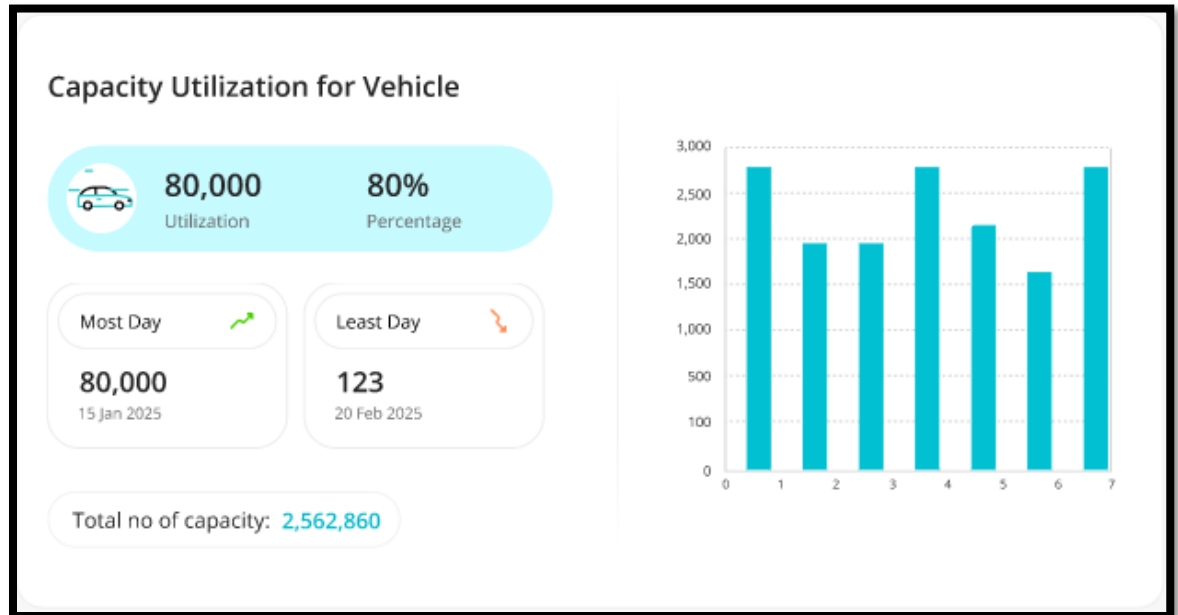


Capacity Utilization for Vehicle



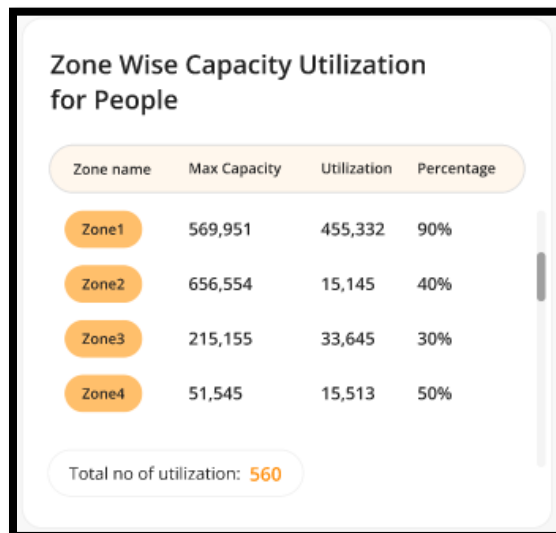
Capacity Utilization for Vehicle





6. Zone Wise Capacity Utilization for People:

- This report will display the zone wise max capacity, utilization counts & their percentages for the peoples.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Zone Wise Capacity Utilization for People

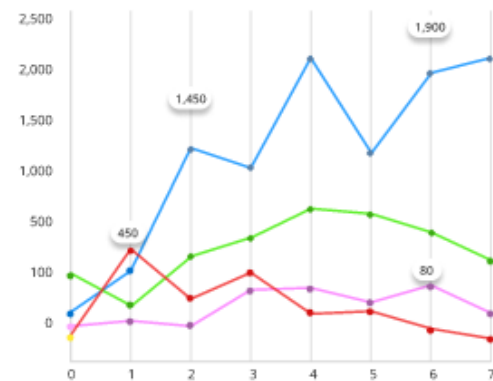
Zone name	Max Capacity	Utilization	Percentage
Zone1	569,951	455,332	90%
Zone2	656,554	15,145	40%
Zone3	215,155	33,645	30%
Zone4	51,545	15,513	50%

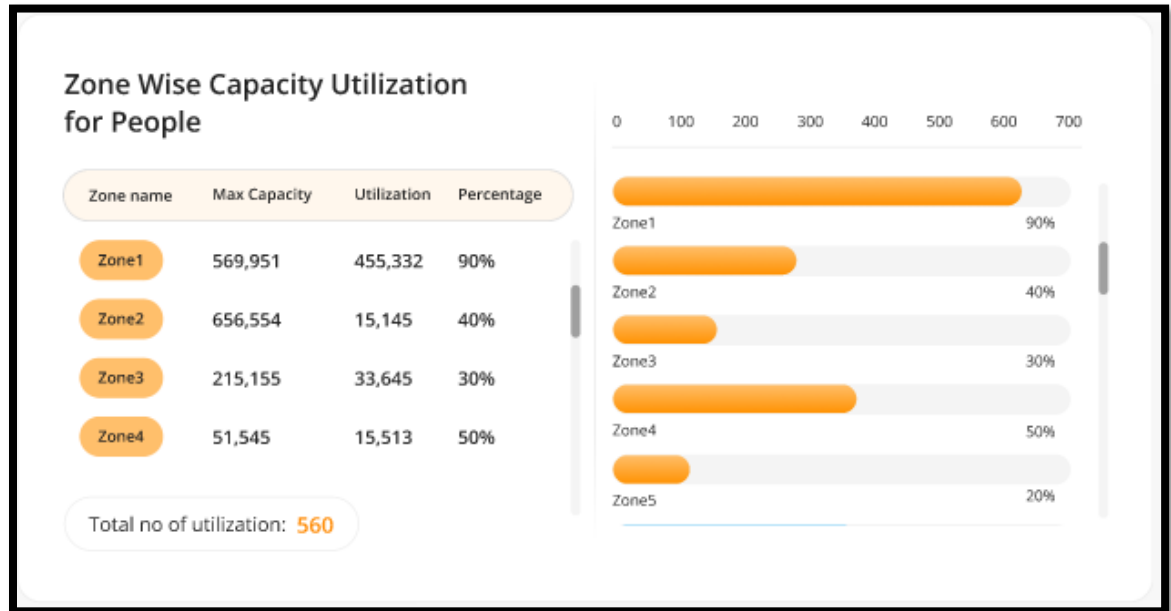
Total no of utilization: 560

Zone Wise Capacity Utilization for People

Zone name	Max Capacity	Utilization	Percentage
Zone1	569,951	455,332	90%
Zone2	656,554	15,145	40%
Zone3	215,155	33,645	30%
Zone4	51,545	15,513	50%

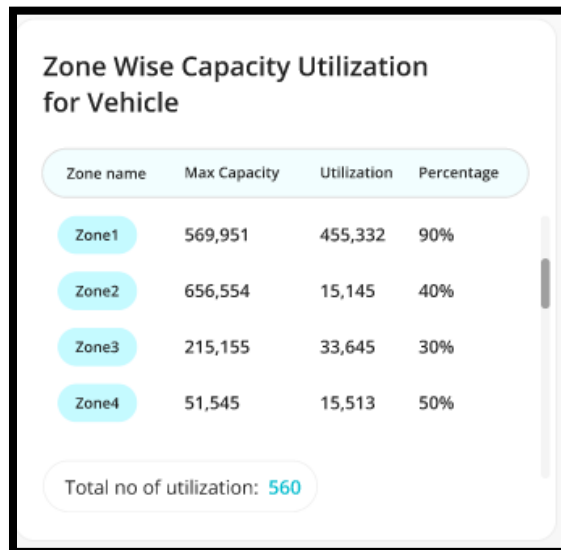
Total no of utilization: 560





7. Zone Wise Capacity Utilization for Vehicle:

- This report will display the zone wise max capacity, utilization counts & their percentages for the vehicles.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Zone Wise Capacity Utilization for Vehicle

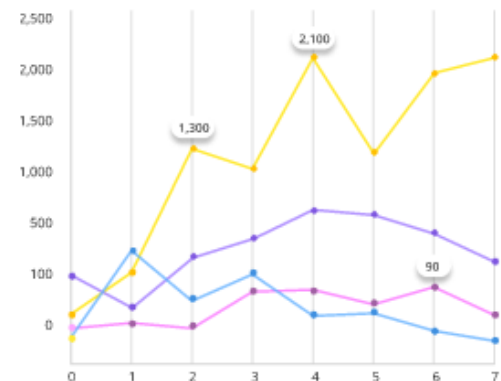
Zone name	Max Capacity	Utilization	Percentage
Zone1	569,951	455,332	90%
Zone2	656,554	15,145	40%
Zone3	215,155	33,645	30%
Zone4	51,545	15,513	50%

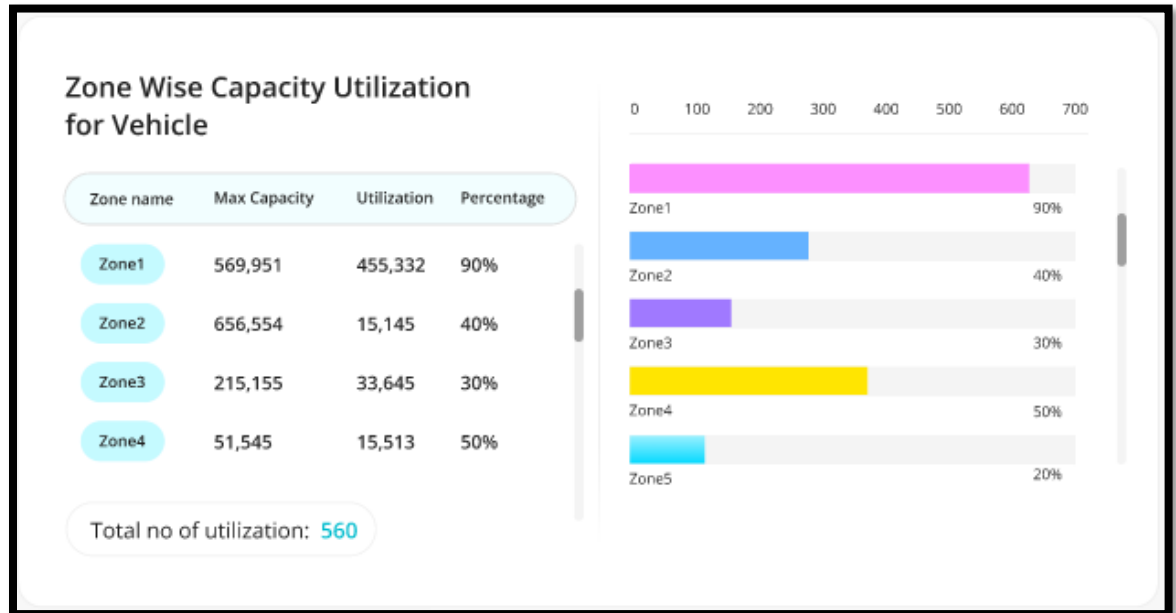
Total no of utilization: 560

Zone Wise Capacity Utilization for Vehicle

Zone name	Max Capacity	Utilization	Percentage
Zone1	569,951	455,332	90%
Zone2	656,554	15,145	40%
Zone3	215,155	33,645	30%
Zone4	51,545	15,513	50%

Total no of utilization: 560

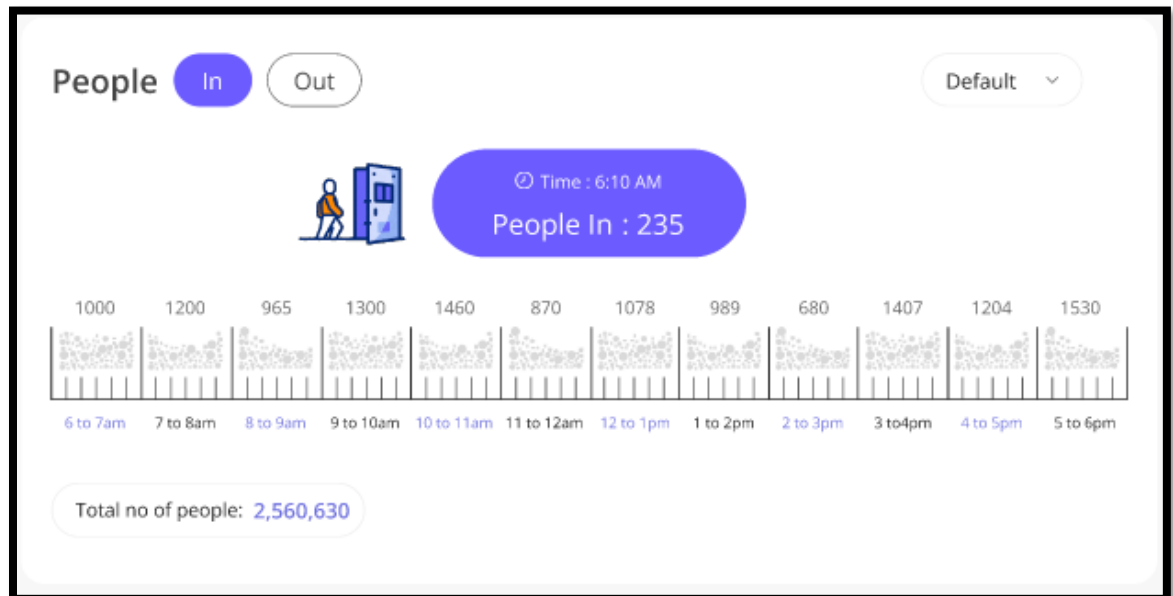




8. People In & Out:

- This report will display counts of total people IN & OUT.
- The setup page will provide the ability to select the specific camera & system will reflect the data accordingly on the dashboard.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:





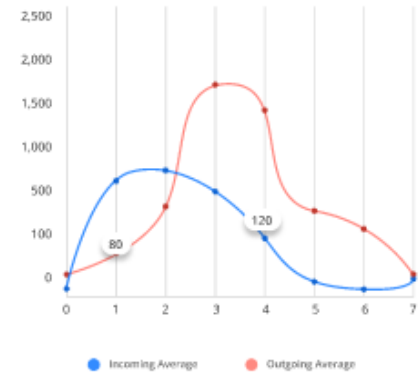


9. Average People Counting:

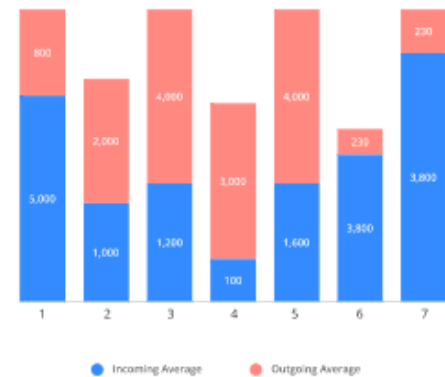
- This report will display counts of total average people incoming & outgoing with their Most & Least day and counts.
- It will be generated from the configured camera in the system only.
- It will provide only two charts with 2*1 option & 1 chart with 3*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:



Average People Counting



Average People Counting

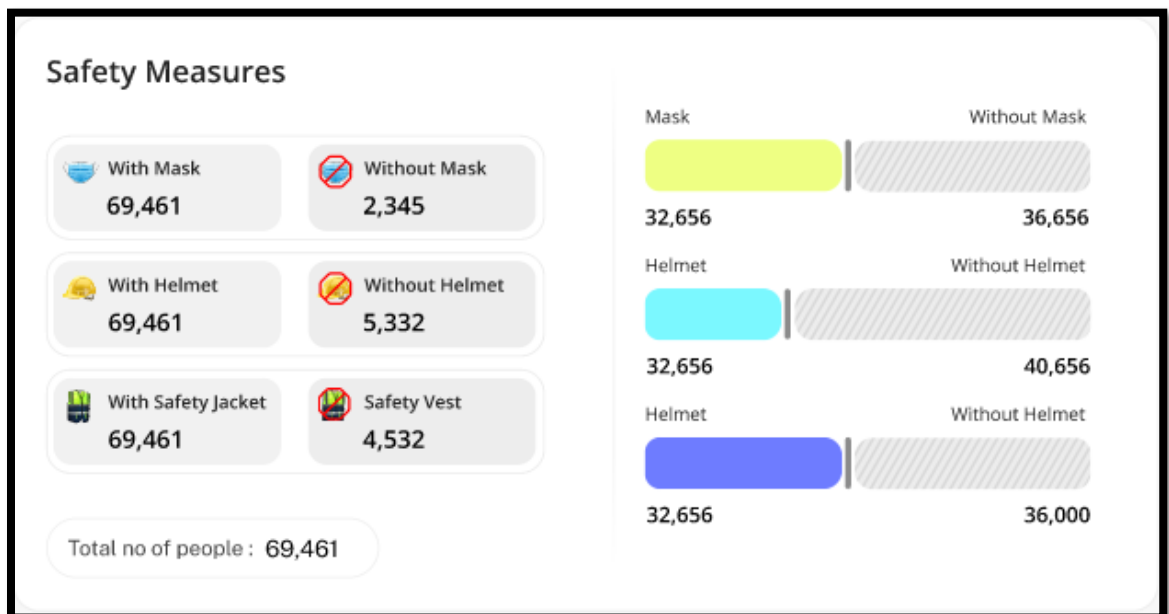
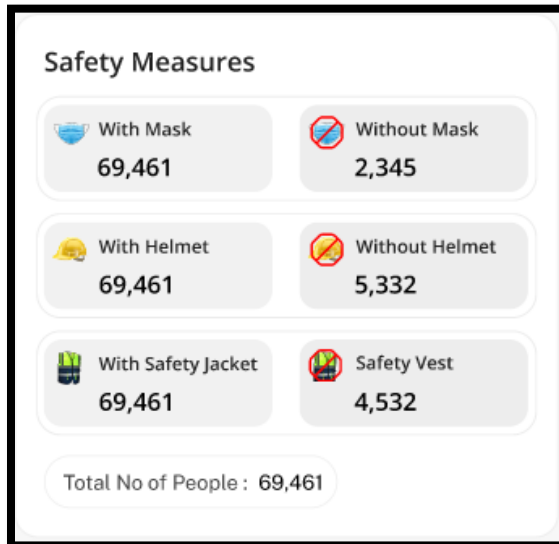


Average People Counting

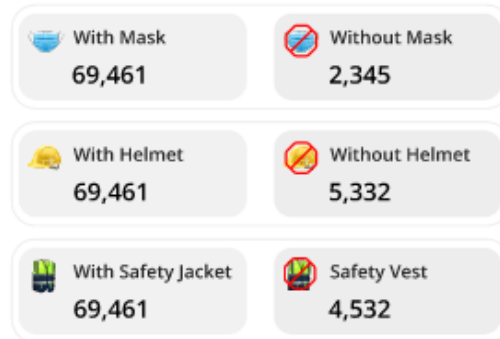


10. Safety Measures:

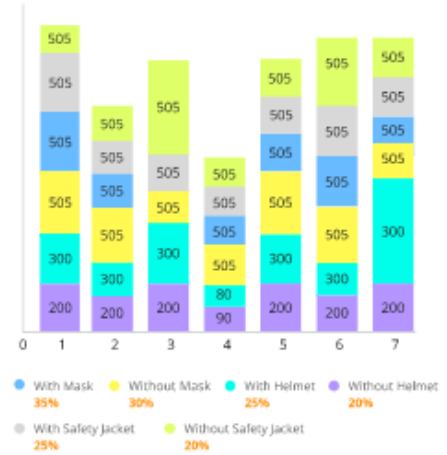
- This report will display counts of the detected peoples who is following or not following safety measures. These will include Mask, Helmet & Jacket Vest only.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



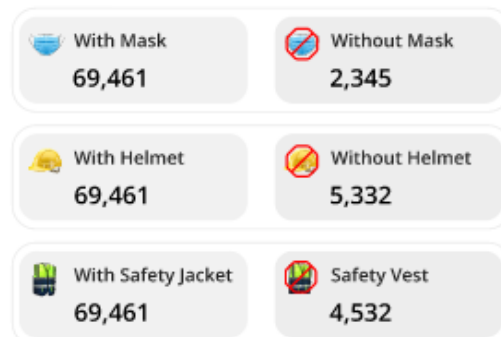
Safety Measures



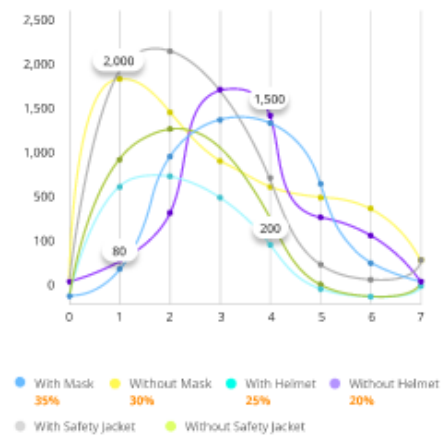
Total no of people : 69,461



Safety Measures



Total no of people : 69,461



11. Slip & Fall Detection:

- This report will display counts of the detected peoples who is fall on the surface.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Slip & Fall Detection

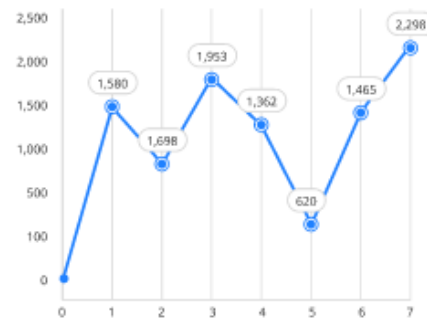


25,600

Slip & Fall Detection



25,600



Slip & Fall Detection

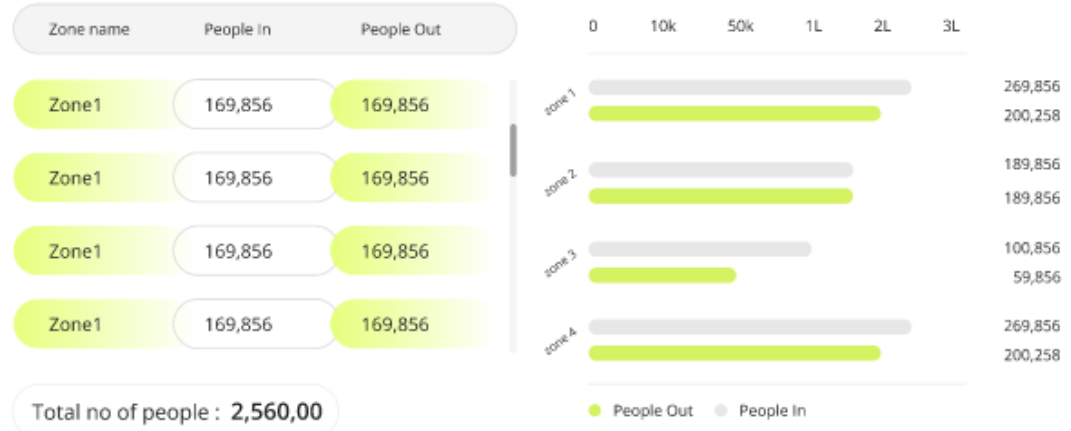


12. Zone Wise People Counting:

- This report will display the zone wise counts of the detected IN & OUT peoples.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Zone wise People Counting



Zone wise People Counting



13. People Count by Gender:

- This report will display the gender wise people counts along with their Most & Least day Date & counts.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option & 1 charts with 3*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Gender



Male
78.5K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025



Female
64.5K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025



Undefined
28.7K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025

Total number of people: 25,600

Gender



Male
78.5K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025



Female
64.5K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025

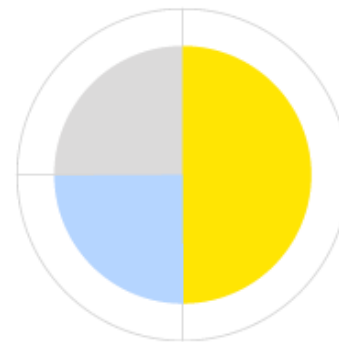


Undefined
28.7K

Least: 20,060
07-03-2025

Most: 20,060
07-03-2025

Total number of people: 25,600



Male 2,006,325 Female 1,003,335 Undefined 30,000



14. New Vs Total Visitors:

- This report will display the counts of the people for New Vs Total Visitors.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

New vs Total Visitors



New Visitors

Total Visitors

2.5L

6.2L

Total new vs total visitors : 2,560,000

New vs Total Visitors



New Visitors

Total Visitors

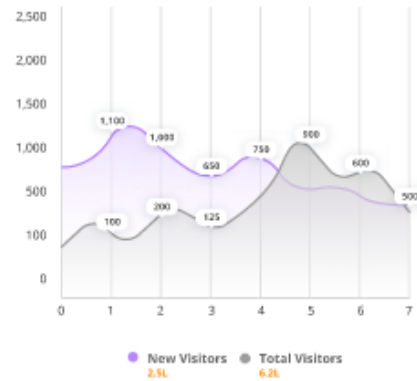
2.5L

6.2L

Total new vs total visitors : 2,560,000



New vs Total Visitors



New vs Total Visitors



15. Cumulative People Count:

- This report will display the total number of cumulative people counts.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

Cumulative People Count



Cumulative
People Count

2,560,000

Total no of cumulative p... : **2,560,000**

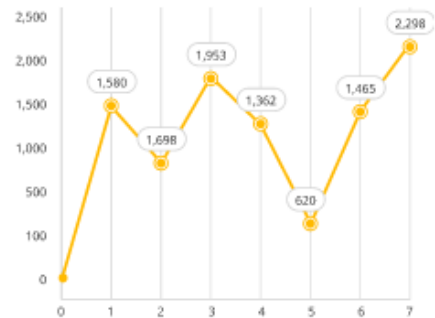
Cumulative People Count



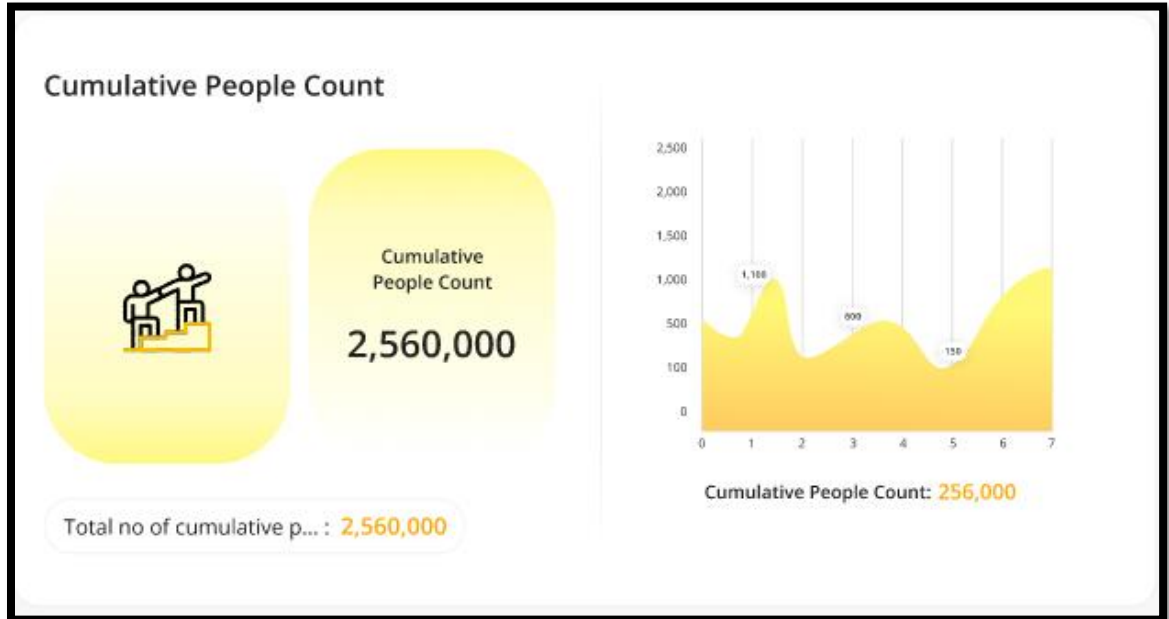
Cumulative
People Count

2,560,000

Total no of cumulative p... : **2,560,000**

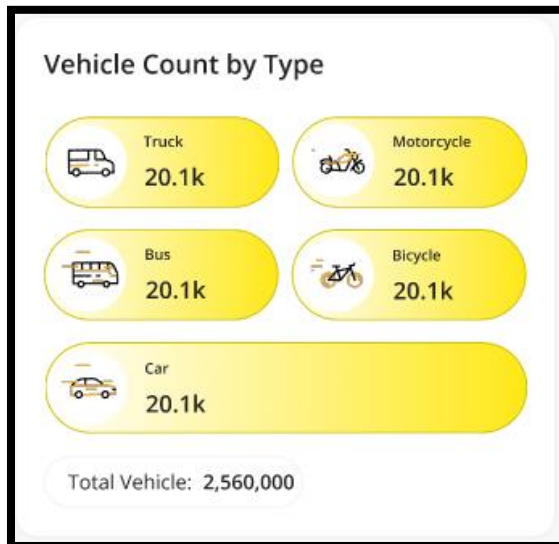


Cumulative People Count: **2,560,000**



16. Vehicle Count by Type:

- This report will display the total number of detected vehicles counts by its type.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



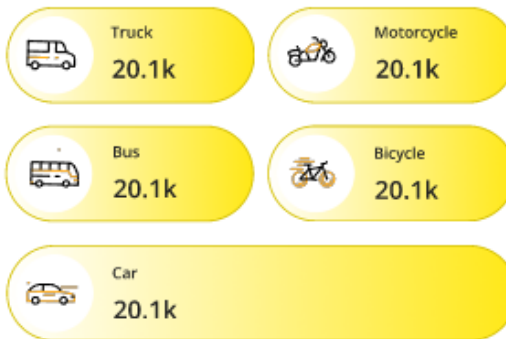
Vehicle Count by Type



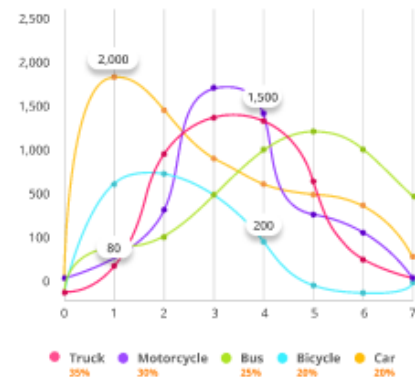
Total Vehicle: 2,560,000

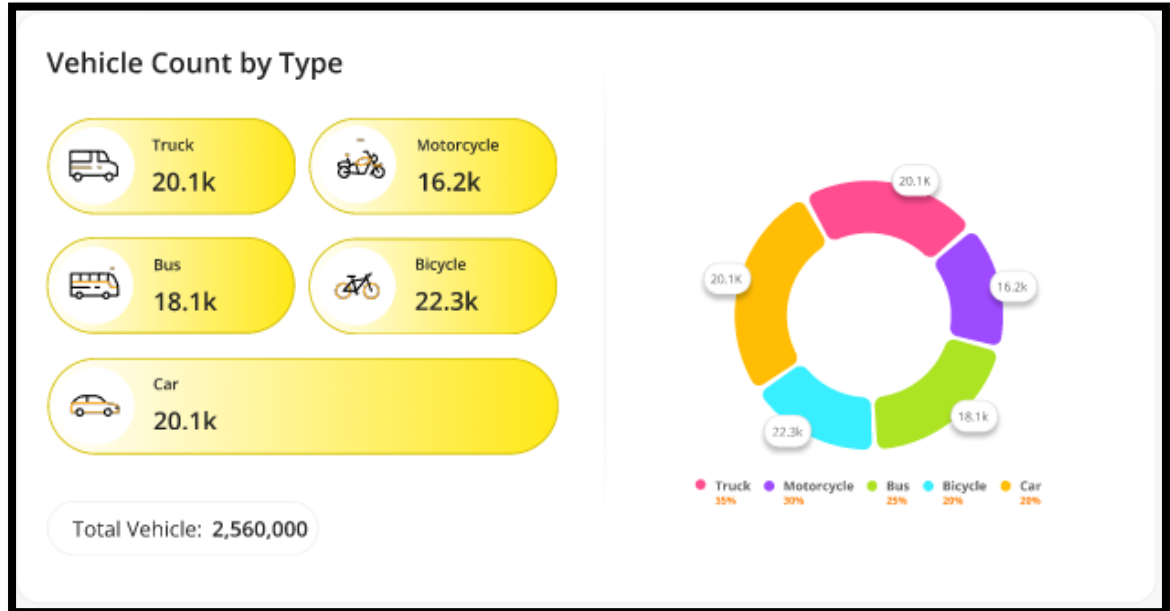


Vehicle Count by Type



Total Vehicle: 2,560,000



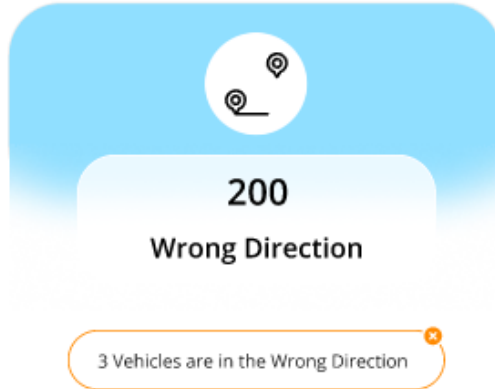


17. Vehicle in Wrong Direction:

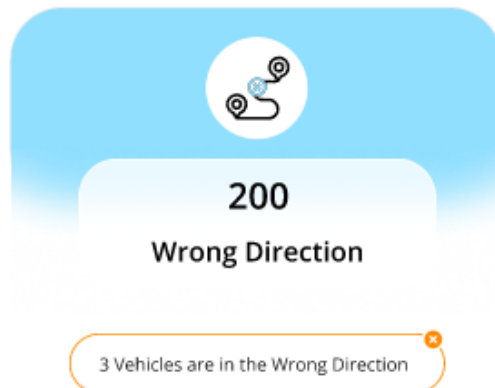
- This report will display the counts of total number of detected vehicles in wrong direction.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:



Vehicle in Wrong Direction



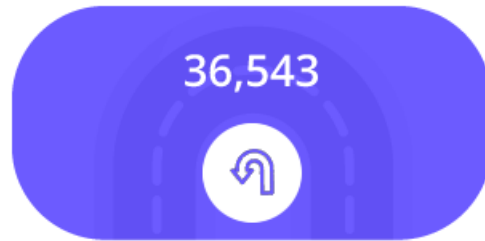
Vehicle in Wrong Direction



18. Vehicle U Turn Detection:

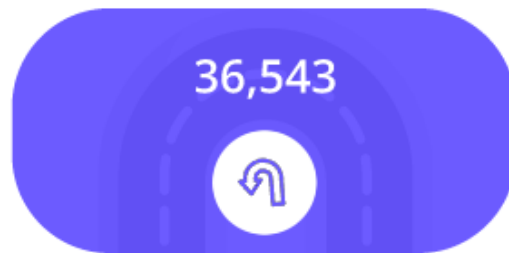
- This report will display the counts of total number of detected vehicles who has taken u turns.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Vehicle U Turn detection

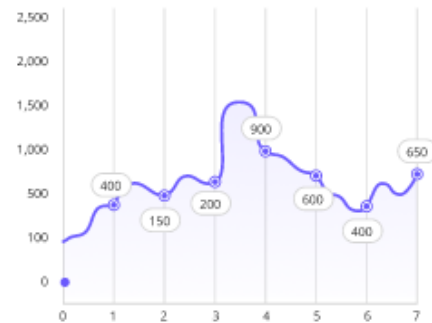


Total U Turn detection: 36,543

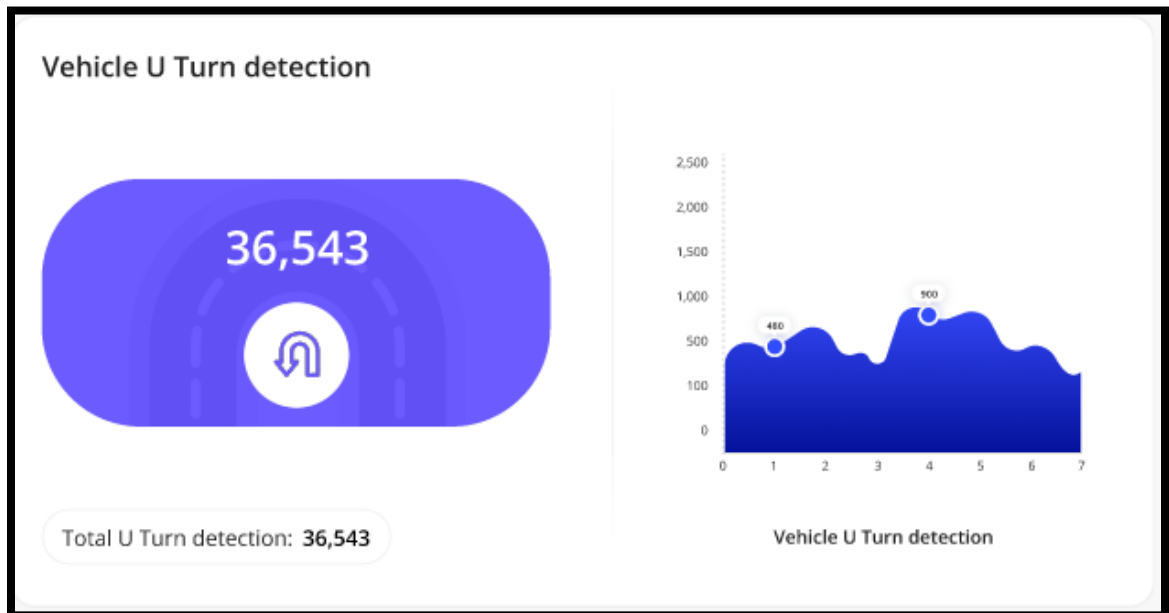
Vehicle U Turn detection



Total U Turn detection: 36,543



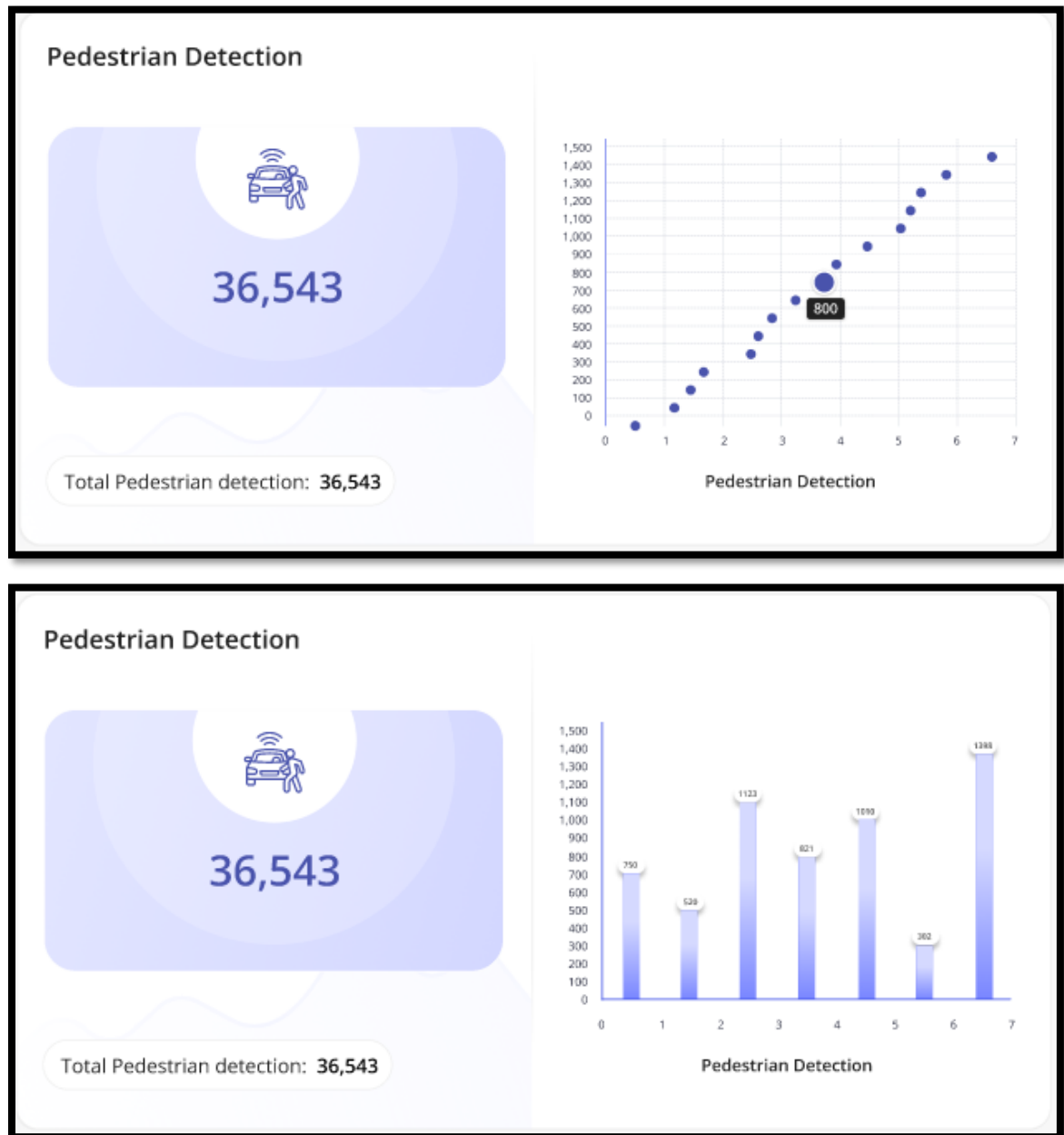
Vehicle U Turn detection



19. Pedestrian Detection:

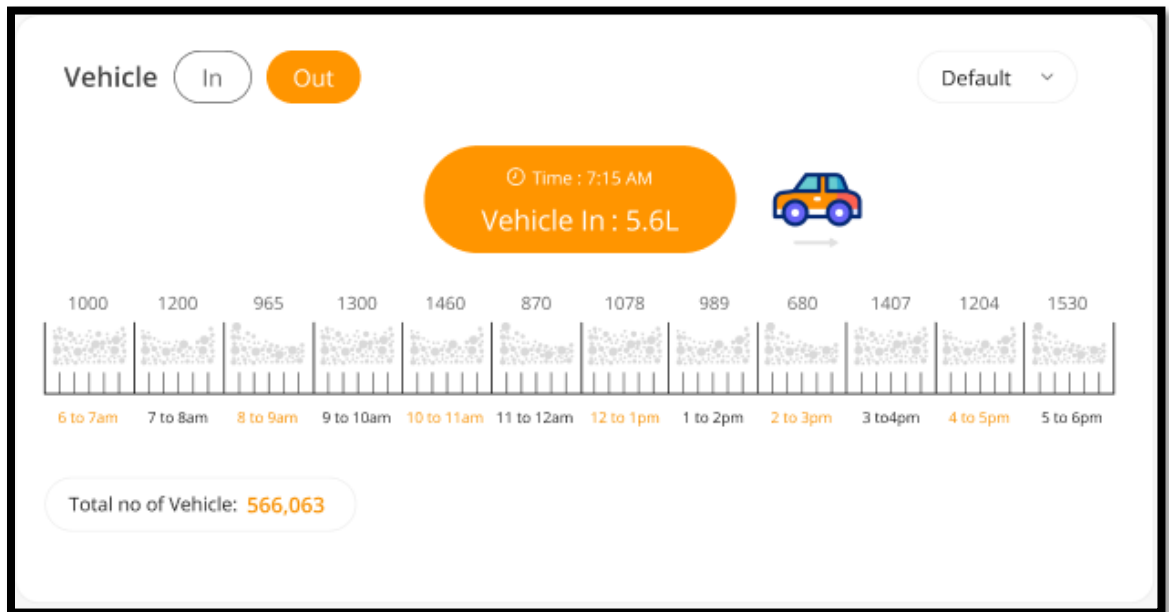
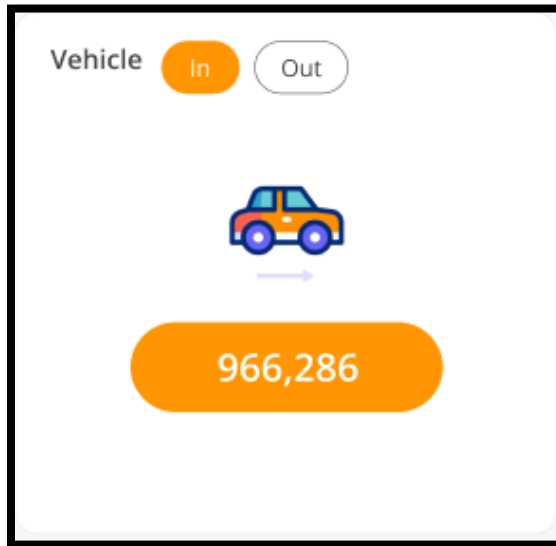
- This report will display the counts of total number of detected pedestrians.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

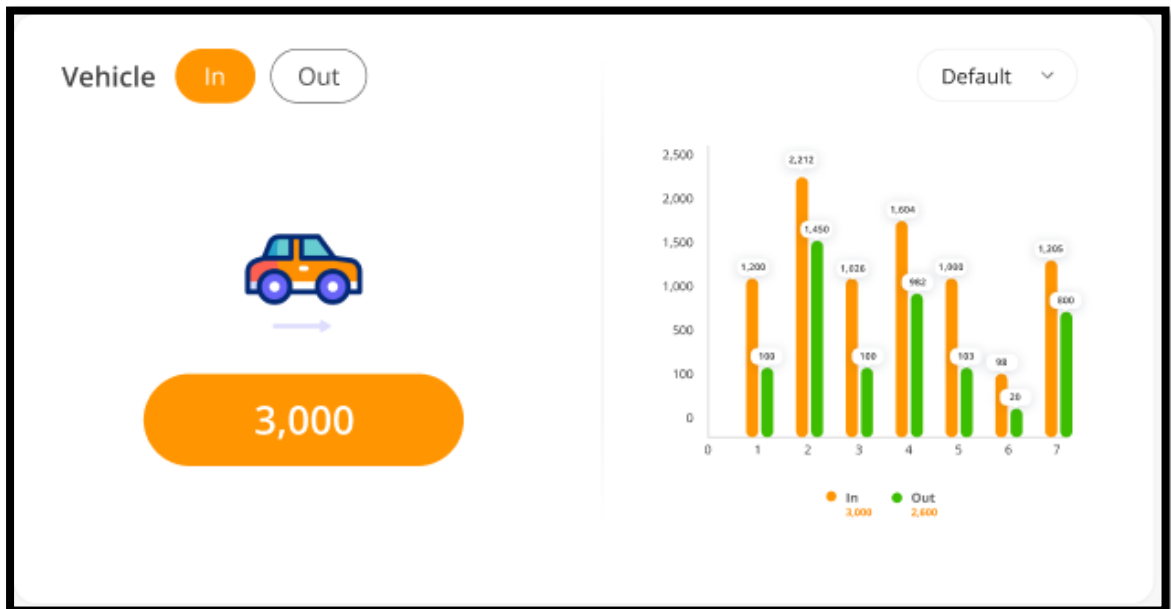
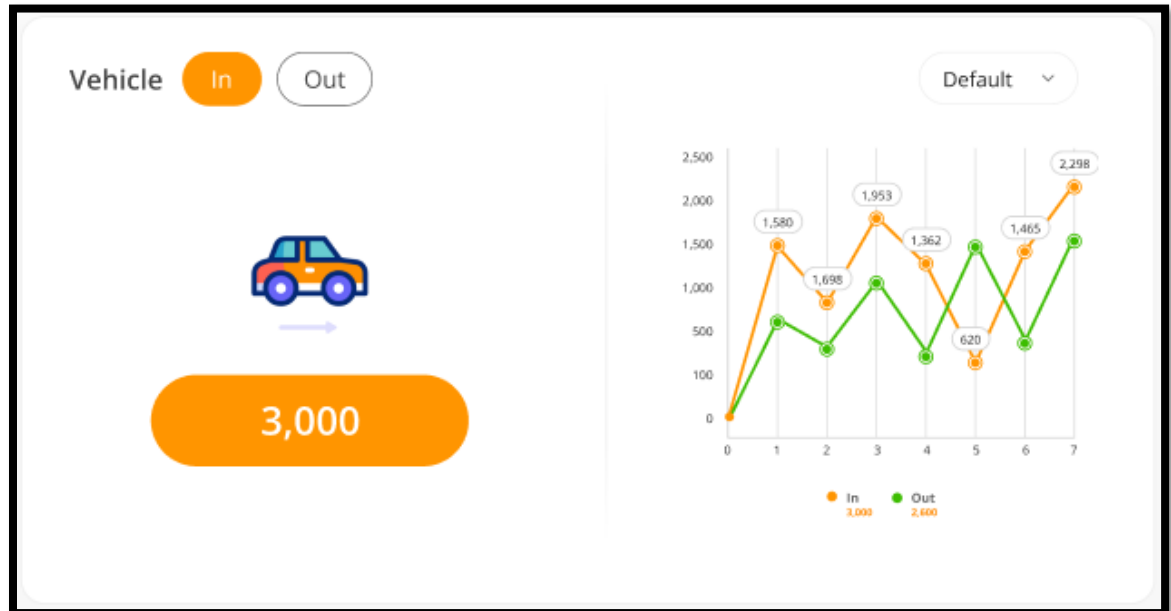




20. Vehicle In & Out:

- This report will display the IN & OUT counts for the total number of detected vehicles.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Please find the attached screenshot below for your reference:

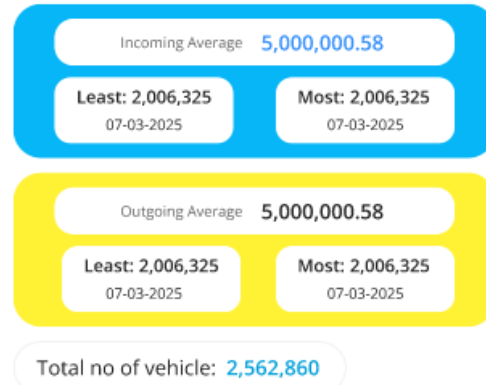




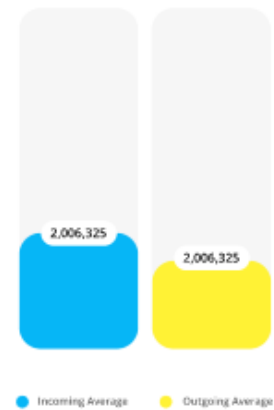
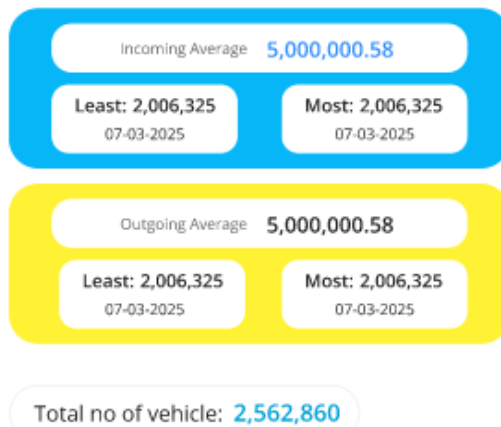
21. Average Vehicle Counting:

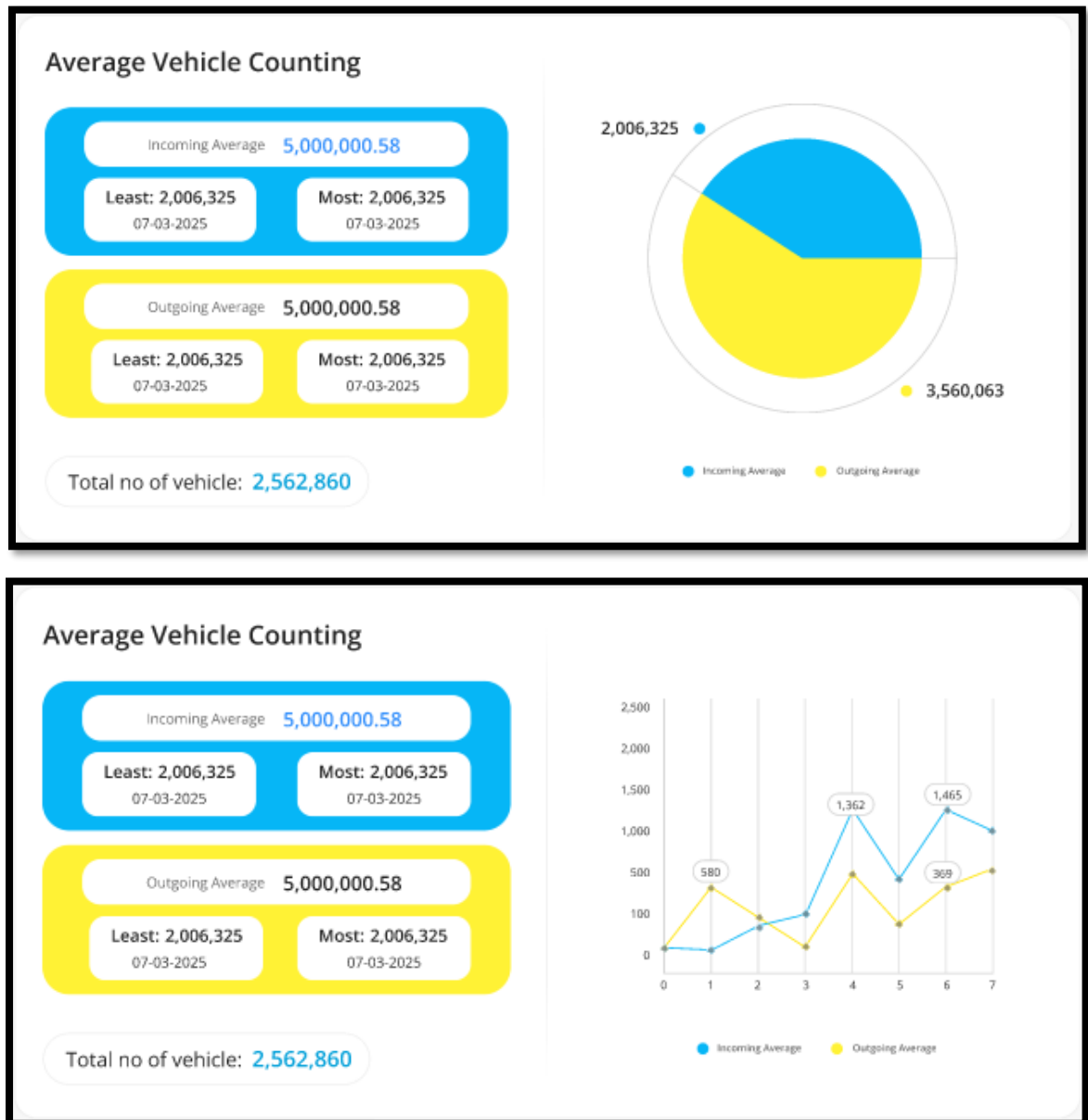
- This report will display the average of total incoming & outgoing vehicles with their most & least day & counts.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Average Vehicle Counting



Average Vehicle Counting





Average Vehicle Counting

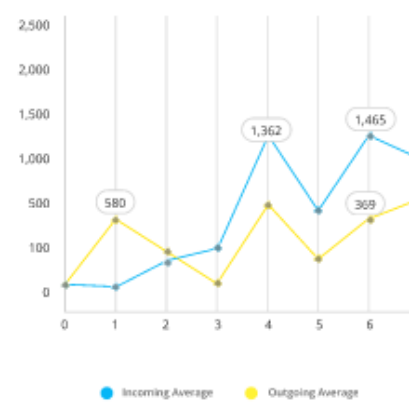
Incoming Average **5,000,000.58**

Least: 2,006,325 Most: 2,006,325
07-03-2025 07-03-2025

Outgoing Average **5,000,000.58**

Least: 2,006,325 Most: 2,006,325
07-03-2025 07-03-2025

Total no of vehicle: **2,562,860**

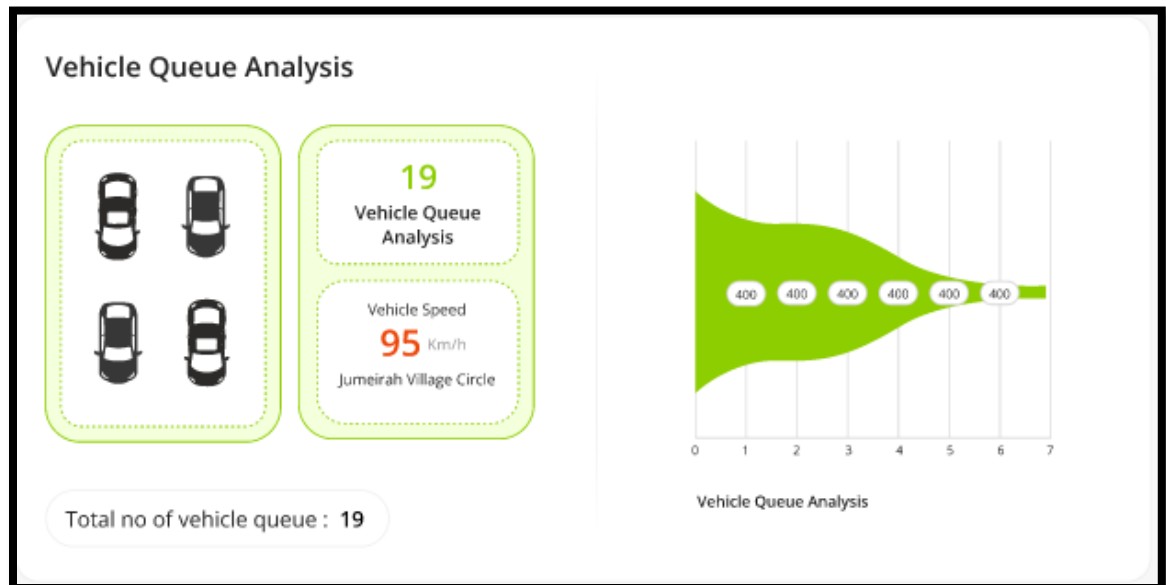
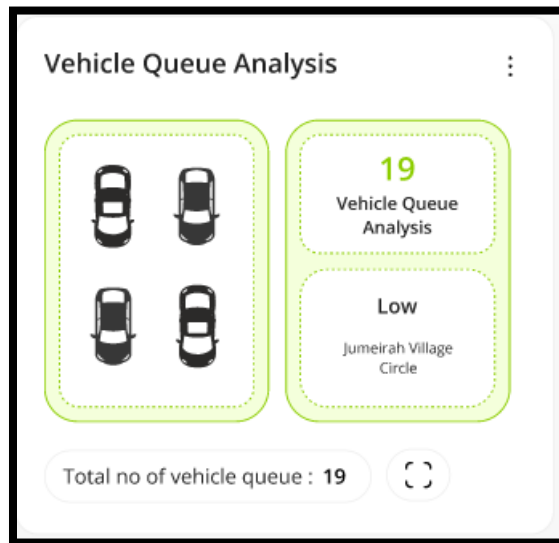


580 1,362 1,465 369

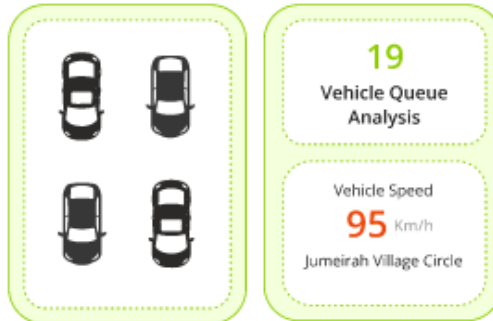
● Incoming Average ● Outgoing Average

22. Vehicle Queue Analysis:

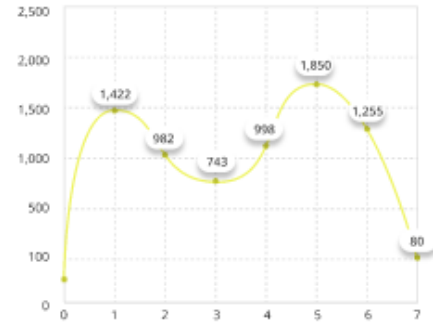
- This report will display the counts of total vehicles which are in the queue within selected time duration.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Vehicle Queue Analysis

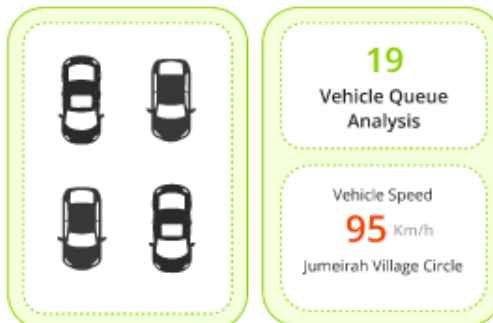


Total no of vehicle queue : 19

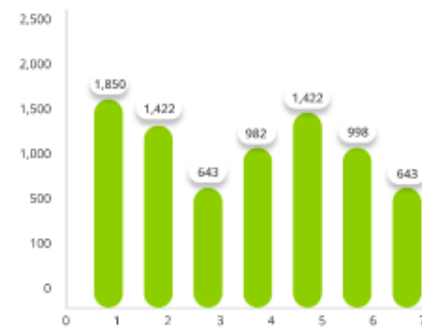


Vehicle Queue Analysis

Vehicle Queue Analysis



Total no of vehicle queue : 19



Vehicle Queue Analysis

23. Stopped Vehicle Count Type:

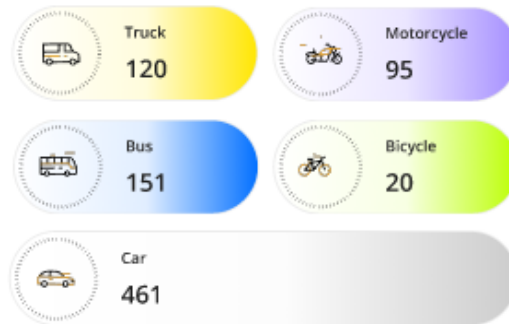
- This report will display the counts of total vehicles stopped by its type.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Stopped Vehicle Count by Type

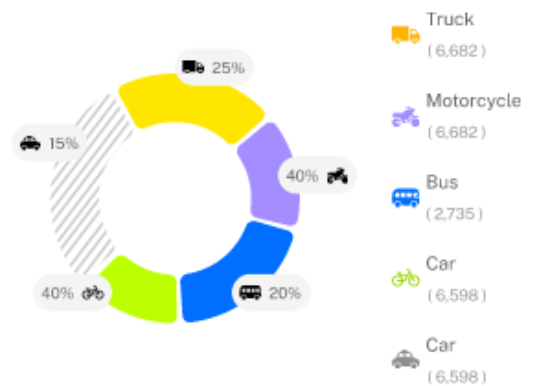


Total no of Stopped vehicle: 1,835

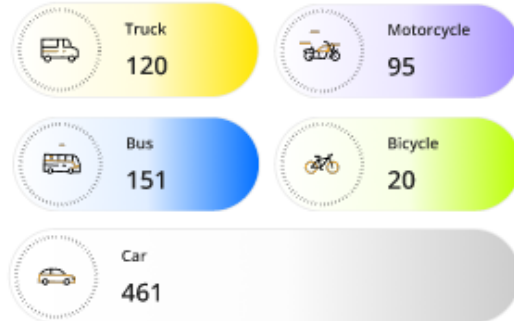
Stopped Vehicle Count by Type



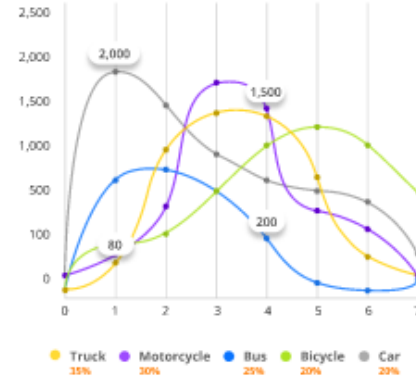
Total no of Stopped vehicle: 1,835



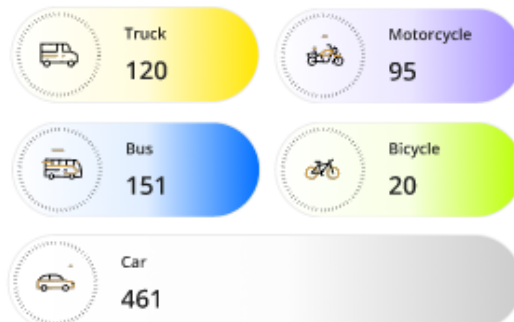
Stopped Vehicle Count by Type



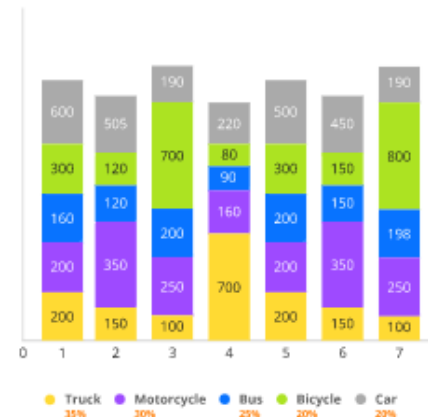
Total no of Stopped vehicle: 1,835



Stopped Vehicle Count by Type



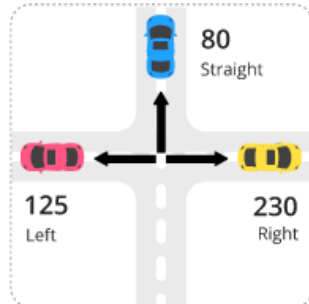
Total no of Stopped vehicle: 1,835



24. Vehicle Turning Movement Counts:

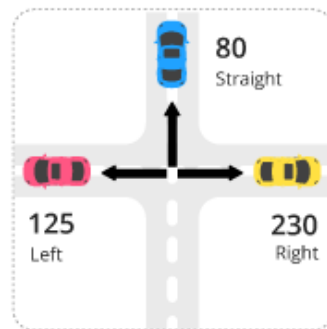
- This report will display the counts vehicles turning movements.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Vehicle Turning Movement counts

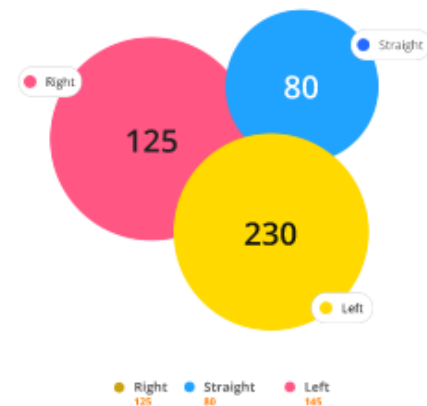


Total no of vehicle queue : 1,200

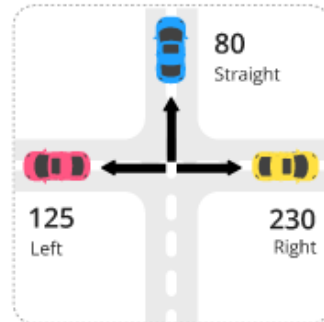
Vehicle Turning Movement counts



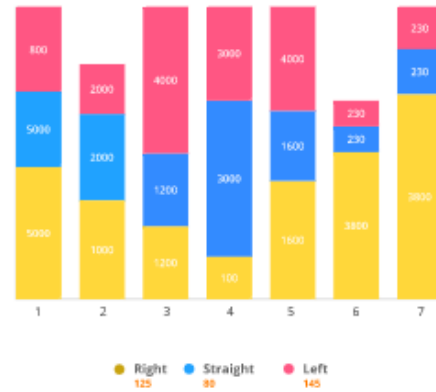
Total no of vehicle queue : 1,200



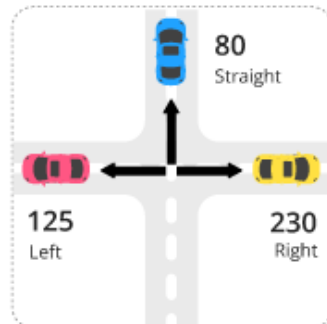
Vehicle Turning Movement counts



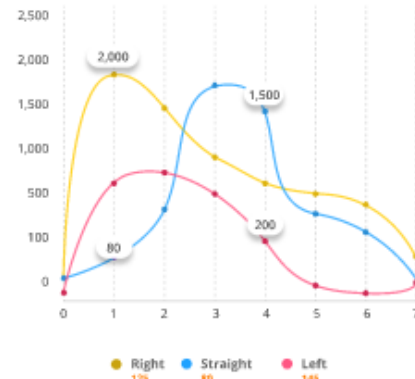
Total no of vehicle queue : 1,200



Vehicle Turning Movement counts



Total no of vehicle queue : 1,200



25. Shopping Cart Counting:

- This report will display the counts of total shopping carts detected.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Shopping Cart Counting



23,564

Total shopping cart counting : 23,564

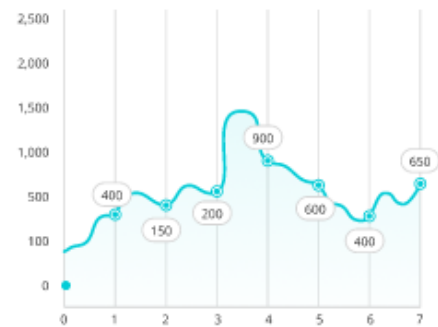


Shopping Cart Counting



23,564

Total shopping cart counting : 23,564

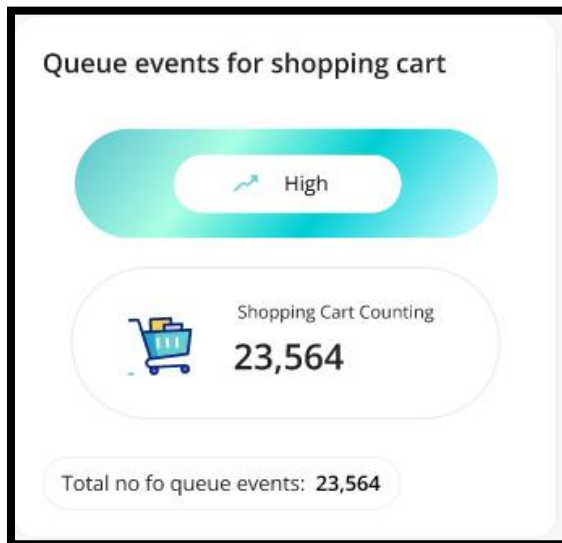


Shopping Cart Counting

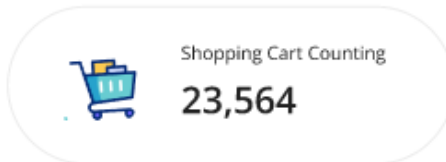


26. Queue Events for Shopping Cart:

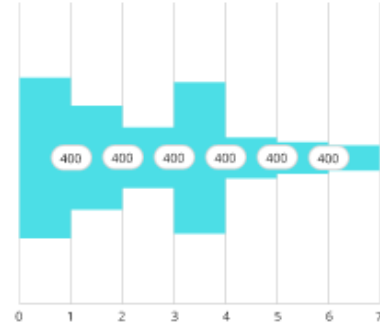
- This report will display the counts of total shopping carts detected in the queue.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Queue events for shopping cart

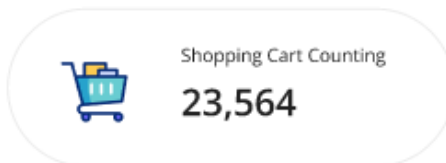


Total no fo queue events: **23,564**

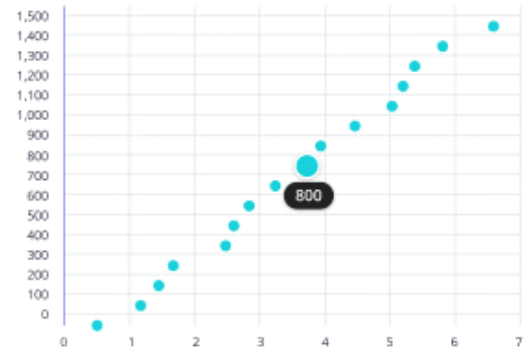


Queue events for shopping cart

Queue events for shopping cart



Total no fo queue events: **23,564**

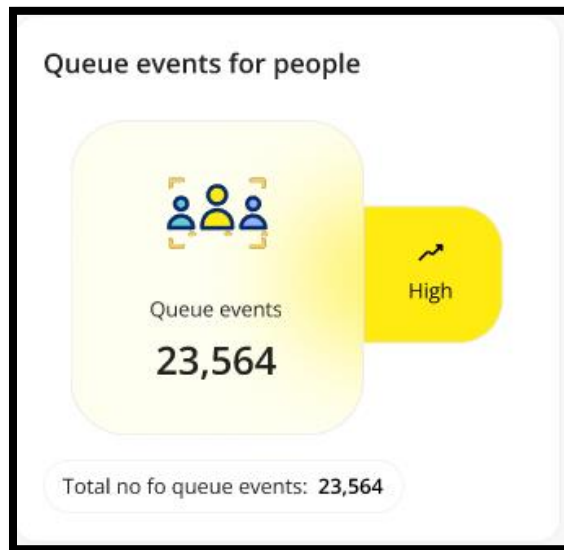


Queue events for shopping cart

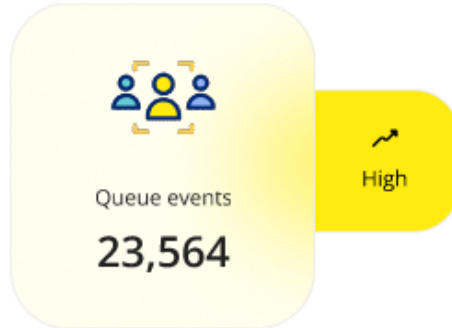


27. Queue Events for People:

- This report will display the counts of total peoples detected in the queue.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Queue events for people

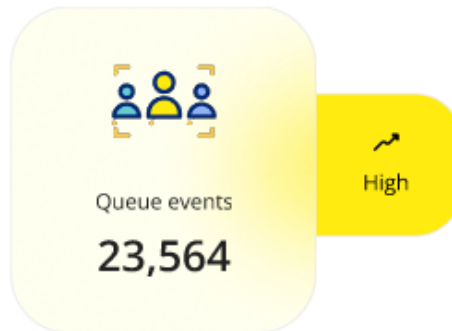


Total no fo queue events: 23,564

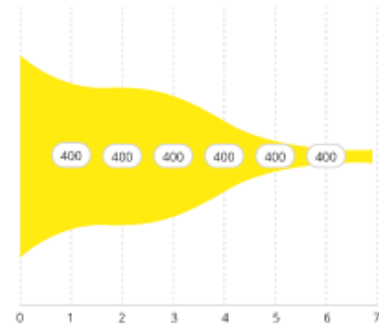


Queue events for people

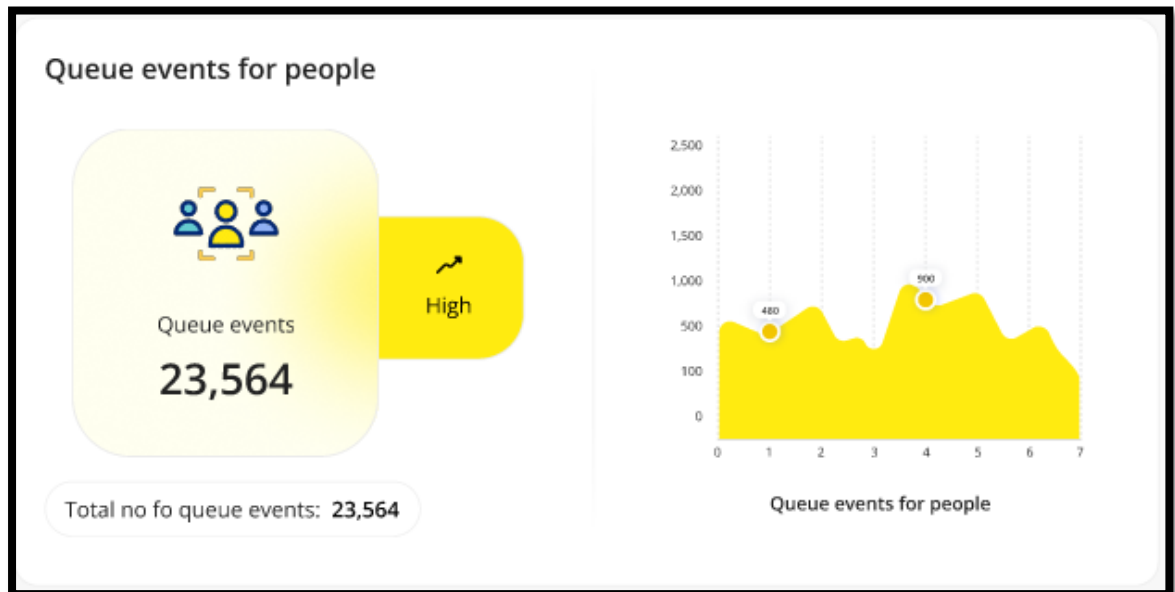
Queue events for people



Total no fo queue events: 23,564

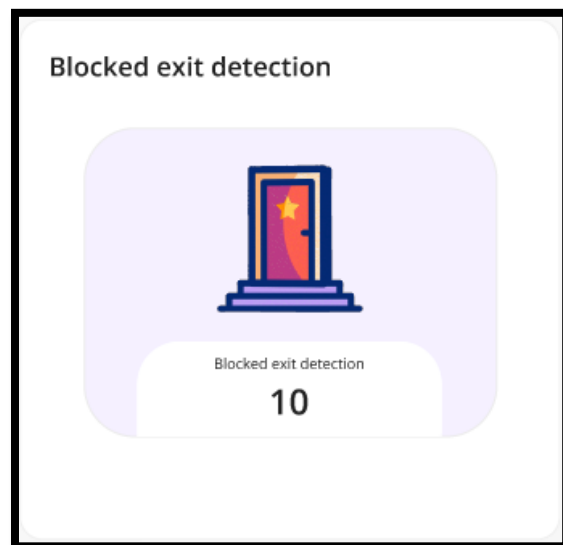


Queue events for people

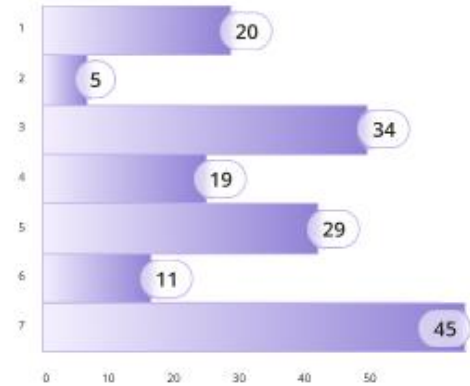


28. Block Exit Detection:

- This report will display the counts of total exit block detected.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

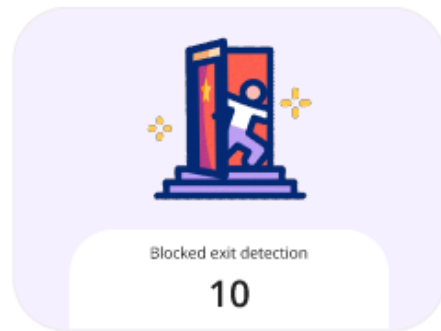


Blocked exit detection

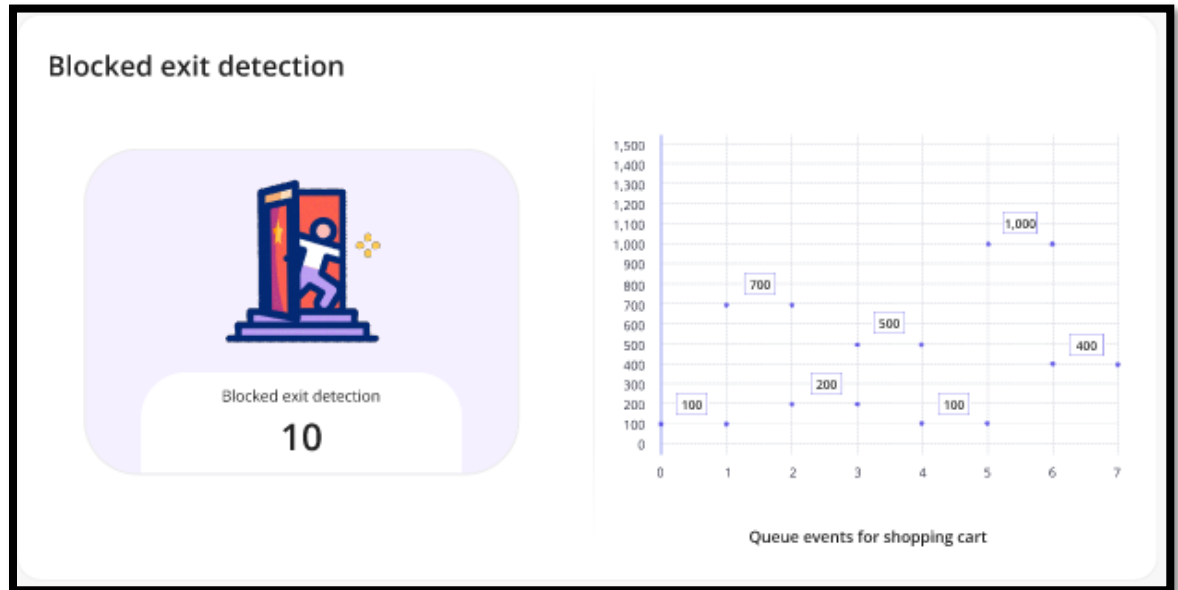


Queue events for shopping cart

Blocked exit detection

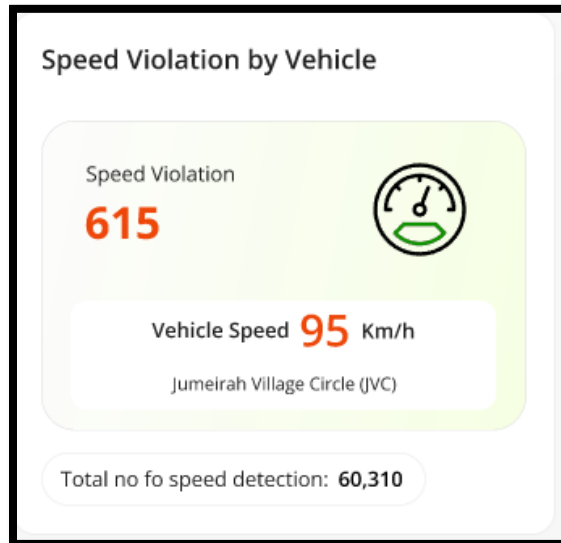


Queue events for shopping cart



29. Speed Violation by Vehicle:

- This report will display the counts of total detected speed violations.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:



Speed Violation by Vehicle

Speed Violation

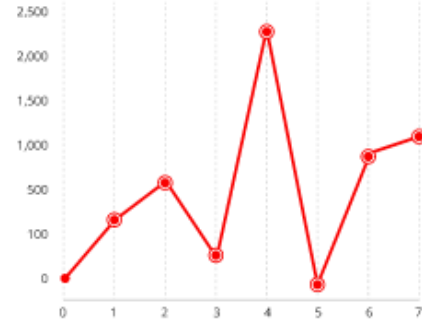
615



Vehicle Speed **95** Km/h

Jumeirah Village Circle (JVC)

Total no fo speed detection: **60,310**



Speed Detection by Vehicle

Speed Violation by Vehicle

Speed Violation

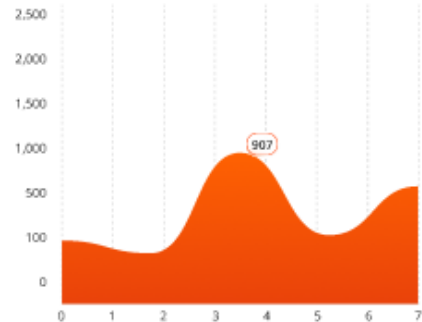
615



Vehicle Speed **95** Km/h

Jumeirah Village Circle (JVC)

Total no fo speed detection: **60,310**



Speed Detection by Vehicle

30. Traffic Jam by Day:

- This report will display the counts of total detected traffic jams.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Traffic Jam by Day

10

Least: 24
07-03-2025

Most: 20
07-03-2025



Traffic Jam

Jumeirah Village Circle (JVC)

11:10 AM

Traffic Jam by Day

10

Least: 24
07-03-2025

Most: 20
07-03-2025

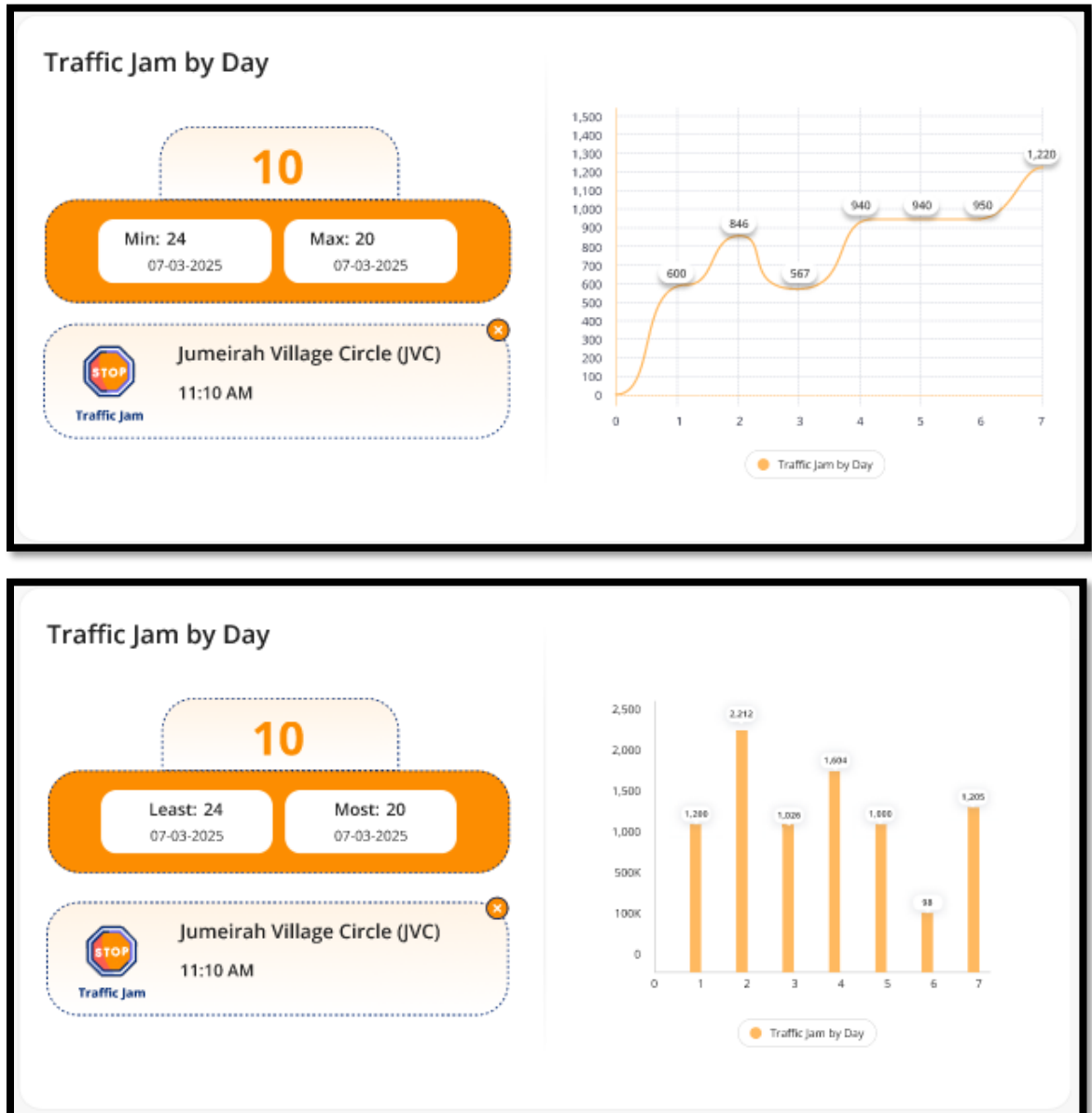


Traffic Jam

Jumeirah Village Circle (JVC)

11:10 AM





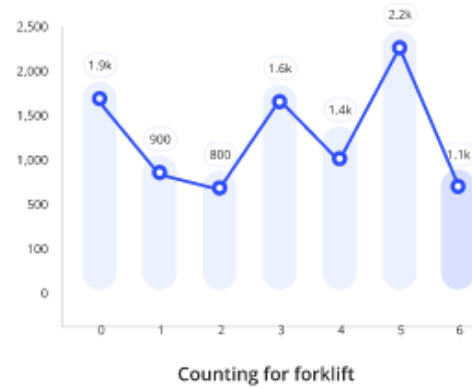
31. Counting for Forklift:

- This report will display the counts of total detected forklifts.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

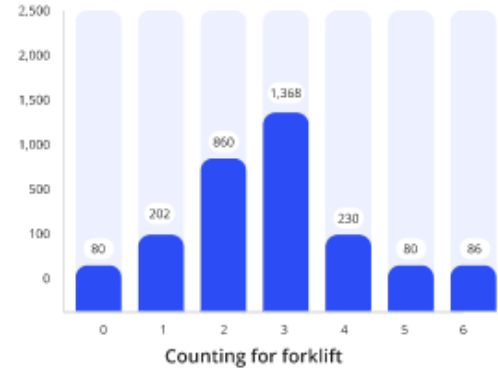
Counting for forklift



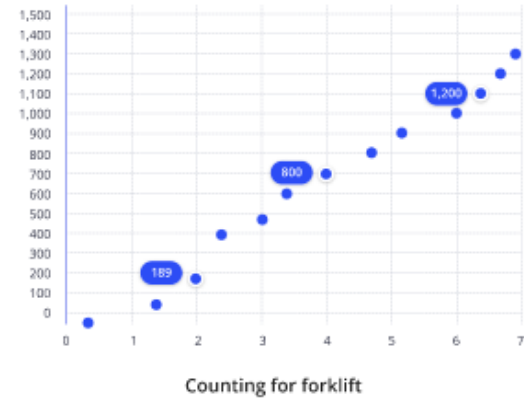
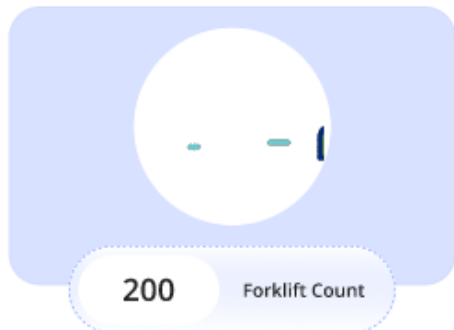
Counting for forklift



Counting for forklift



Counting for forklift



32. Queue Events for Forklift:

- This report will display the counts of total detected forklifts in the queue.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Queue events for forklift



200
Forklift

Low

Total queue events for forklift: 350

Queue events for forklift



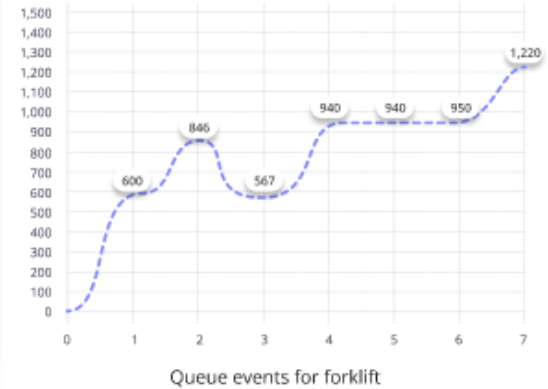
200
Forklift

Low

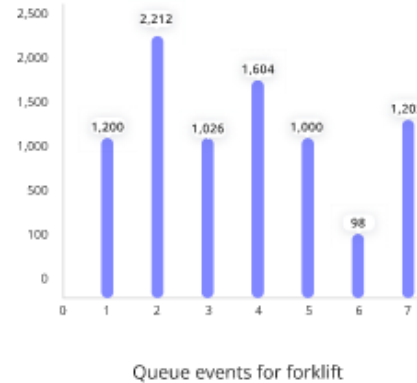
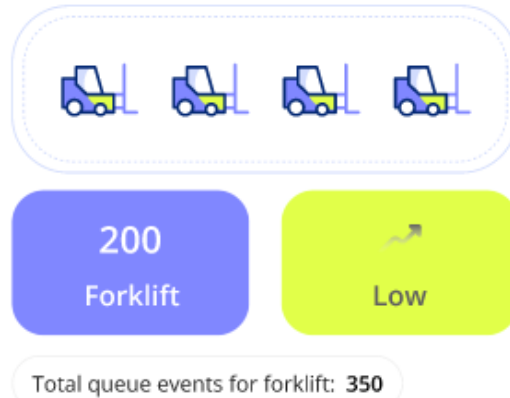
Total queue events for forklift: 350



Queue events for forklift



Queue events for forklift



33. Vehicle Detection Heatmap:

- This report will display the heatmap for the detected vehicles.
- It will be generated from the configured camera in the system only.
- It will provide only 1 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Vehicle Detection Heatmap

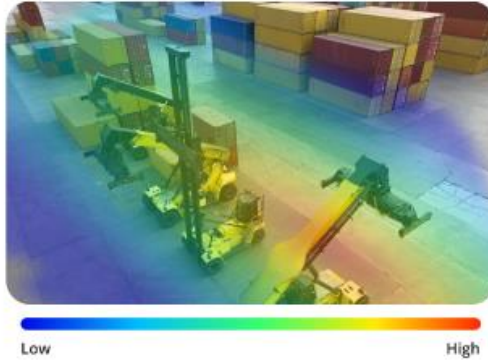


Vehicle Detection Heatmap



34. Forklift Heatmap:

- This report will display the heatmap for the detected forklifts.
- It will be generated from the configured camera in the system only.
- It will provide only 1 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Forklift Heatmap

Forklift Heatmap

35. Detected Forklifts:

- This report will display the counts of total detected forklifts & proximately of forklifts.
- It will be generated from the configured camera in the system only.
- It will provide only 3 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Detect Forklifts



Detect Forklifts

345



Proximity detection

125

Total no fo detect forklifts : 470

Detect Forklifts



Detect Forklifts

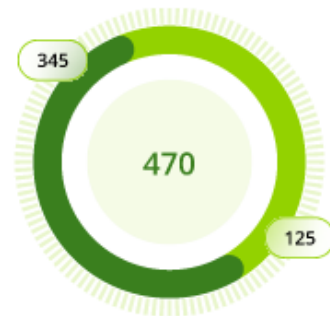
345



Proximity detection

125

Total no fo detect forklifts : 470



● Detect Forklifts
 ● Proximity detection

Detect Forklifts



Detect Forklifts

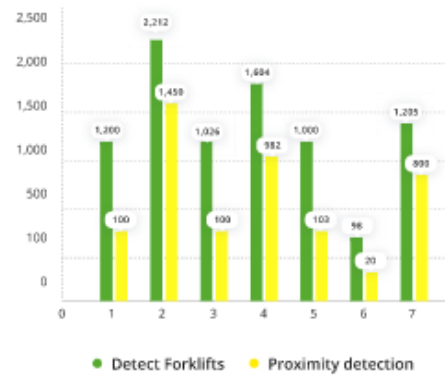
345



Proximity detection

125

Total no fo detect forklifts : 470



Detect Forklifts



Detect Forklifts

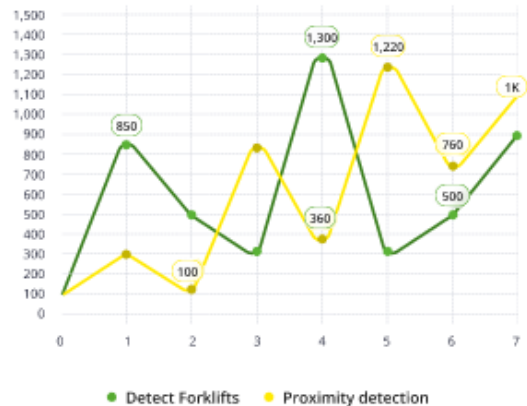
345



Proximity detection

125

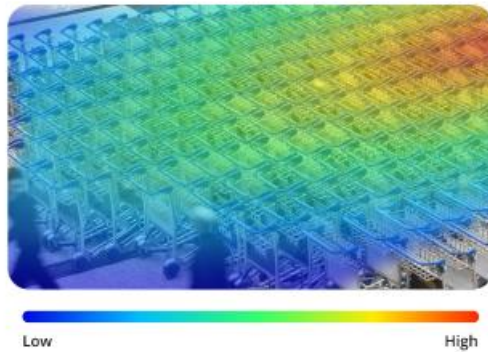
Total no fo detect forklifts : 470



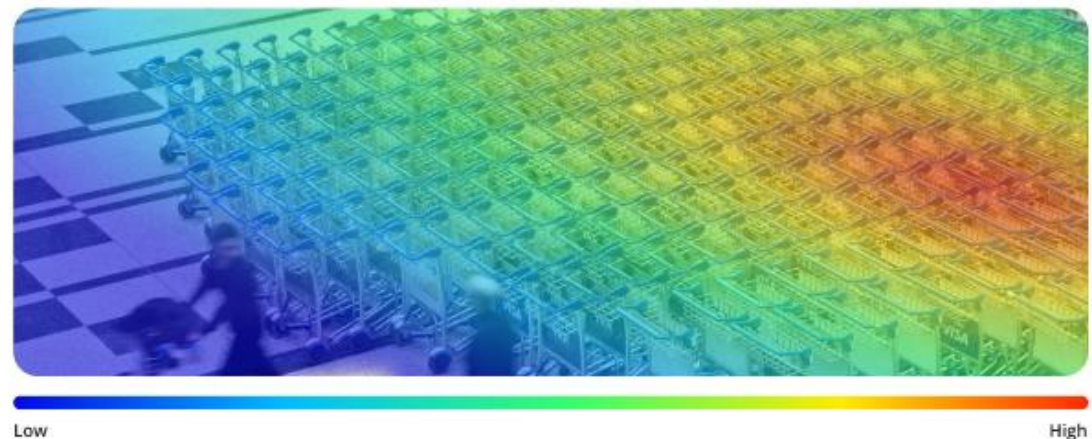
36. Shopping Cart Heatmap:

- This report will display the heatmap of detected shopping carts.
- It will be generated from the configured camera in the system only.
- It will provide only 1 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Shopping Cart Heatmap



Shopping Cart Heatmap



37. Forklift Speed Detection:

- This widget will display the counts of total speed violation by detected forklifts.
- It will be generated from the configured camera in the system only.
- It will provide only 2 charts with 2*1 option. By default, the system will display 1*1 only.
- Pleased find the attached screenshot below for your reference:

Forklift Speed Detection



Speed Violation

615

Forklift Speed **95** Km/h

Jumeirah Village Circle (JVC)

Total Forklift Speed Detection: **60,310**

Forklift Speed Detection



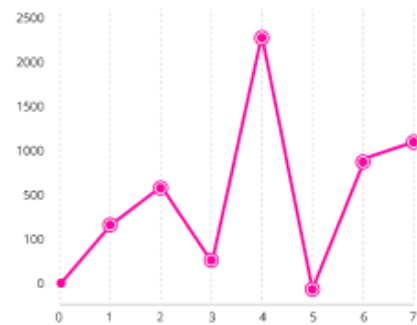
Speed Violation

615

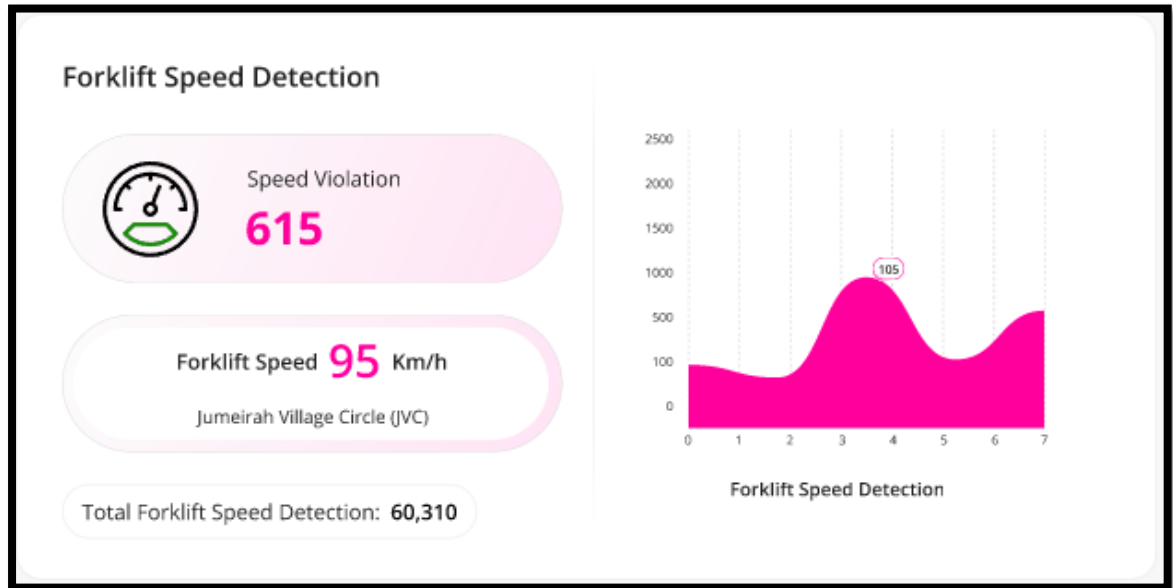
Forklift Speed **95** Km/h

Jumeirah Village Circle (JVC)

Total Forklift Speed Detection: **60,310**

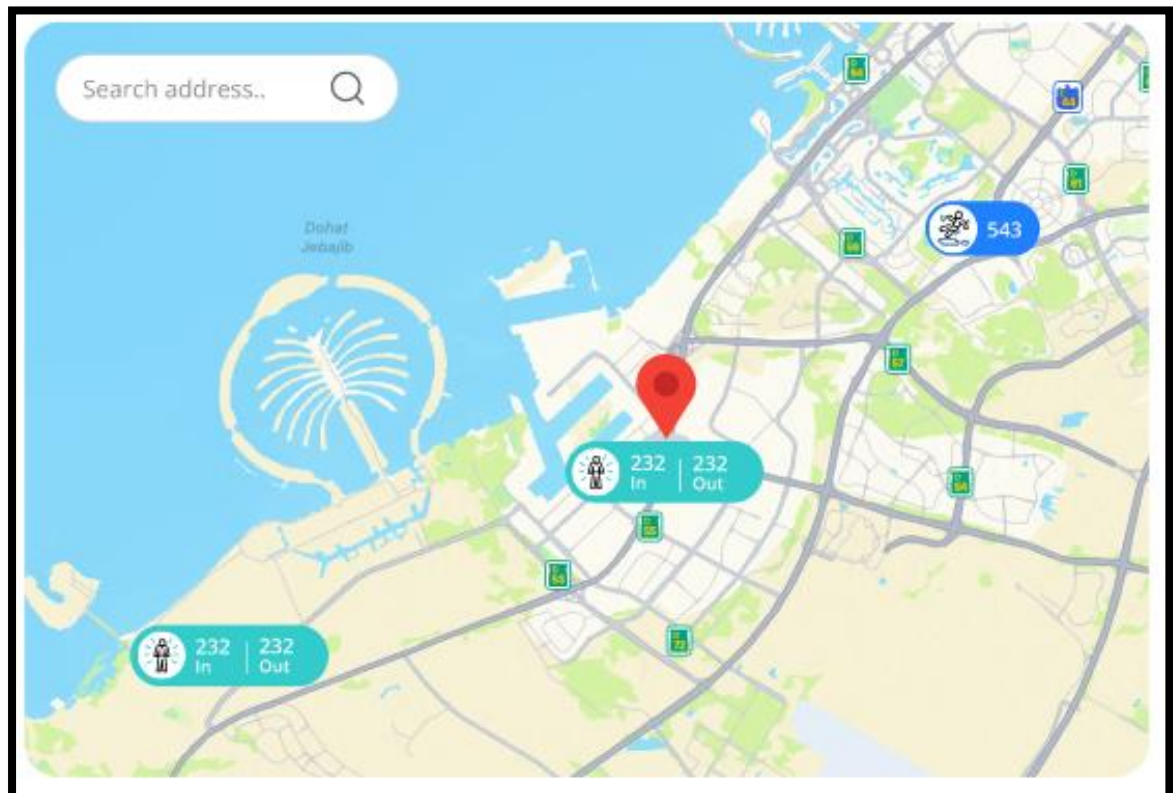


Forklift Speed Detection



38. Map Plan:

- This widget will display the counts on the map based on the selected features.
- It will be generated from the configured camera in the system only.
- Please find the attached screenshot below for your reference:



39. Floor Plan:

- This report will display the counts on the floor plan image based on the selected features.
- It will be generated from the configured camera in the system only.
- Please find the attached screenshot below for your reference:



Business Rules:

- The authorized users will be able to configure up to 40 [Refer 3.18.1] widget in the specific dashboard from the system.
- The **authorized users** will be able to perform **following actions** on each widget:
 - ➔ Can apply **duration filter** to view the data on the chart.
 - ➔ Can **expand** the widget.
 - ➔ Can **Delete** the added widget.
 - ➔ Can **export** it as **CSV**.
 - ➔ Can **Setup** the widget preview.
 - ➔ Can **move** added widget from one place to another place as per their needs within the **dashboard area**.
- The system will allow users to perform **above-mentioned actions** based on the **assigned permission** to them.
- The system will display following features only under the dropdown list:
 - ➔ People In & Out Count
 - ➔ Slip & Fall Count
 - ➔ Vehicle In & Out Count
 - ➔ Pedestrian Detection Count

- ➔ Vehicle Queue Analysis Count
- ➔ Speed Violation by Vehicle Count
- ➔ Traffic Jam by Day
- ➔ Shopping Cart Count
- ➔ Forklift Count

Data Dictionary:

forkliftCount					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	Objectid	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceId	Objectid		674daaf7d34a2b901c847454	Device id of the data
3	channelNo	Int32		0	channel no of the device
4	lines	Object[]		[{}]	List of each lines
4.1	> lineIndex	Int32		1	Line index number
4.2	> name	String		Rule1	Name of the line
4.3	> inCount	Int32		20	In count of line
4.4	> outCount	Int32		10	Out count of line
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

ShoppingCartCount					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	Objectid	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceId	String	No	674daaf7d34a2b901c847454	Device id form where shopping cart data is found
3	cameraIp	String	No	Manage entire system	Camera ip form where shopping cart data is found
4	channelNo	Int32	No	0	Channel no of the device
5	lines	Object[]	No	[{"lineIndex": 1, "name": "name1", "inCount": 0, "outCount": 0}]	Collection of all configured line for the shopping cart counting
5.1	> lineIndex	Int32	No	1	Index number of the line
5.2	> name	String	No	rule1	Name of the line
5.3	> inCount	Int32	No	0	In count from this line
5.4	> outCount	Int32	No	0	Out count from this line
6	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
7	createdBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
9	updatedBy	Objectid	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

VehicleCount					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	vehicleCounts	Object[]	No	[{}]	Vehicle in and out data
2.1	> channel	Int32	No	0	Channel no of the device
2.2	> lines	Object[]	No	[{}]	Line wise chennel data
2.3	>> LineIndex	Int32	No	1	Line index number for counting
2.4	>> Name	String	No	Line1	Line name for counting
2.5	>> InCount	Int32	No	50	Total in count for line
2.6	>> In	Object	Yes	{}	In counts by vehicle types
2.7	>>> Car	Number	No	10	In Count by Car
2.8	>>> Bus	Number	No	10	In Count by Bus
2.9	>>> Truck	Number	No	10	In Count by Truck
2.1	>>> Motorcycle	Number	No	10	In Count by Motorcycle
2.11	>>> Bicycle	Number	No	10	In Count by Bicycle
2.12	>> OutCount	Number	No	50	Total out count for line
2.13	>> Out	Object	Yes	{}	Out counts by vehicle types
2.14	>>> Car	Number	No	10	Out Count by Car
2.15	>>> Bus	Number	No	10	Out Count by Bus
2.16	>>> Truck	Number	No	10	Out Count by Truck
2.17	>>> Motorcycle	Number	No	10	Out Count by Motorcycle
2.18	>>> Bicycle	Number	No	10	Out Count by Bicycle
3	deviceId	String	No	674daaf7d34a2b901c847454	Device id form where shopping cart data is found
4	cameraIp	String	No	Manage entire system	Camera ip form where shopping cart data is found
5	channelNo	Int32	No	0	Channel no of the device
6	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
7	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
8	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
9	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

queueManagement					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceId	ObjectId	No	674daaf7d34a2b901c847454	Device id of the data
3	channelNo	Int32	No	0	channel no of the device
4	ruleIndex	Int32	No	1	Rule index of the features
5	ruleName	String	No	rule1	Rule name of the features
6	count	Int32	No	10	count of the features
7	eventName	String	No	OpenSDK.WiseAI.QueueCountChanged	name of the event
8	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
9	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
10	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
11	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
12	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

SOFTWARE REQUIREMENT SPECIFICATION

SOFTWARE DEVELOPMENT

TEMPLATE: SD:TP:101 REV- 00
EFFECTIVE DATE: 01/04/2024

peopleCount					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceId	ObjectId	No	674daaf7d34a2b901c847454	Device id for the people count data
3	channelNo	Int32	No	0	Channel no of the device
4	lines	Object[]	No	[{}]	Collection of the lines handled in the channel
4.1	> lineIndex	Int32	No	1	Line index number
4.2	> name	String	No	Rule1	Name of the line
4.3	> inCount	Int32	No	5	In count on the line
4.4	> outCount	Int32	No	4	Out count on the line
4.5	> ageInfo	Object[]	Yes	[{}]	Age wise detail of the people count
4.5.1	>> ageType	String	No	Young	Age type
4.5.2	>> count	Int32	No	3	People count by type
4.6	> genderInfo	Object[]	Yes	[{}]	Gender wise detail of the people count
4.6.1	>> genderType	String	No	Male	Gender detail
4.6.2	>> count	Int32	No	4	Gender wise people count
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

multiLaneVehicleCount					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	_id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	deviceId	ObjectId	No	674daaf7d34a2b901c847454	Device id of the data
3	channelNo	Int32	No	0	channel no of the device
4	directionCount	Object[]	No	[{}]	Direction list of counts based on lines
4.1	> direction	String	No	Left	Direction name, Line name
4.2	> count	Int32	No	200	Vehicle count in specific lane
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

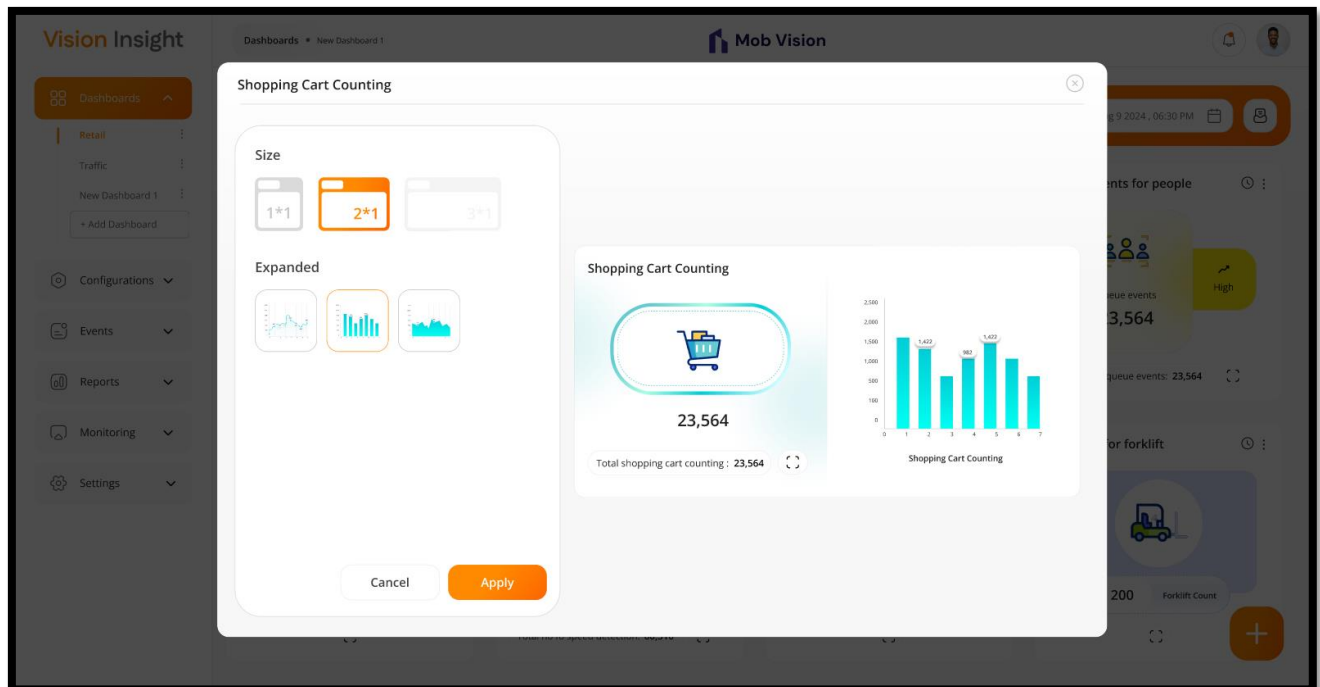
3.18.4. Dashboards > Dashboard Page > Widget > Setup Page

The system provides a feature for **authorized users** to **setup widget preview** as per their needs through the **BI Dashboard Web Application**.

This option will be available under **each widget** which are **configured / added** under the **specific dashboard only**.

Prototype:

→ The screen below illustrates a **sample setup page**, including options to select **Size** and **Expanded view**:



Business Rules:

- The system will allow **authorized users** to access this option.
- The system will open **setup popup** once user clicks on the **“Setup”** option located under each widget which are configured on the specific dashboard.
- Under that popup, the users will be able to view & configure **followings**:
 - Size
 - It is the **size of the selected widget**.
 - It will provide following options:
 - 1*1
 - 2*1
 - 3*1
 - The system will **enable or disable** any above option based on the **predefined design** for each widget implemented in the **attached Figma**.
 - Expanded
 - This option will provide **up to 3 charts option based on the needs**.

- This option will **get appeared** when user will select the **2×1 or 3×1** option from the above size.
- The system will display **preview** at **right-hand side section** based on the selected options under the same popup.
- The system will **render the actual data** & display that specific widget on the dashboard screen based on the **selected setup**.
- The users must have to click on **“Apply” button** to reflect the **updated changes** in the setup popup to specific dashboard.
- Users can **re-setup** the things by clicking on the same **“Setup” option**.

Data Dictionary:

➔ The system will utilize below highlighted column for the setup configurations:

dashboardPreference					
Sr No	Column_name	Type	Nullable	Sample Data	Comments
1	id	ObjectId	No	674daaf7d34a2b901c847454	Autogenerated id
2	userId	String	No	674daaf7d34a2b901c847454	Dashboard is mapped with this user
3	dashboardPreferenceDesign	String	Yes	{}	Collection if widgets for this dashboard
4	dashboardName	String	Yes	Retail	General description of the email template
5	IsDeleted	Boolean	Yes	FALSE	The user is Deleted or not, Default value will be "False" only
6	createdBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
7	createdOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.
8	updatedBy	ObjectId	Yes	Logged-in User's Id	This will be automatically inserted by the backend. No user input is required.
9	updatedOn	DateTime	Yes	System Date & Time	This will be automatically inserted by the backend. No user input is required.

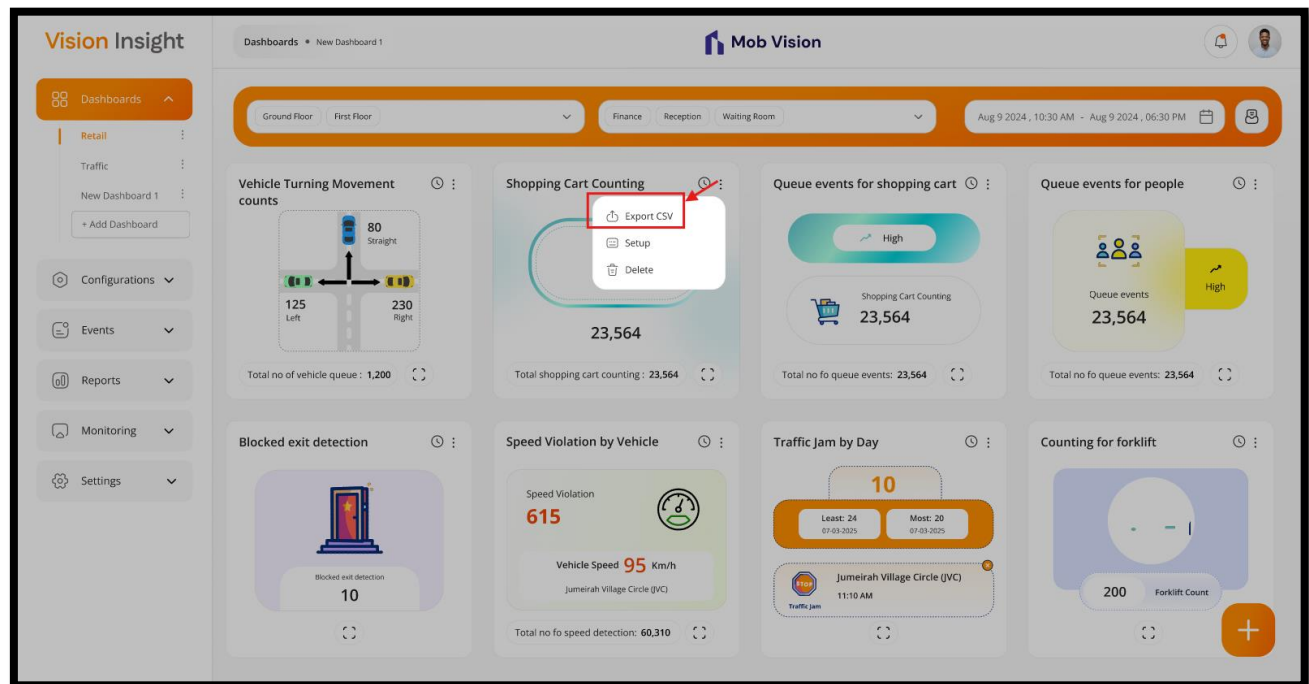
3.18.5. Dashboards > Dashboard Page > Widget > Export as CSV Option

The system provides a feature for **authorized users** to **export a report in .csv format** based on the **selected time frame & date range** through the **BI Dashboard Web Application**.

This option will be available under **each widget** (*whenever it's applicable only*) which are **configured / added** under the **specific dashboard only**.

Prototype:

➔ The screen below illustrates the **option to select a duration** and **export the report as CSV format only** based on the selected time range:



Business Rules:

- The system will allow **authorized users** to access this option.
- Once user clicks on the **“Export CSV” option** located under each widget which are configured on the specific dashboard.
- The **CSV** will include following information in the exported report:
 - ➔ Exported By
 - ➔ Exported On (Date & Time)
 - ➔ Selected Date Range & Time
 - ➔ **Data of all the selected widgets** with **whatever filters** they have applied on the specific widget & showcased on the specific dashboard.

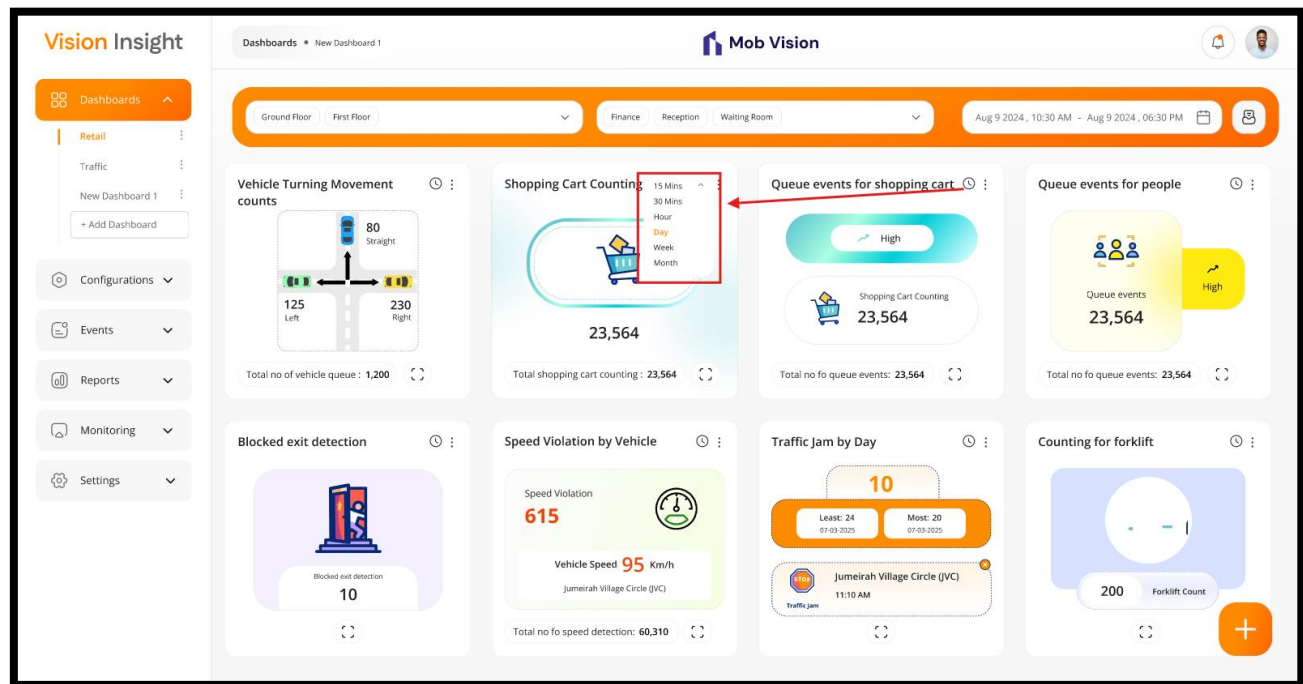
3.18.6. Dashboards > Dashboard Page > Widget > Duration Selection Filter Option

The system provides a feature for **authorized users** to **filter out the data** based on the **selected duration (time)** through the **BI Dashboard Web Application**.

This option will be available under **each widget (whenever it's applicable only)** which are **configured / added under the specific dashboard only**.

Prototype:

➔ The screen below illustrates the **option to select a duration**, based on which the system will **dynamically update the chart data**:



Business Rules:

- The system will allow users to select the following **duration filter** located under each **configured widget** on the dashboard:
 - ➔ 15 Mins
 - ➔ 30 Mins
 - ➔ Hour
 - ➔ Day
 - ➔ Week
 - ➔ Month
- The system will **manage & hide or show** the respective filter options based on the **selected date range** on the **specific dashboard only**.
- The **charts** which are associated with specific widget will **render & display the data** based on the **selected time filter option**.
- System will **automatically select the default** from these options based on the **selected date range**.

- Users can also change & select any **enabled value** from this filter option.

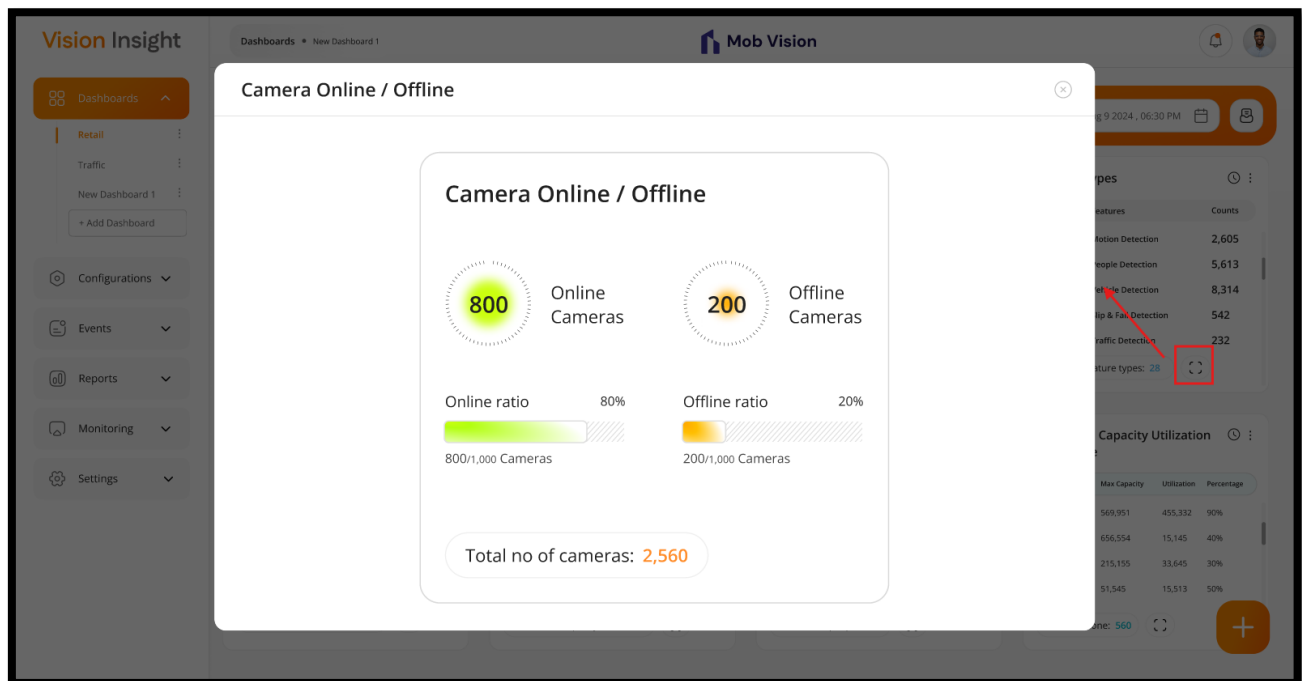
3.18.7. Dashboards > Dashboard Page > Widget > Expand Option

The system provides a feature for **authorized users** to **expand the view of any widget** associated / configured under the **specific dashboard** through the **BI Dashboard Web Application**.

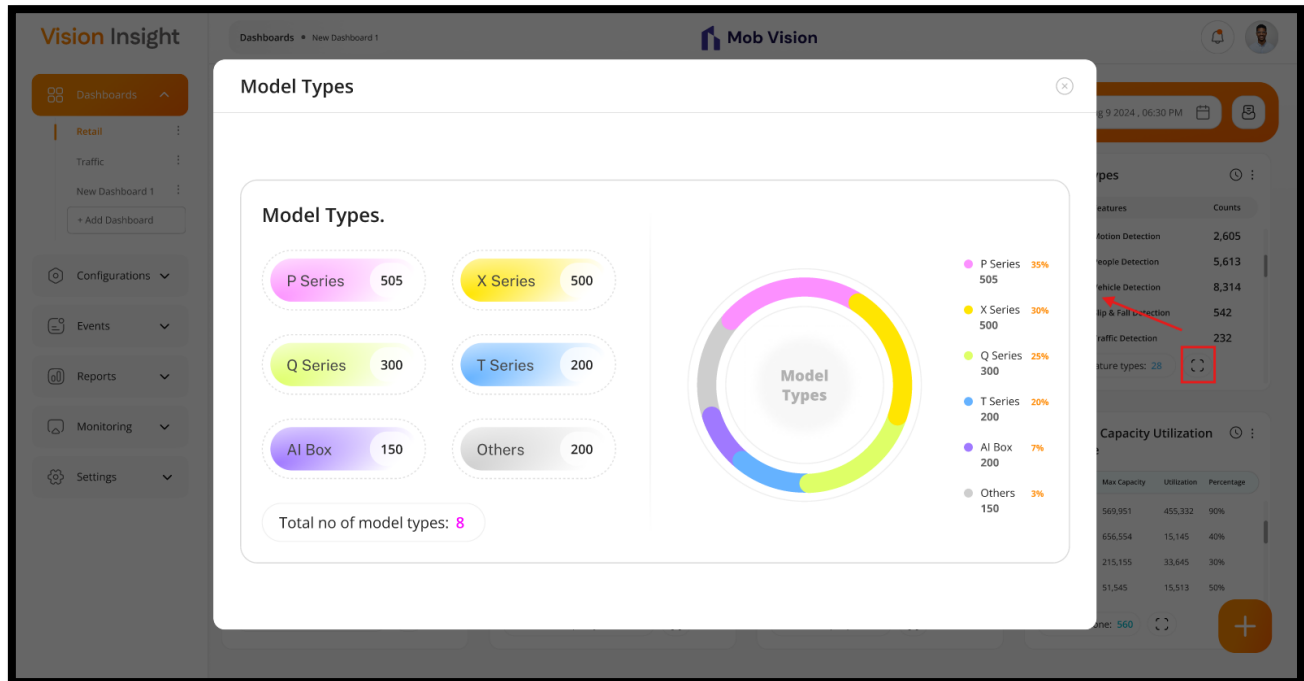
This option will be available under **each widget** (*whenever it's applicable only*) which are **configured / added under the specific dashboard only**.

Prototype:

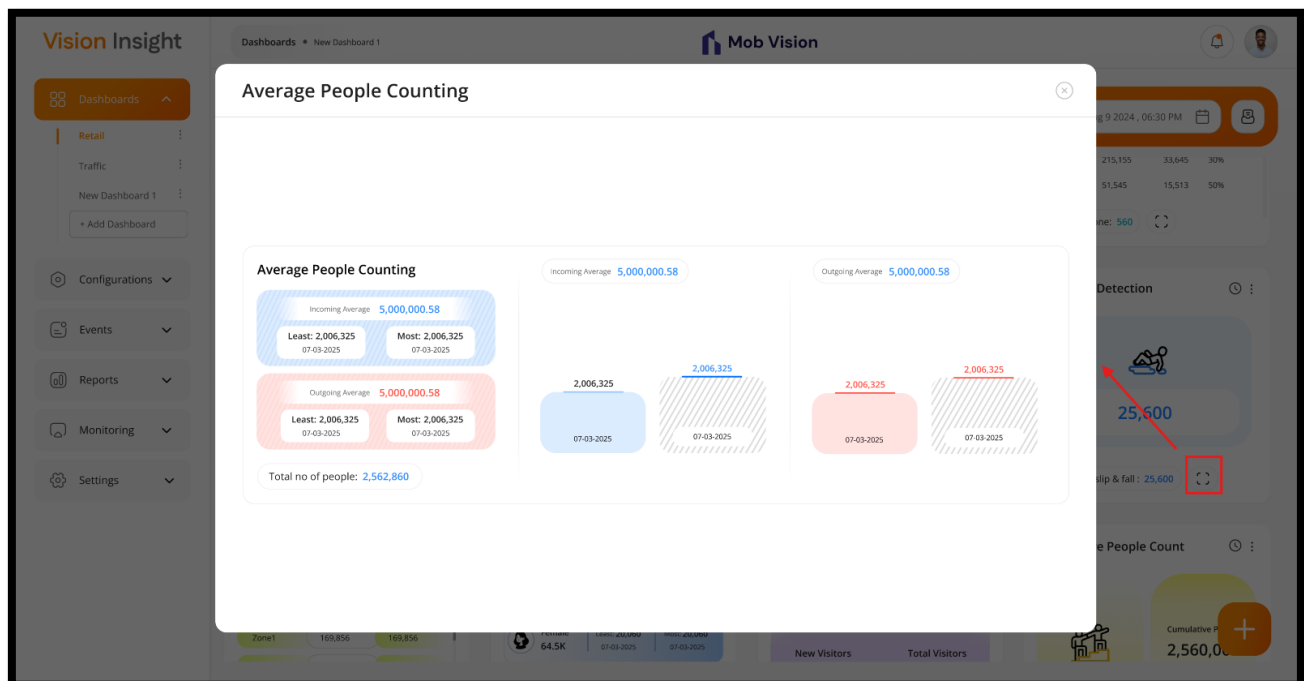
- The screen below illustrates the **option to open a configured widget preview** in an **expanded view** when the widget is set to **1*1 size**:



→ The screen below illustrates the **option to open a configured widget preview** in an **expanded view** when the widget is set to **2*1 size**:



→ The screen below illustrates the **option to open a configured widget preview** in an **expanded view** when the widget is set to **3*1 size**:

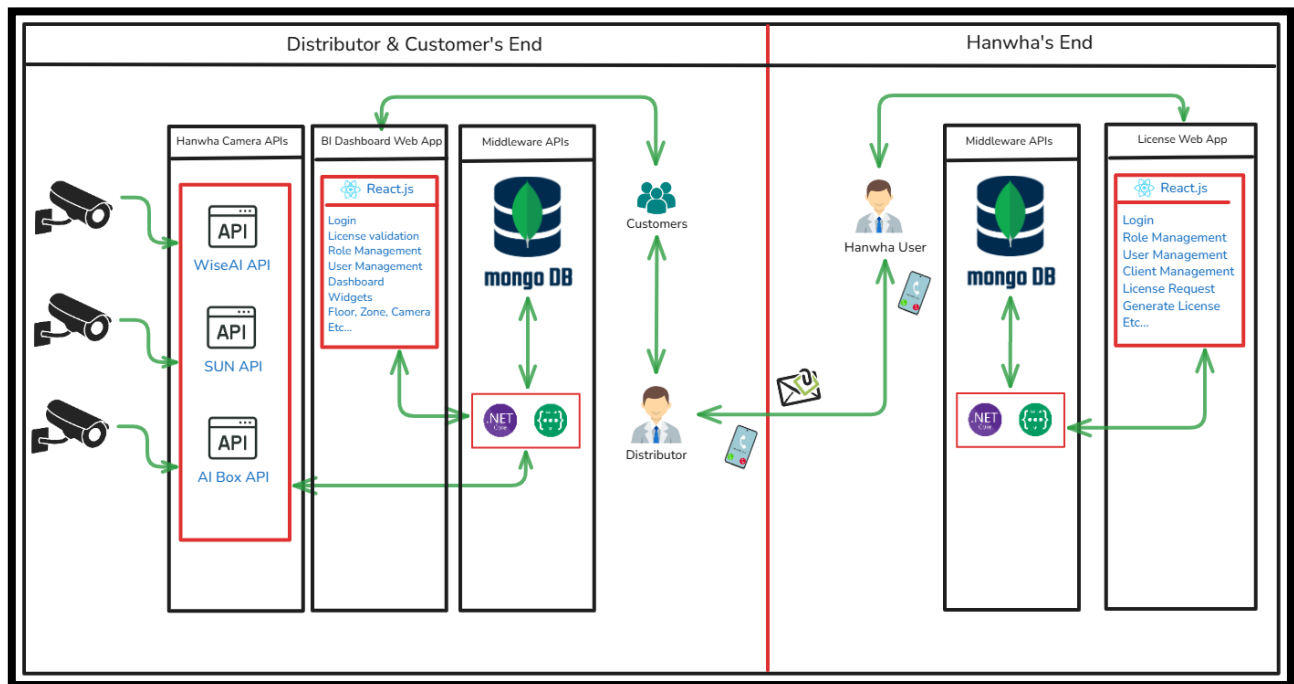


Business Rules:

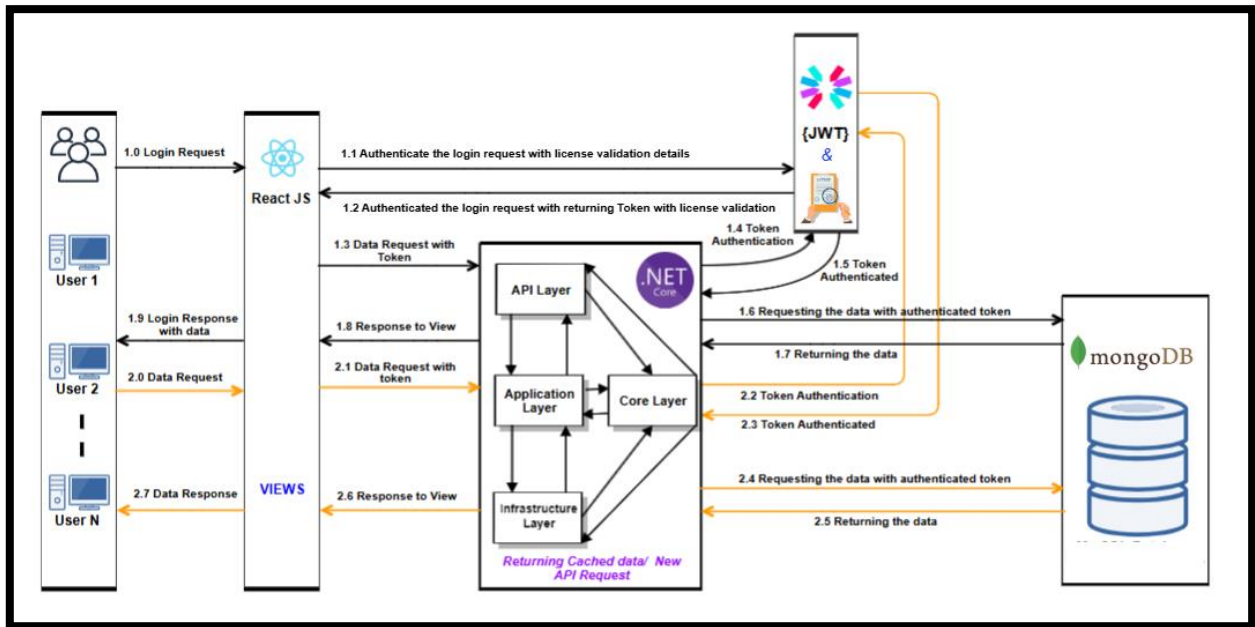
- The system will **open popup** when users will click on the “**Expand**” icon.
- Under that popup, the users will be able to see **widget in expanded view**.
- There will be option to **close the popup window**.

4. System Architecture

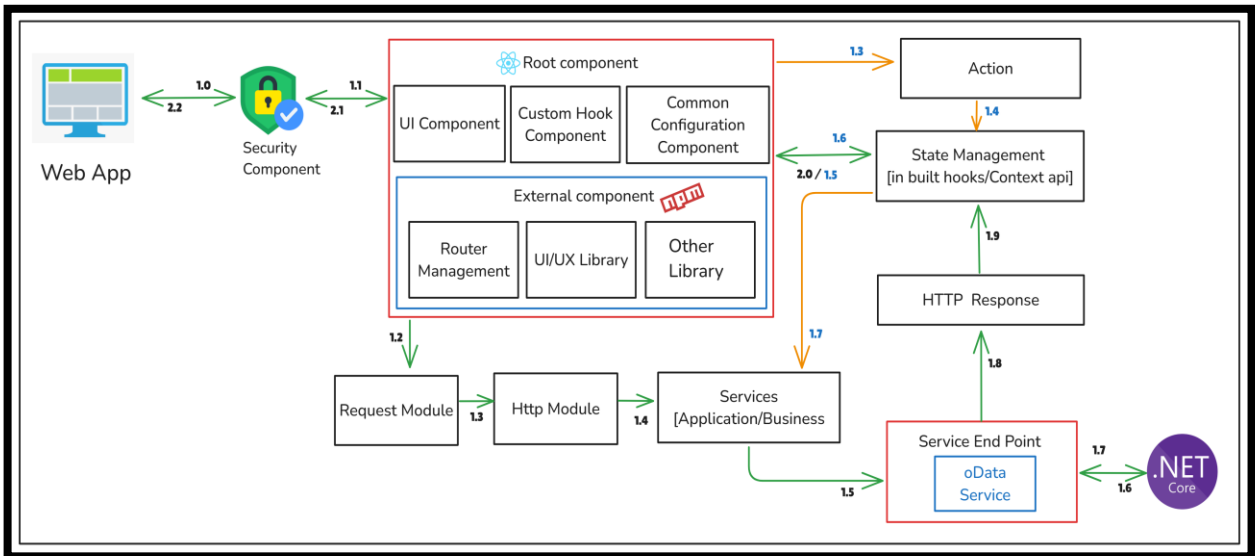
➔ The image below represents the flow of the **entire system architecture**, including all applications such as the **Bi Dashboard Web Application**, **BI Dashboard Web Application**, and the **backend middleware for both applications**:



→ The architecture below represents the flow of **Backend Development Using .NET Core + Web API:**



→ The architecture below represents the flow of **Frontend Development Using React.js Technology:**



5. Assumptions, Dependencies and Constraints

- The above attached screenshots in the document are taken from the **Figma design**. Please note that there may be **minor variations** in the **UI** during actual implementation. However, the **structure, content, and data** presented will remain **consistent** with the **screenshots provided**.
- Any type of Server hosting fees, license cost, any other 3rd party API/SDK fees are not part of development cost and will be procured and paid by client.
- SPEC will have no liability for hardware failure deployed when running the proposed solution. Hence, the client should only take up relevant issues with the provider for the respective system.
- Designs showcased in this SRS document as well as the ones shown to the client during the project query discussion calls will be considered as the final designs.
- SPEC INDIA will not be responsible for any sort of server setup on the client's UAT or Production Environment.
- Database administration activities will not fall under the responsibility of SPEC INDIA team.
- Once the application is live in Production, the database maintenance plan must be configured from the client's end.
- SPEC INDIA will not be responsible for any Point-in-Time Database Recovery/DR Drill/Database Upgrade/Maintenance tasks etc.
- The details are prepared based on the client's discussion; hence, changes may be made after completing the requirements study, which may require discussions under the change management process.
- Any delay in the project schedule due to decision-making, data gathering, or approvals from the client side should be handled at the client's side. SPEC INDIA won't be responsible for adding more resources for early completion in such scenarios.
- The warranty period will automatically start ten days after the UAT rollout of both the applications (Client Web App & License Web App) & the **warranty period will be 3 months**.
- After warranty completion, extra charges will be charged for knowledge transfer or documentation for/to another vendor.
- The client will own the developed product and source code submitted after the final milestone payment.
- A delay during any phase of the project by the customer will not be considered a delay by SPEC INDIA, and any payment milestone will not be delayed due to it.
- Any delay in providing feedback or required dependency for more than two weeks will be escalated by SPEC INDIA.

---- end of document ----